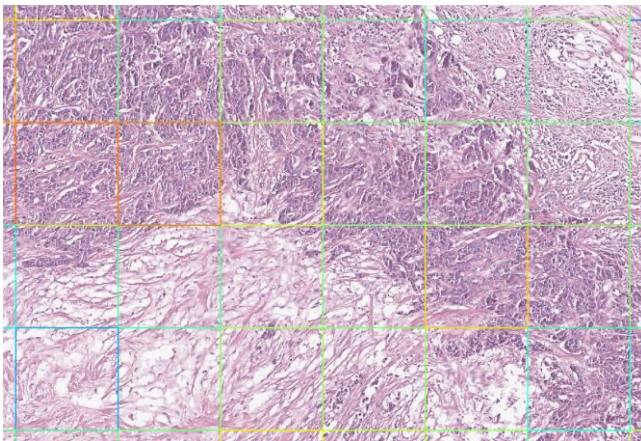


슬라이드 별 ROI 영역 선정 절차 및 참고 사항

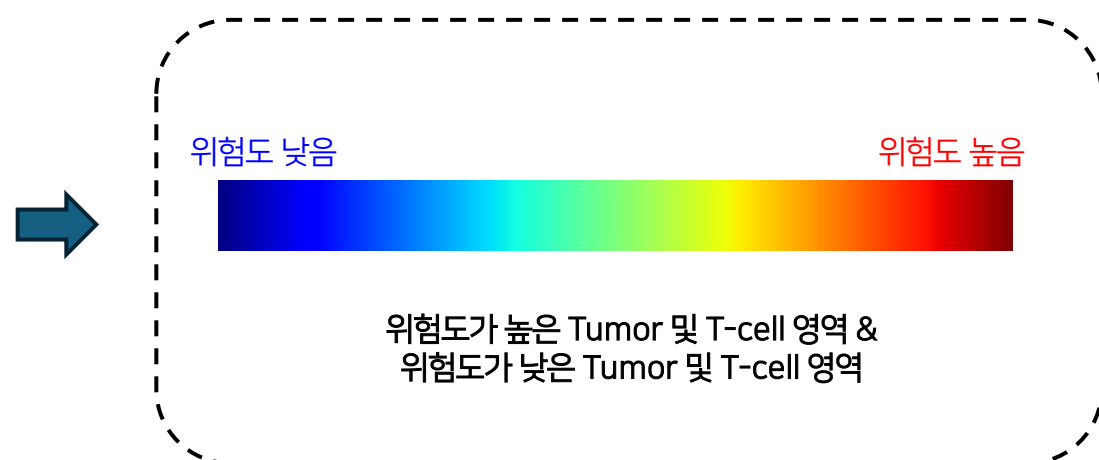
환자 : 1-085_12

- H&E 염색 슬라이드 이미지에서 작은 패치 단위로 분할하여 분석 수행
- 작은 패치의 크기는 개략적으로 약 $270\mu\text{m}$ (정확한 수치는 아니고 근사값, 패치 간에 오차가 있을 수 있음)
- .tmpprotocol 파일을 통해 annotation 파일 전달 ("[환자ID]_ROI_Updated.tmpprotocol")
- 한 튜브에 2개 이상의 패치가 들어감 (평균적으로 4개) -> 이전의 한 개 튜브에 한 패치가 아닌 여러 개의 패치가 한 튜브에 들어감
- 한 튜브에 들어갈 이웃 패치들은 위치적으로 근접하고 (경우에 따라서는 아닐 수도 있음) AI 예측 점수가 유사한 애들끼리 클러스터로 묶음
- 한 환자 당 48개의 튜브를 구성할 수 있는 ROI 영역 선정 완료
- ROI는 Tumor 영역과 T-cell 영역, AI 점수가 높은 곳과 낮은 곳이 구분되는 영역을 위주로 선별 (AI 예측 결과 활용, tiatoolbox)

사각형 사이즈 하나당 약 $270\mu\text{m}$ 크기를 가짐



각 사각형 색상의 의미 -> "AI 재발 예측 위험도"



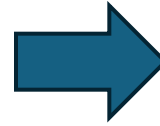
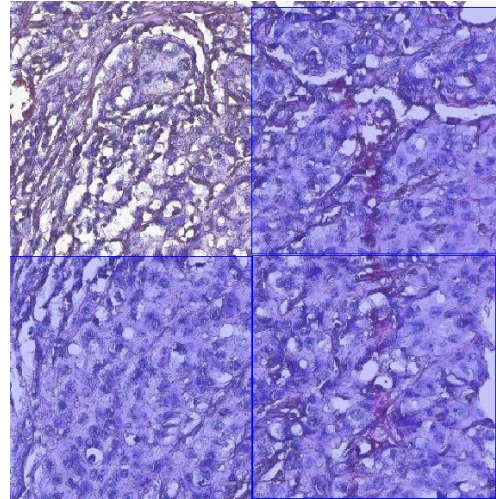
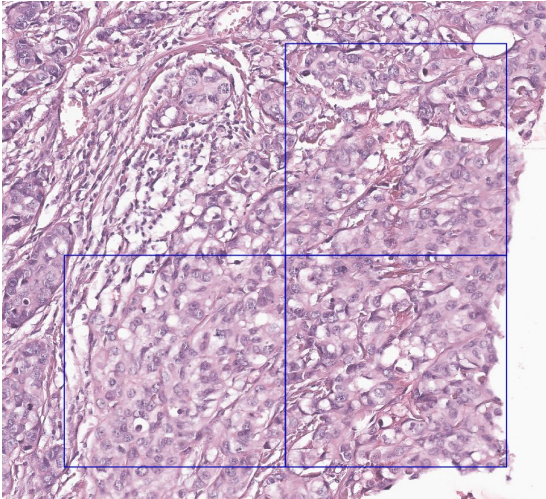
- 1-085-12 환자에서 점수 대에 따른 Region 별 분포

AI 스코어 (위험도)	0.00-0.10	0.10-0.20	0.20-0.30	0.30-0.40	0.40-0.50	0.50-0.60	0.60-0.70	0.70-0.80	0.80-0.90	0.90-1.00	
High score & Tumor (section a)	-	-	-	-	-	-	5	3	2	-	10
Low score & Tumor (section b)	1	11	9	-	-	-	-	-	-	-	21
Tumor Control (section t)	-	-	-	-	2	1	-	-	-	-	3
High score & T-cell (section c)	-	-	-	-	-	3	2	-	-	-	5
Low score & T-cell (section d)	-	-	1	4	4	-	-	-	-	-	9
T-cell Control (section u)	-	-	-	-	-	-	-	-	-	-	0
											48

총 48개의 well 에 들어갈 region 선정!

Naming rule에 따라 구분!

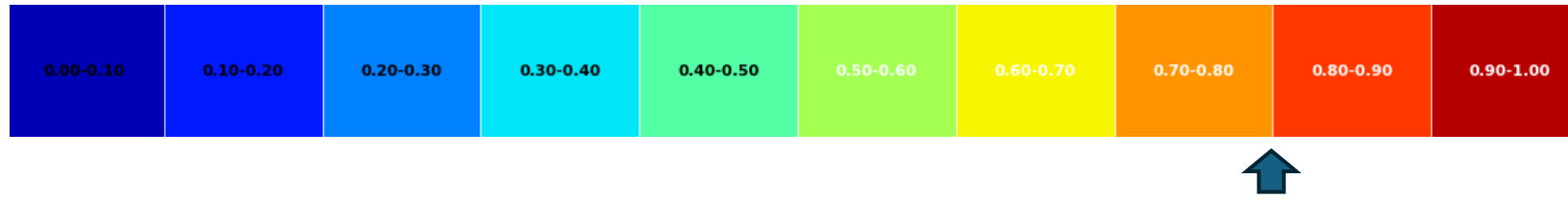
- 아래와 같이 영역이 선정된 경우, 이 경우 b1으로 naming이 되어 있으면, .tmpprotocol 파일에도 annotation 되어 있음



68	☒	b1	B12	☒
69	☒	b1	A12	☒
70	☒	b1	B12	☒

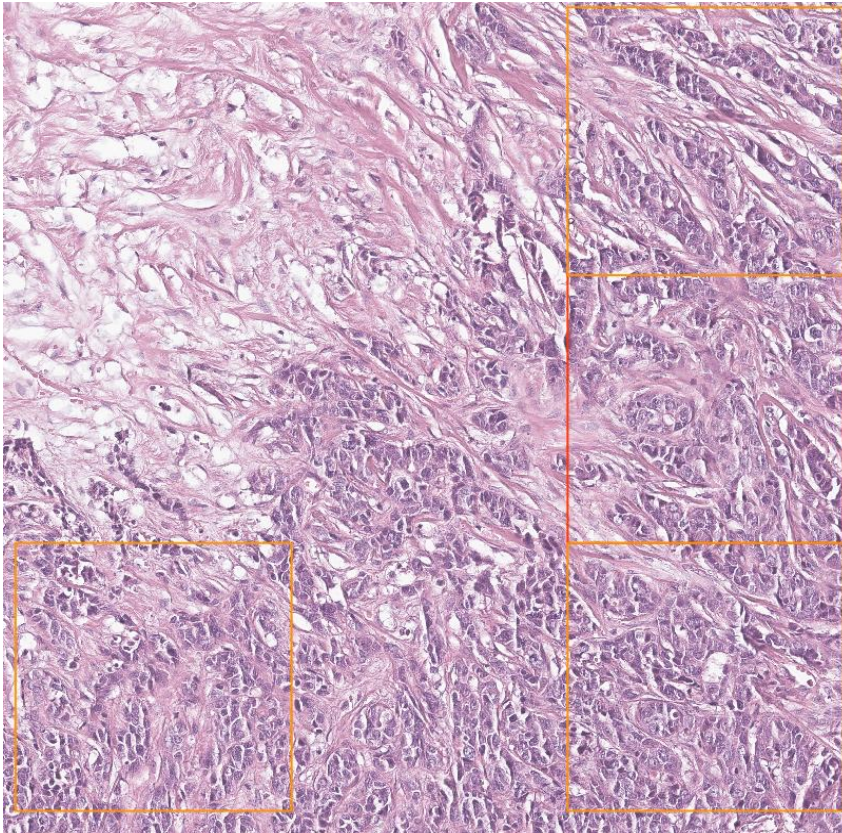
ROI Regions

OmixAI 1-085-12 환자 샘플 high-score tumor regions

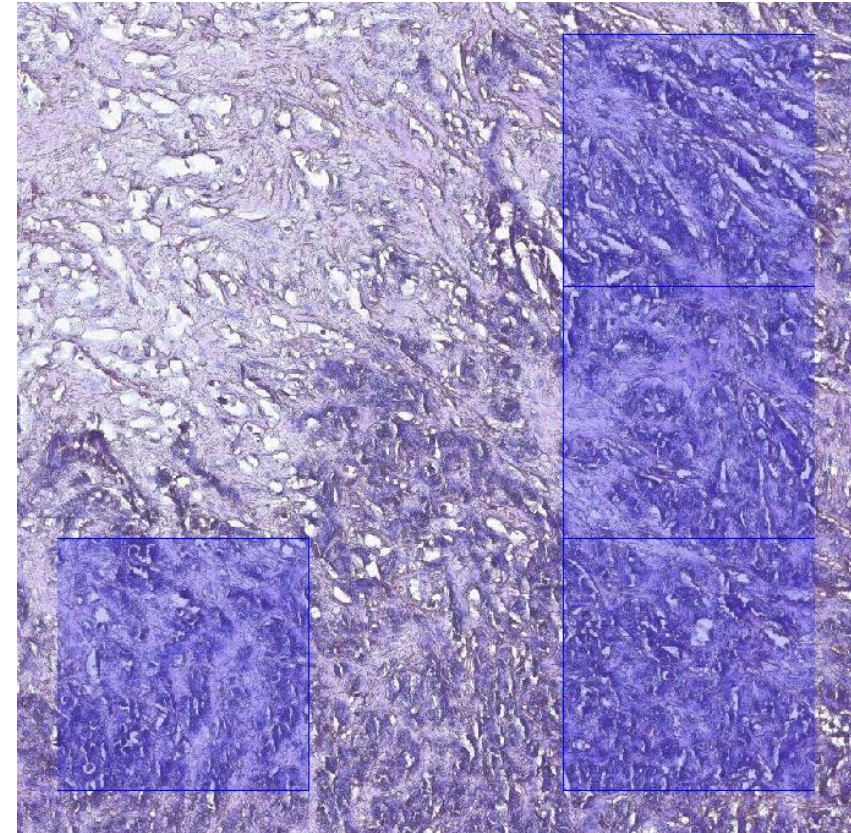


- Section **a1** => High Score & Tumor

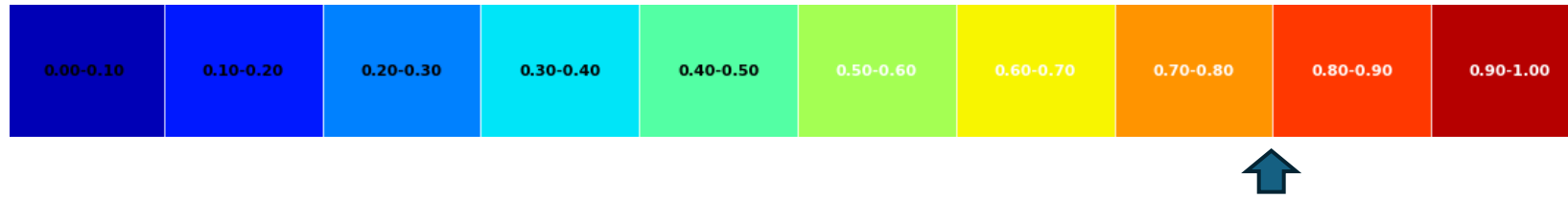
AI 스코어



Astromapper 매핑 결과

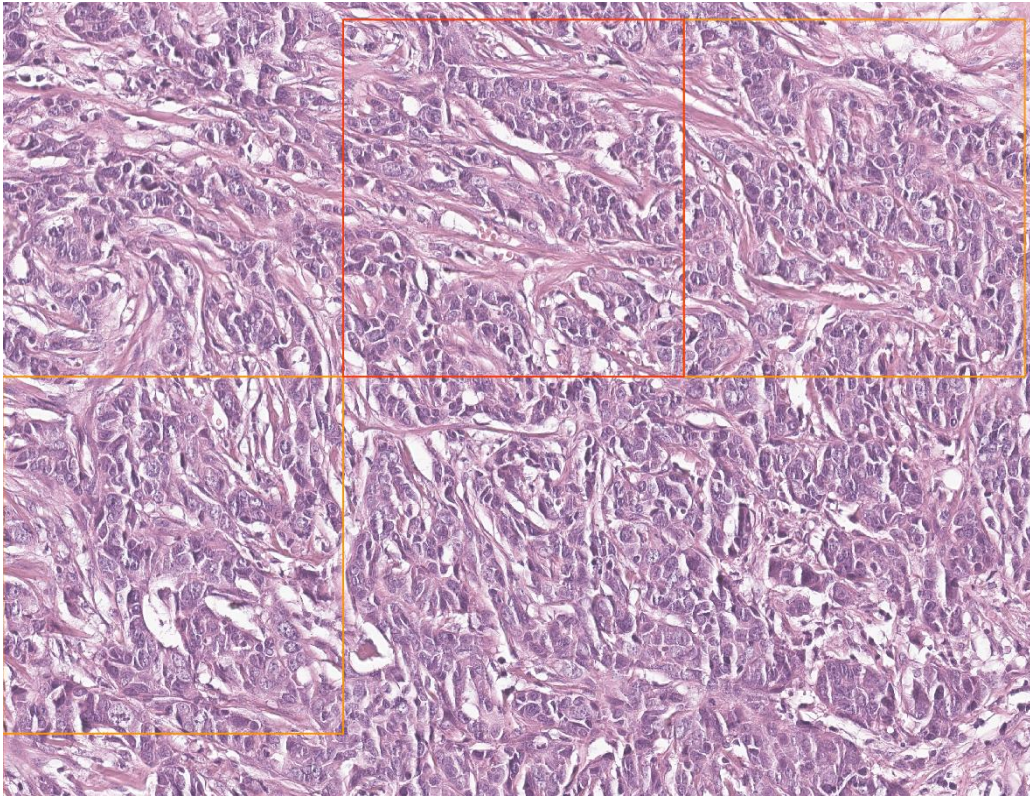


OmixAI 1-085-12 환자 샘플 high-score tumor regions

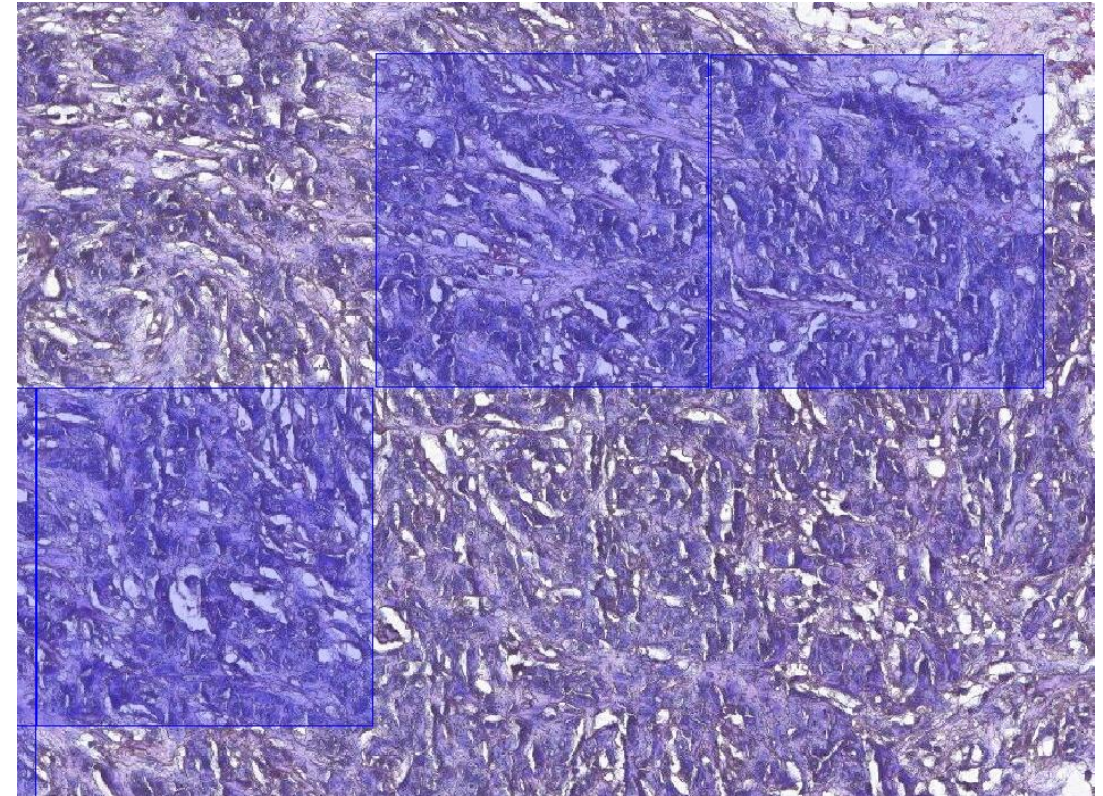


- Section **a2** => High Score & Tumor

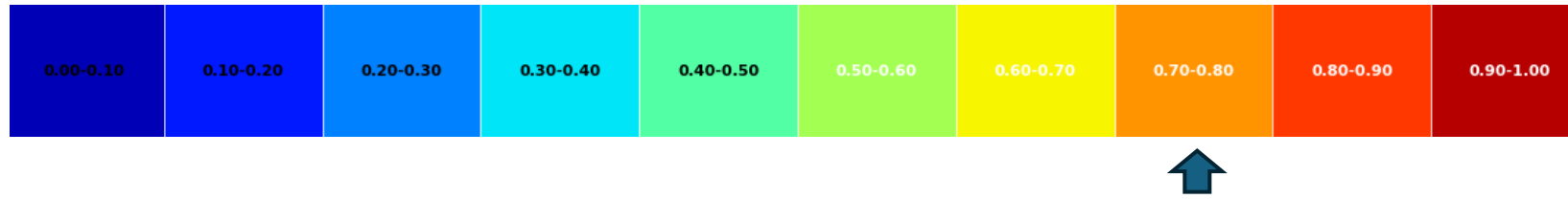
AI 스코어



Astromapper 매핑 결과

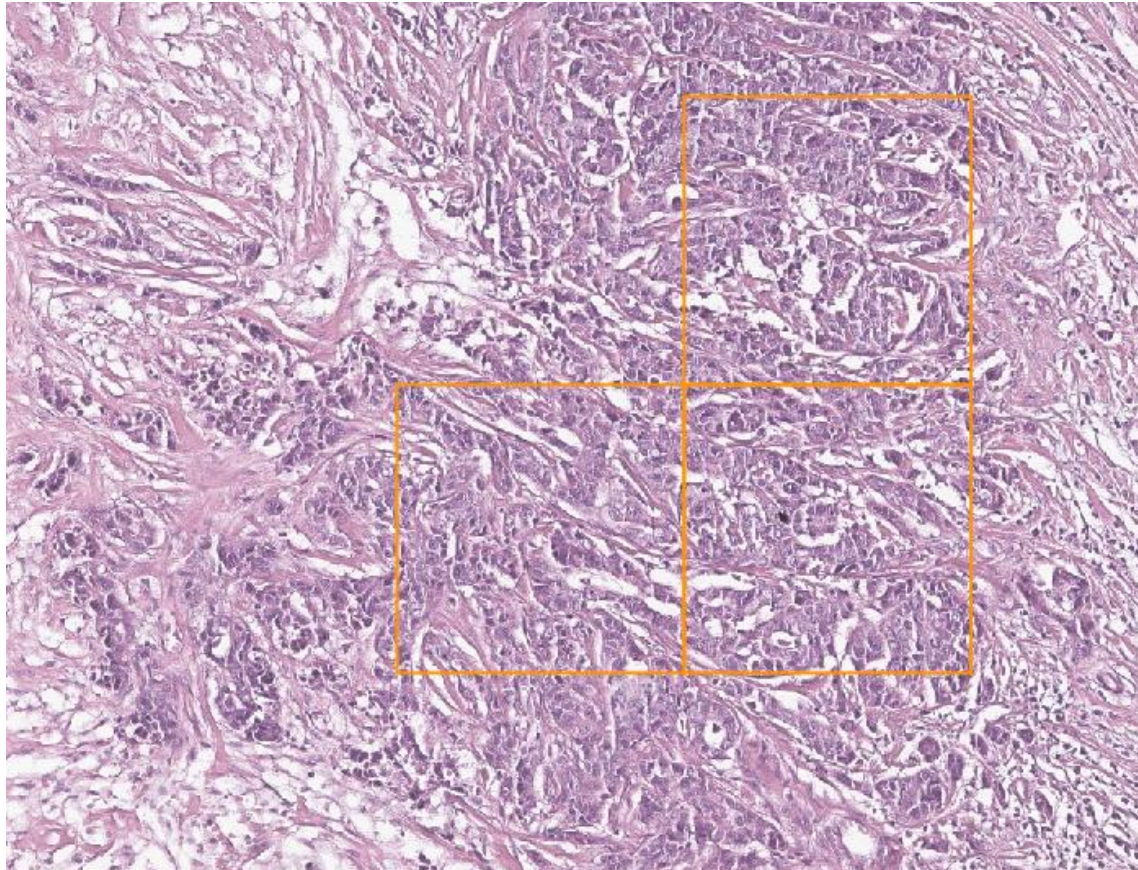


OmixAI 1-085-12 환자 샘플 high-score tumor regions

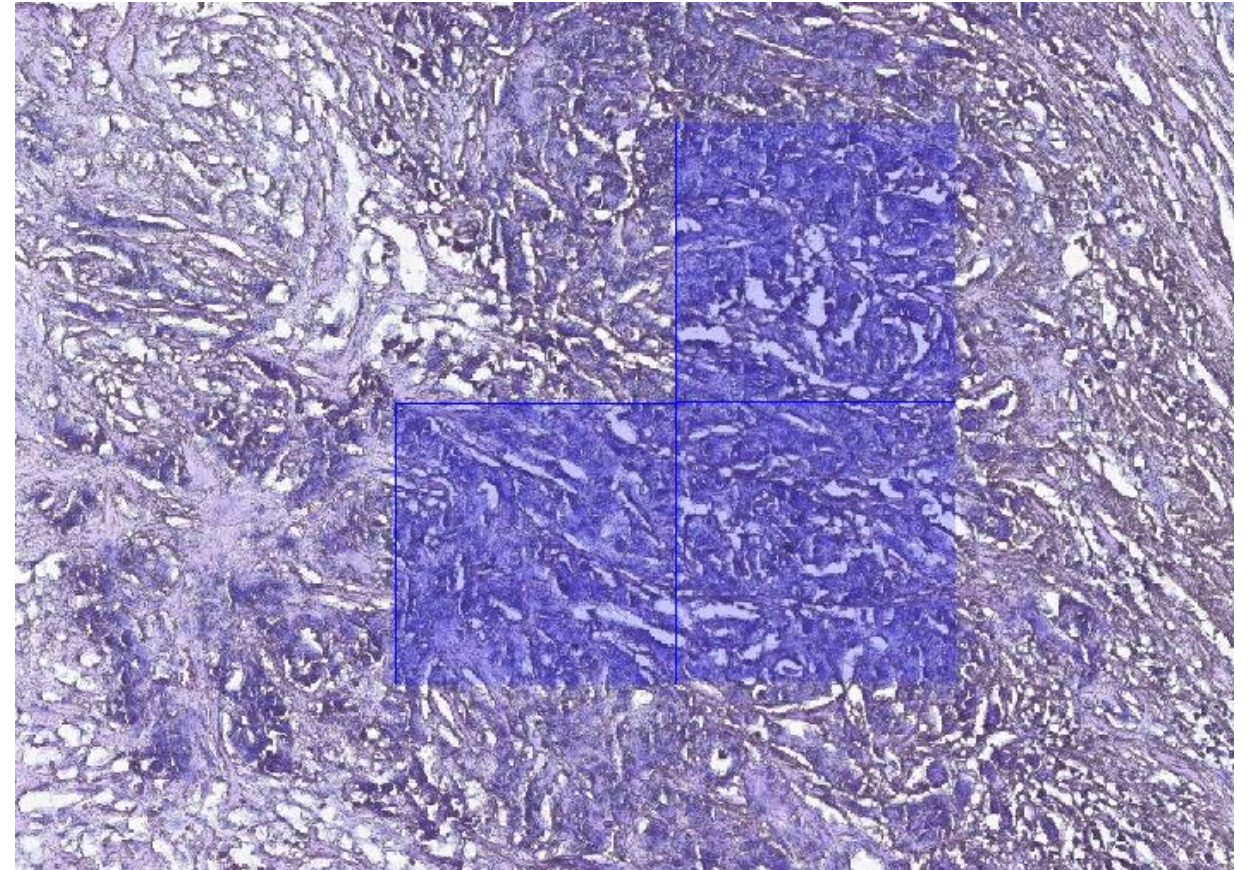


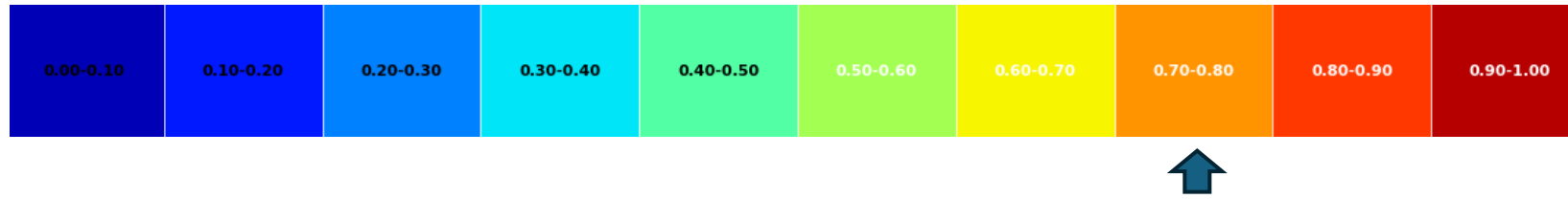
- Section **a3** => High Score & Tumor

AI 스코어



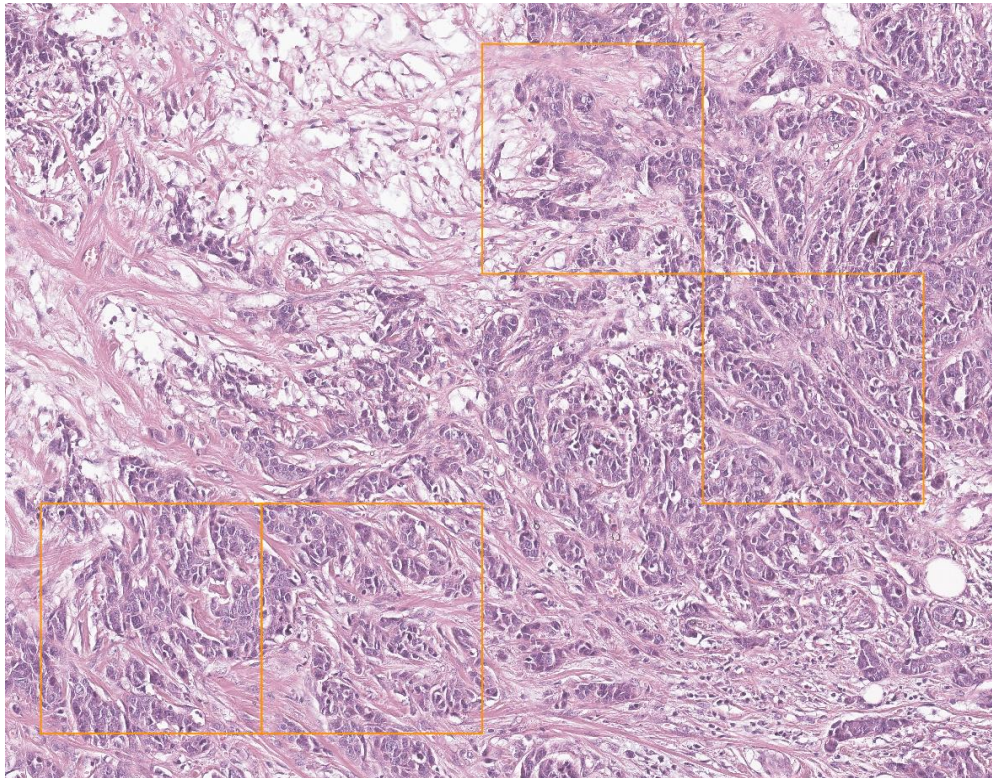
Astromapper 매핑 결과



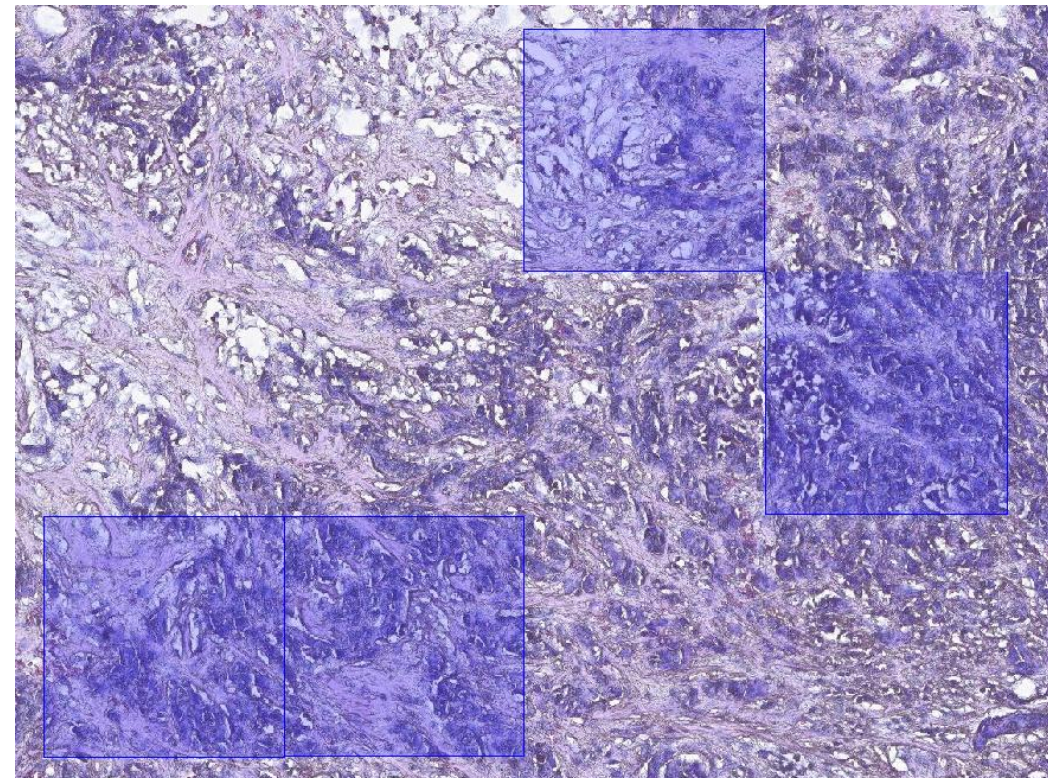


- Section **a4** => **High Score & Tumor**

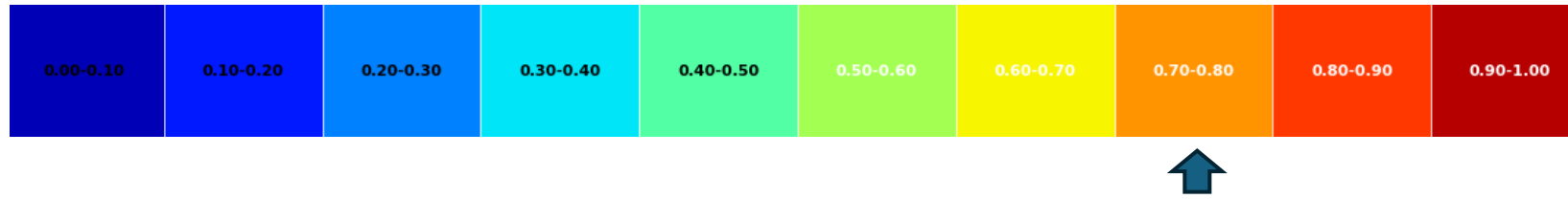
AI 스코어



Astromapper 매핑 결과

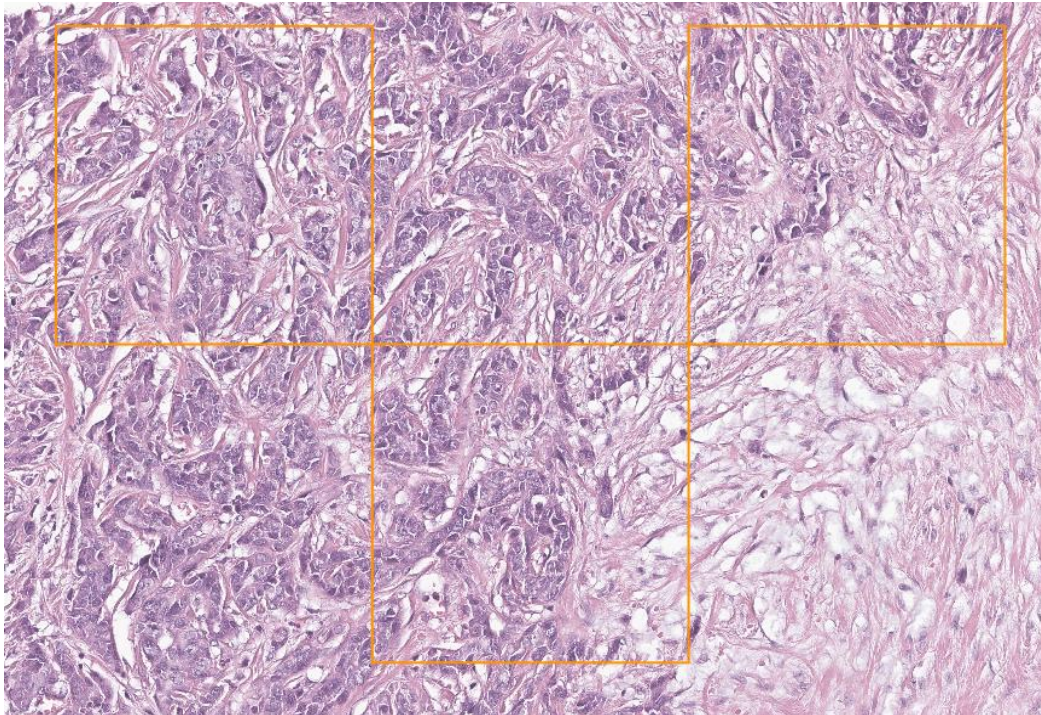


OmixAI 1-085-12 환자 샘플 high-score tumor regions

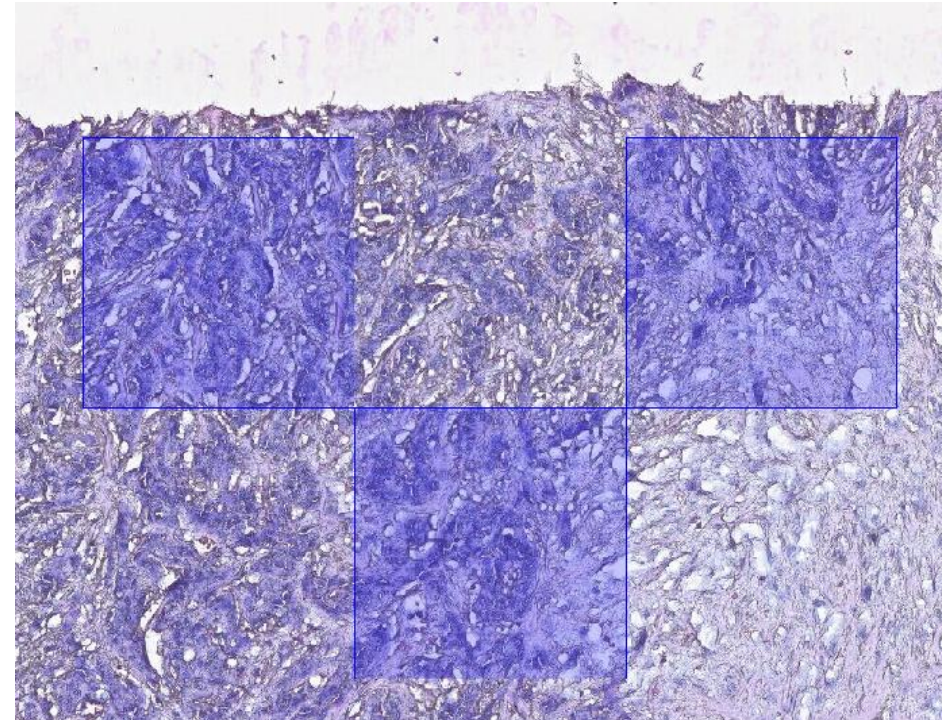


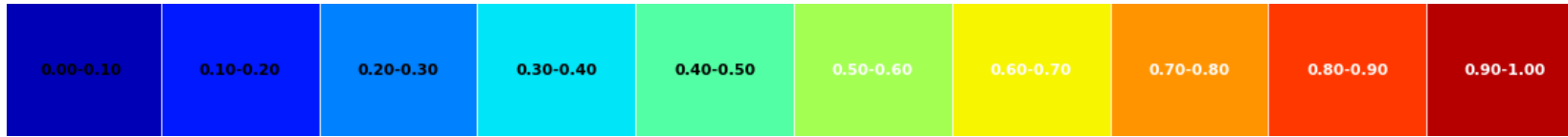
- Section **a5** => High Score & Tumor

AI 스코어



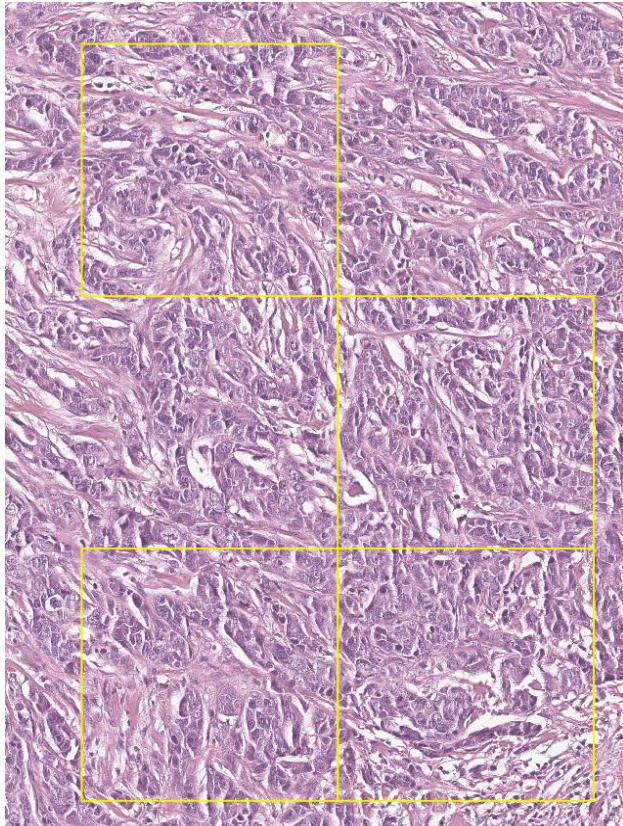
Astromapper 매핑 결과



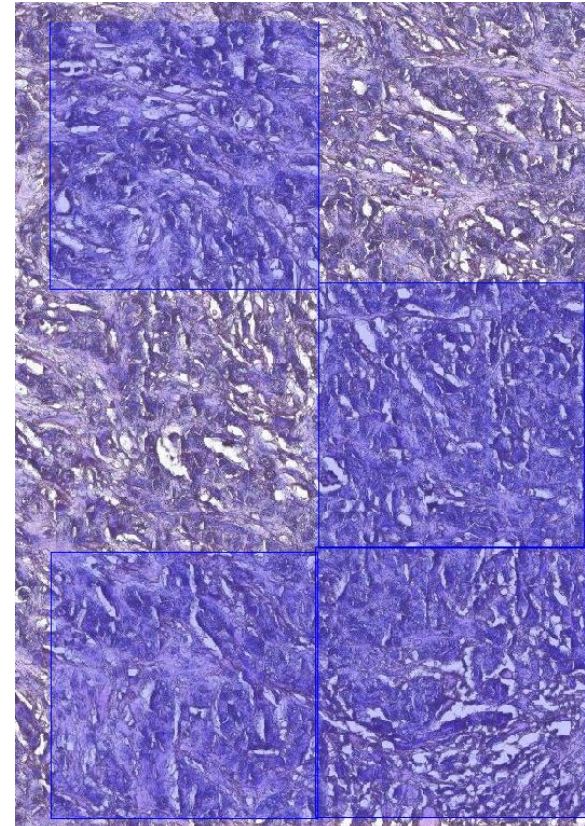


- Section a6 => High Score & Tumor

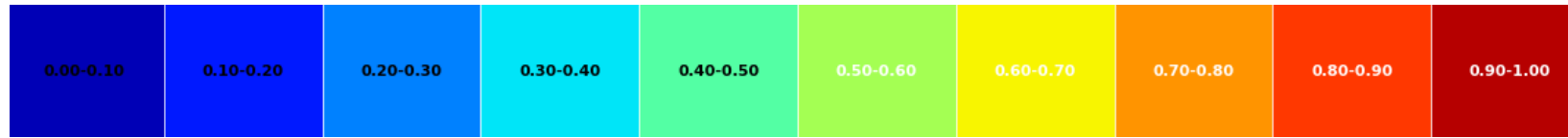
AI 스코어



Astromapper 매핑 결과

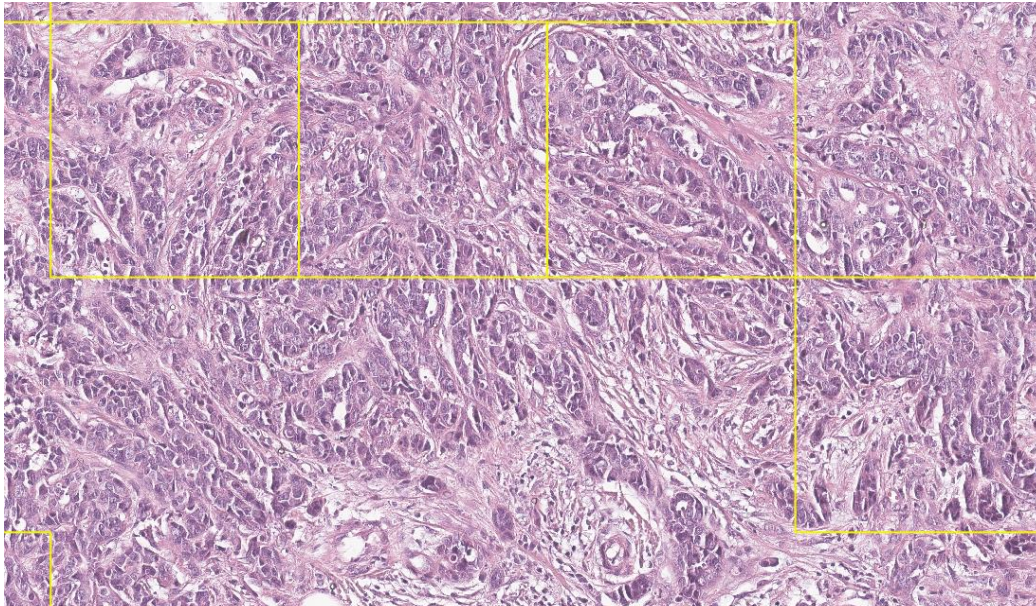


OmixAI 1-085-12 환자 샘플 high-score tumor regions

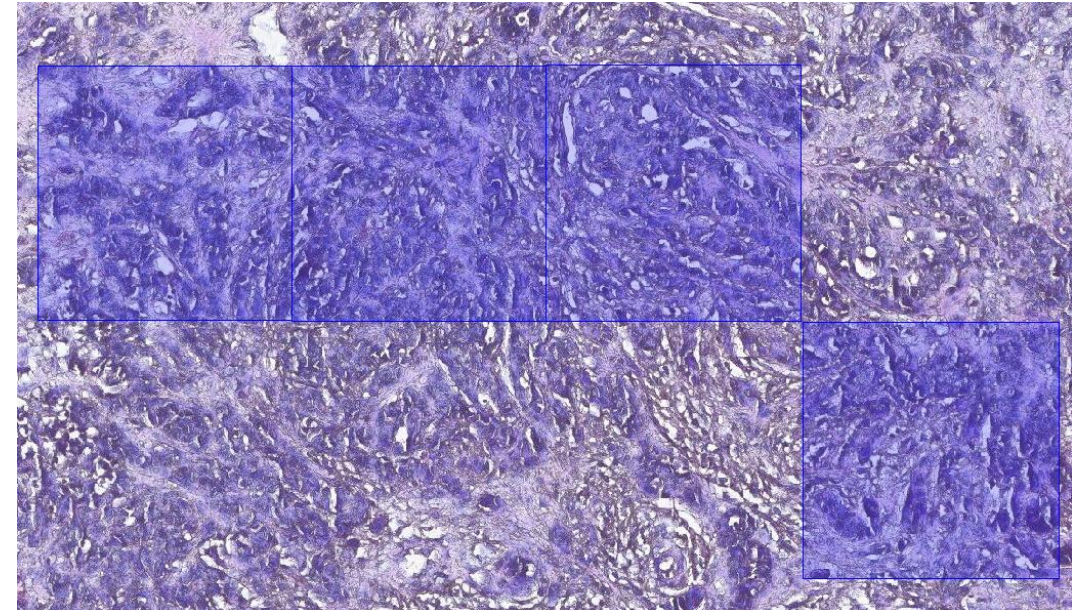


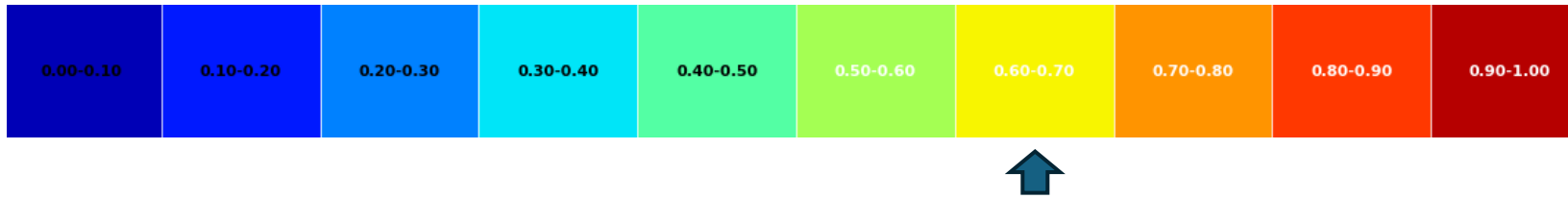
- Section **a7** => High Score & Tumor

AI 스코어



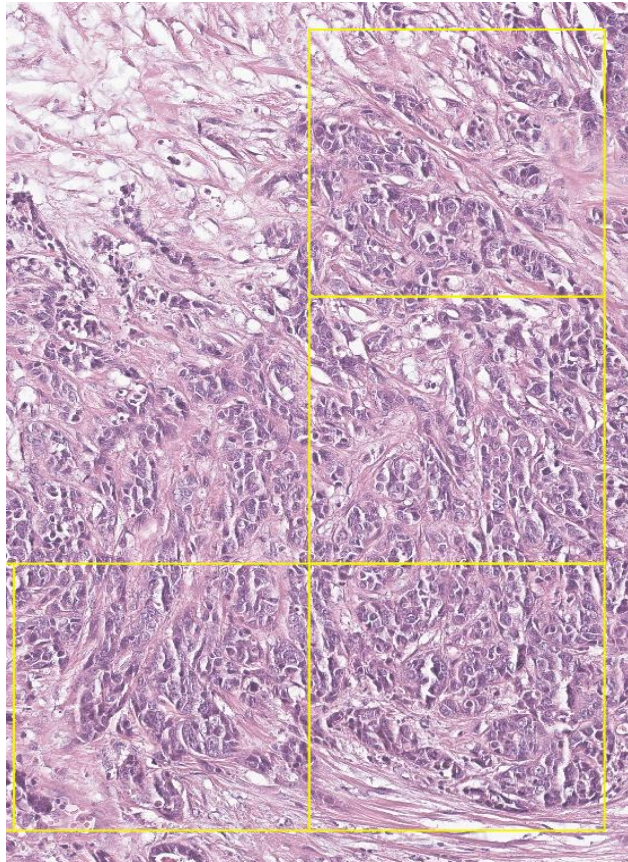
Astromapper 매핑 결과



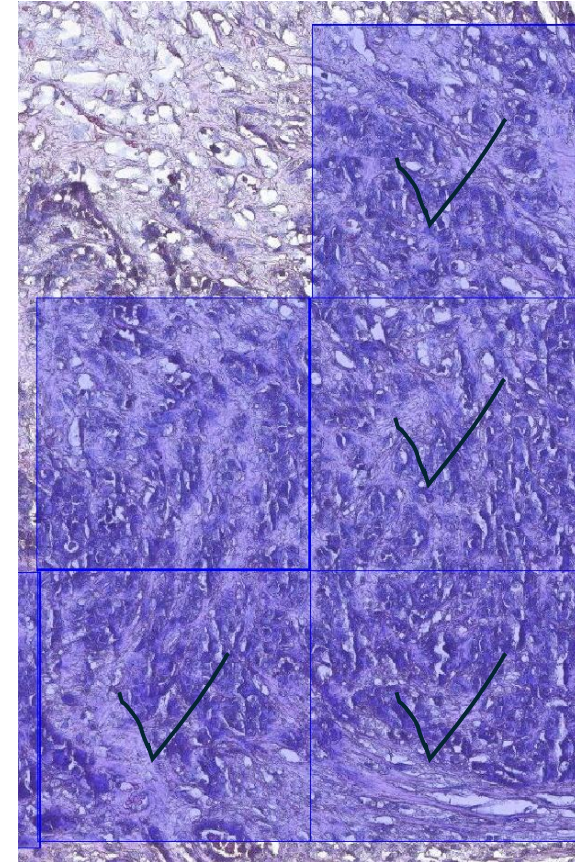


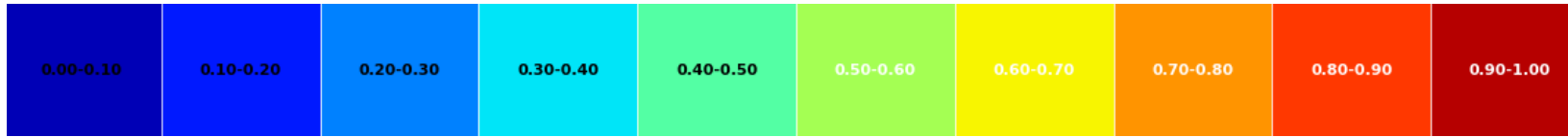
- Section **a8** => High Score & Tumor

AI 스코어



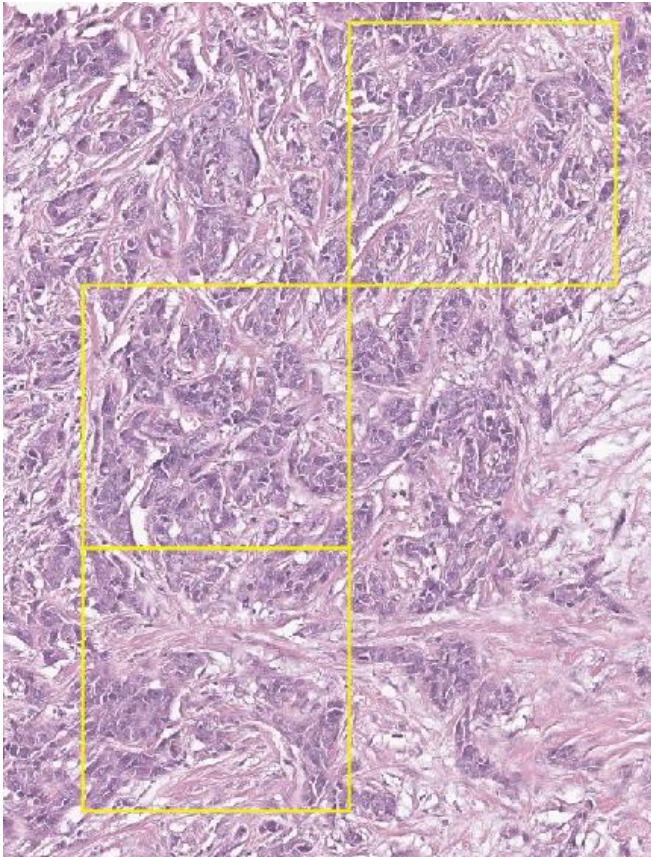
Astromapper 매핑 결과



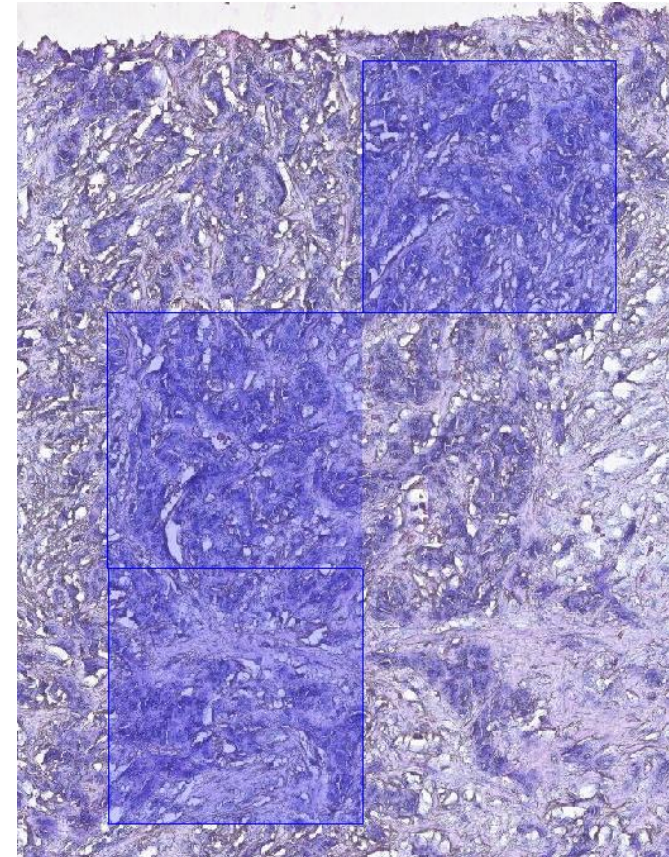


- Section **a9** => High Score & Tumor

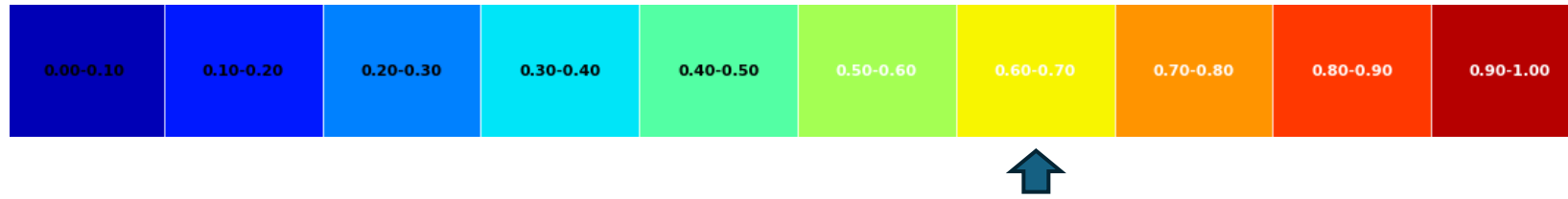
AI 스코어



Astromapper 매핑 결과

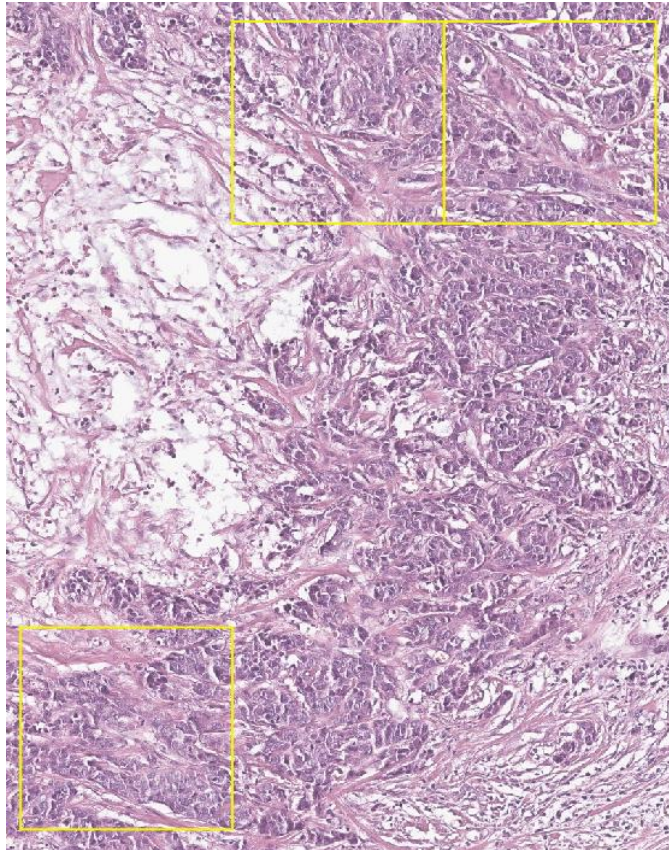


OmixAI 1-085-12 환자 샘플 high-score tumor regions

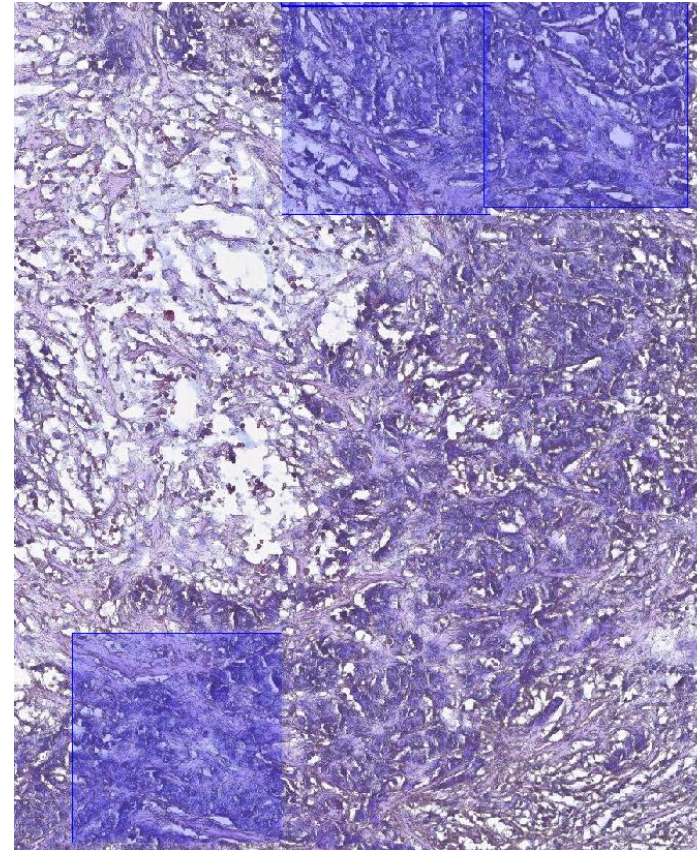


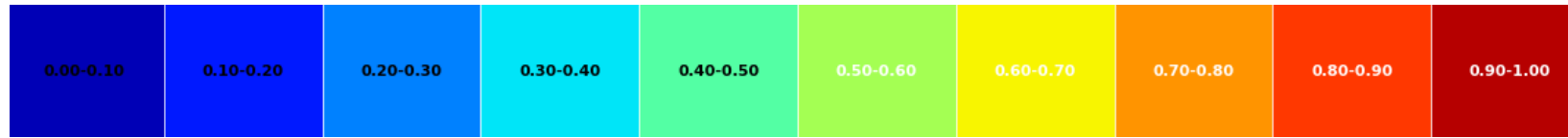
- Section a10 => High Score & Tumor

AI 스코어



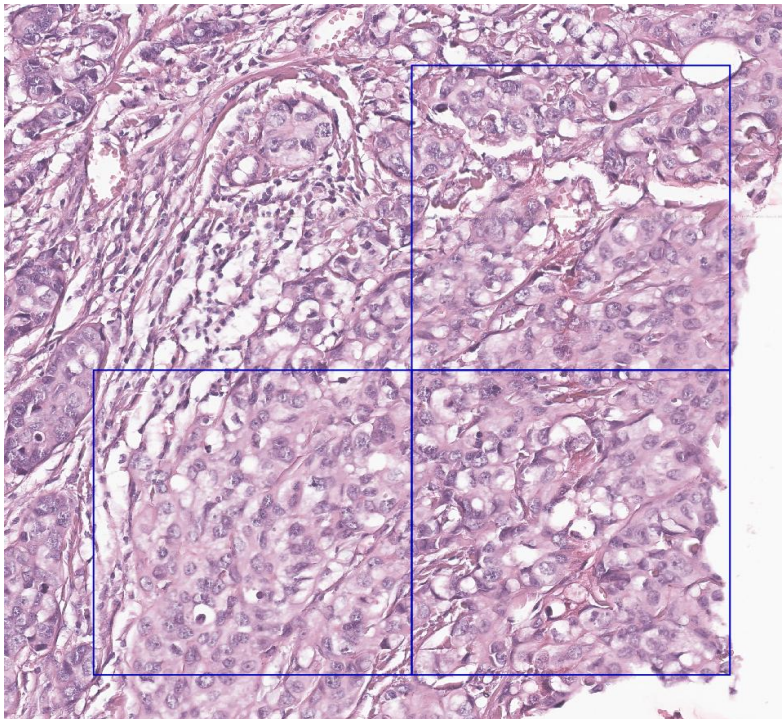
Astromapper 매핑 결과



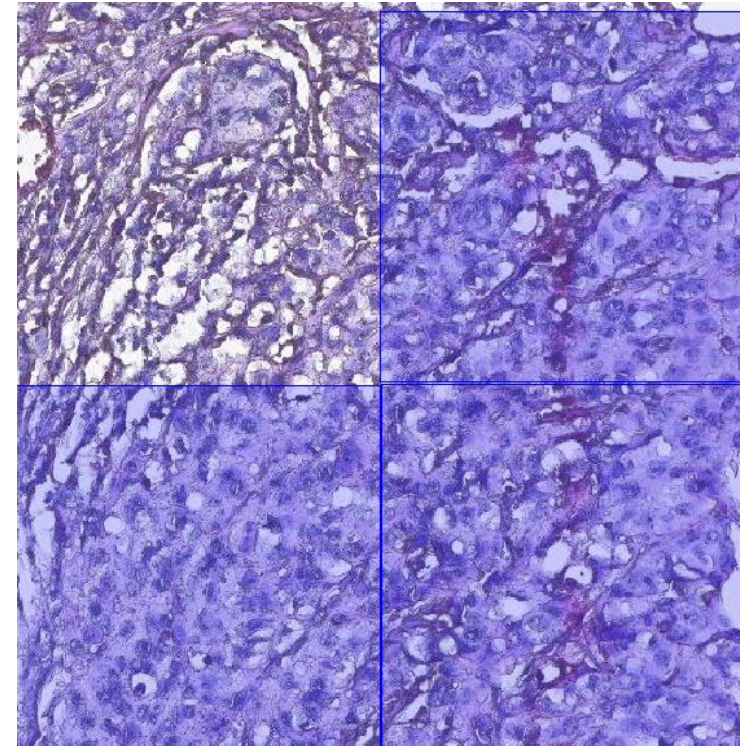


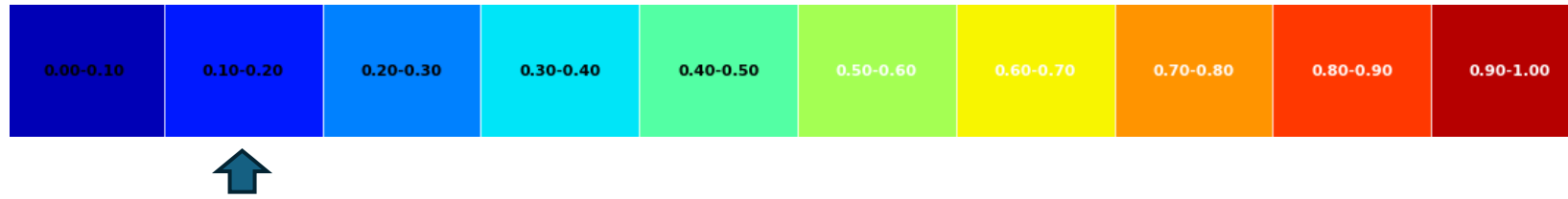
- Section **b1** => Low Score & Tumor

AI 스코어



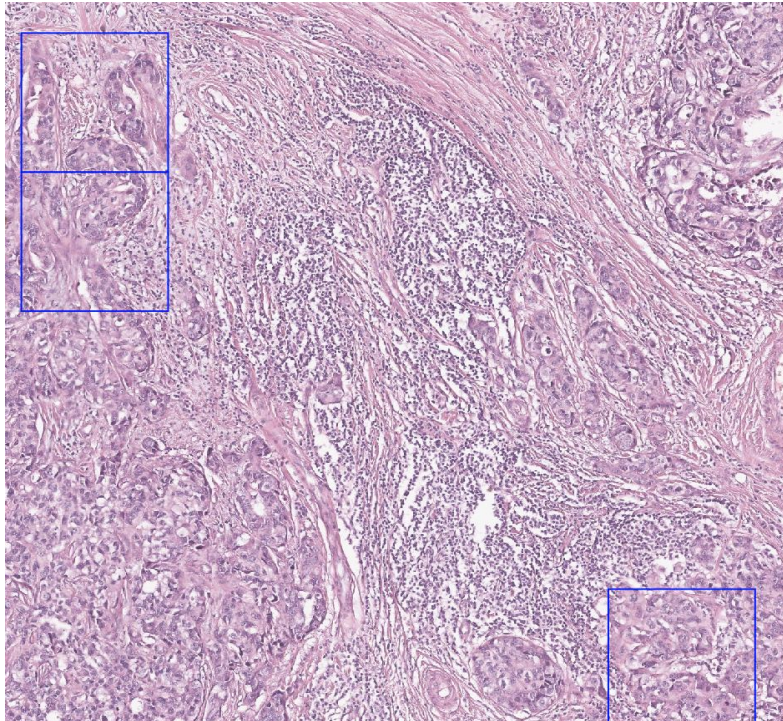
Astromapper 매핑 결과



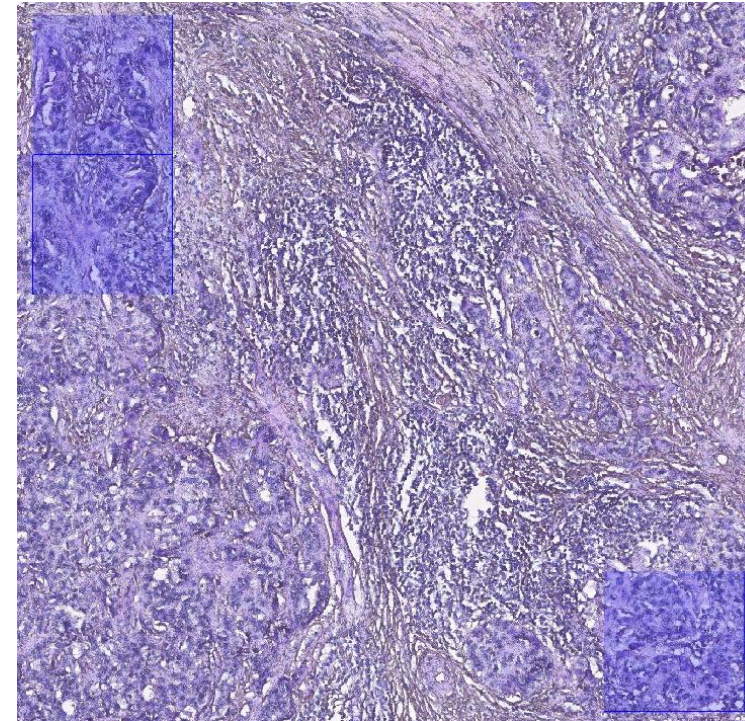


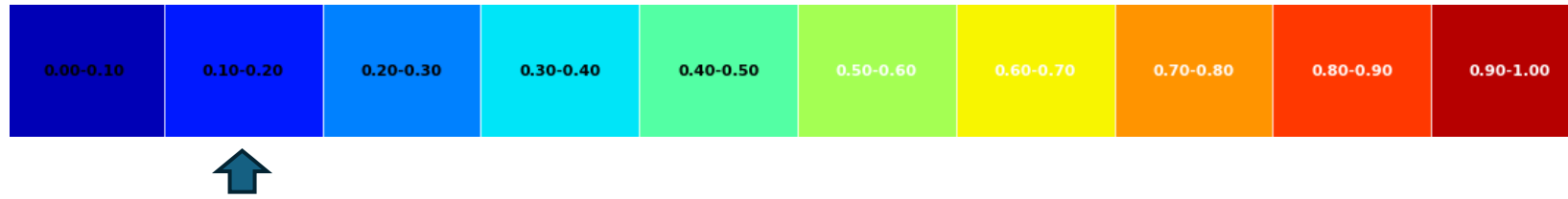
- Section **b2** => Low Score & Tumor

AI 스코어



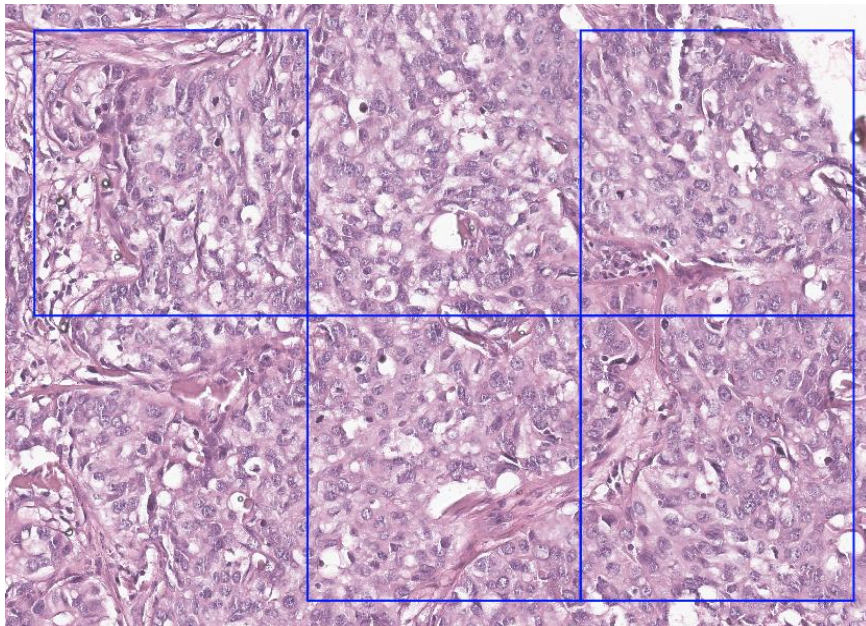
Astromapper 매핑 결과



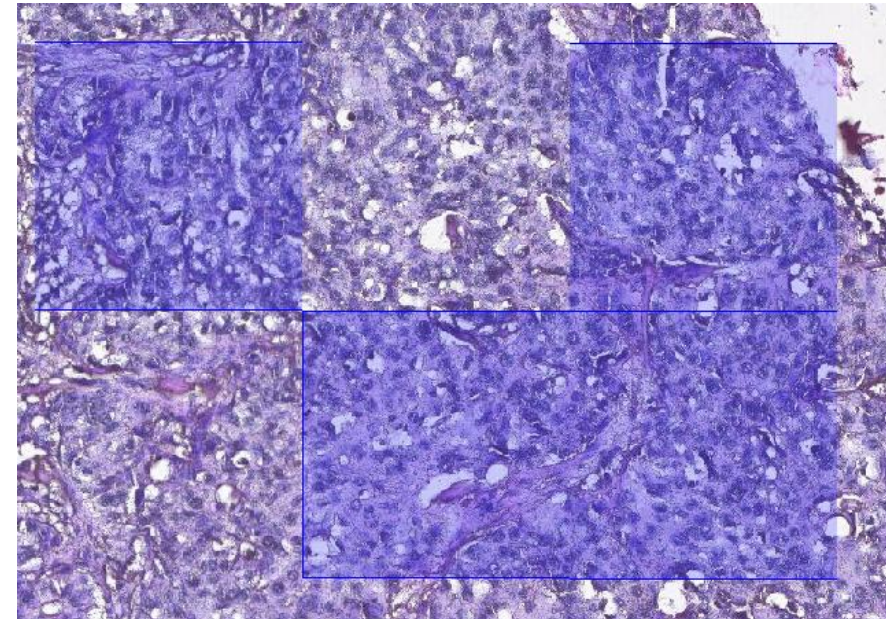


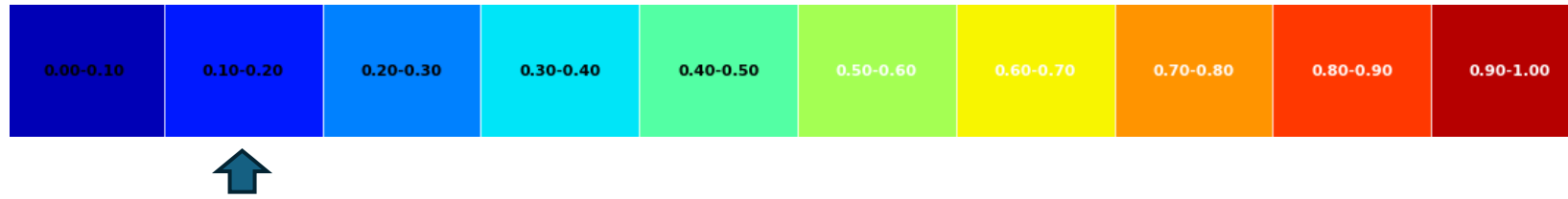
- Section **b3** => Low Score & Tumor

AI 스코어



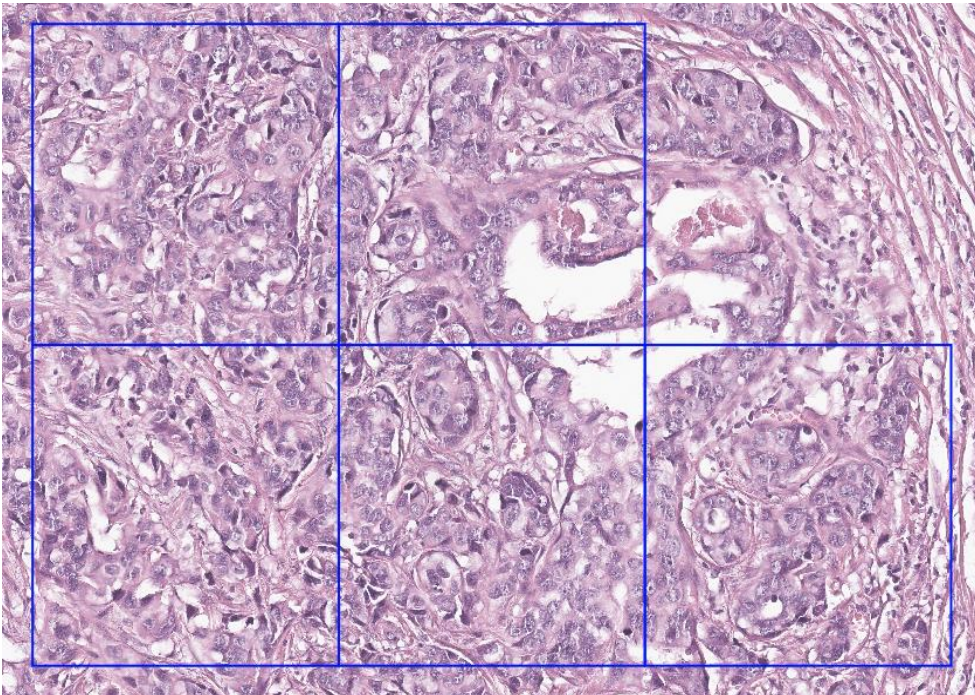
Astromapper 매핑 결과



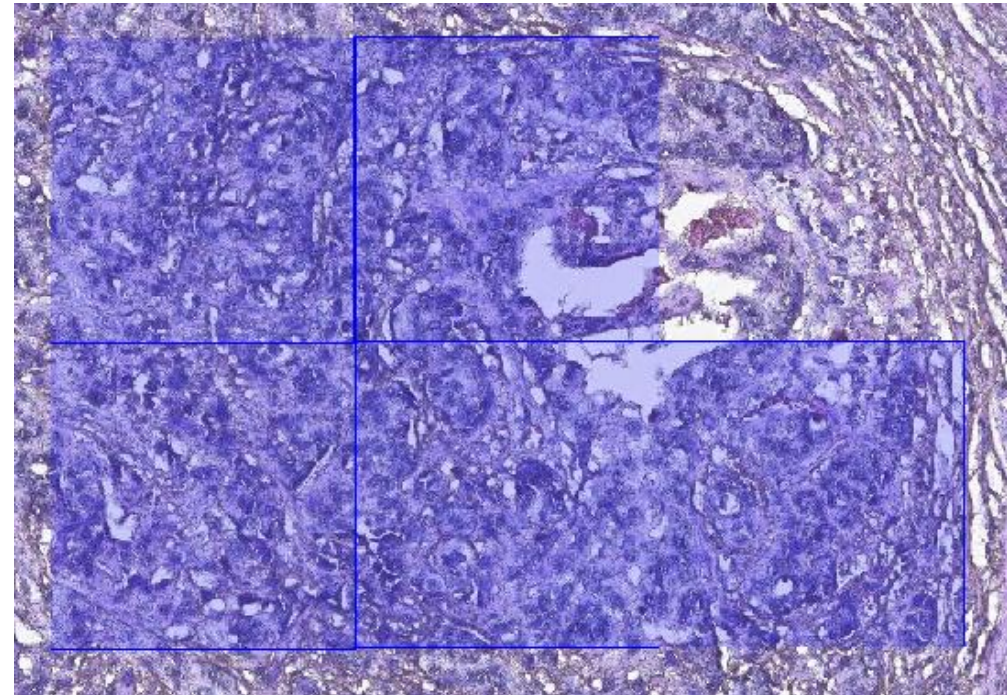


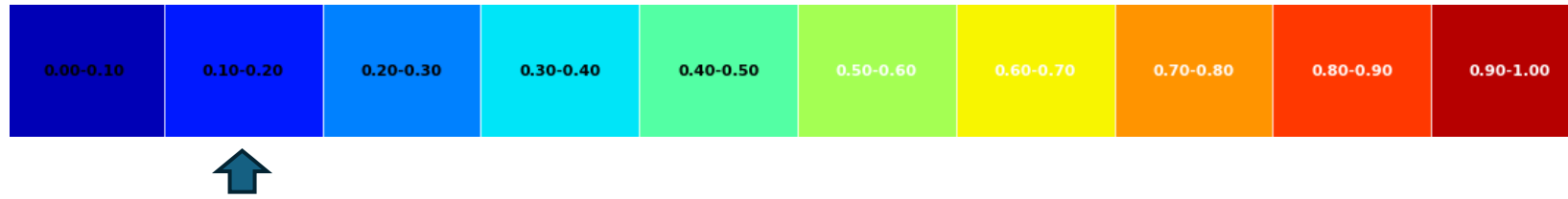
- Section **b4** => Low Score & Tumor

AI 스코어



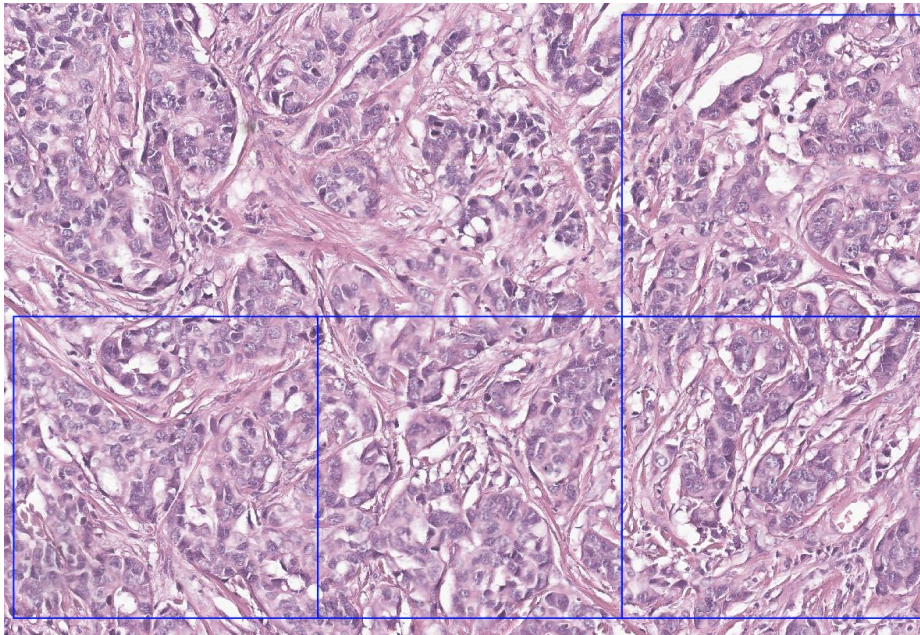
Astromapper 매핑 결과



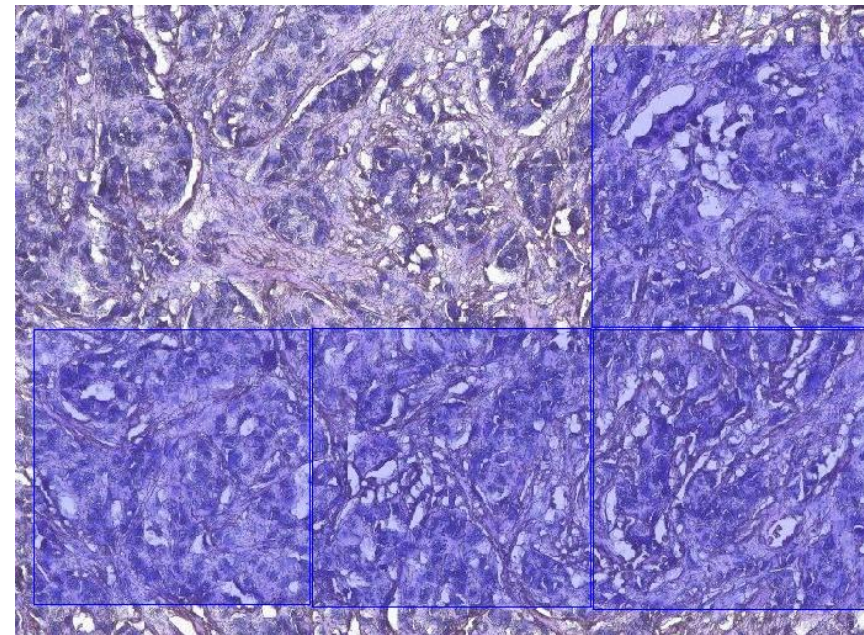


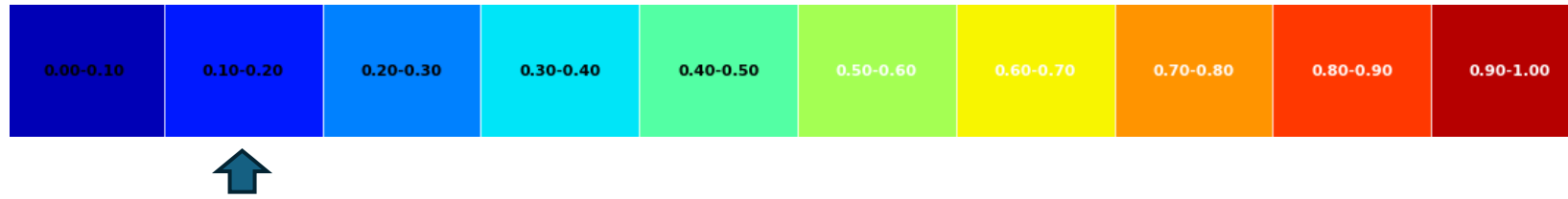
- Section **b5** => Low Score & Tumor

AI 스코어



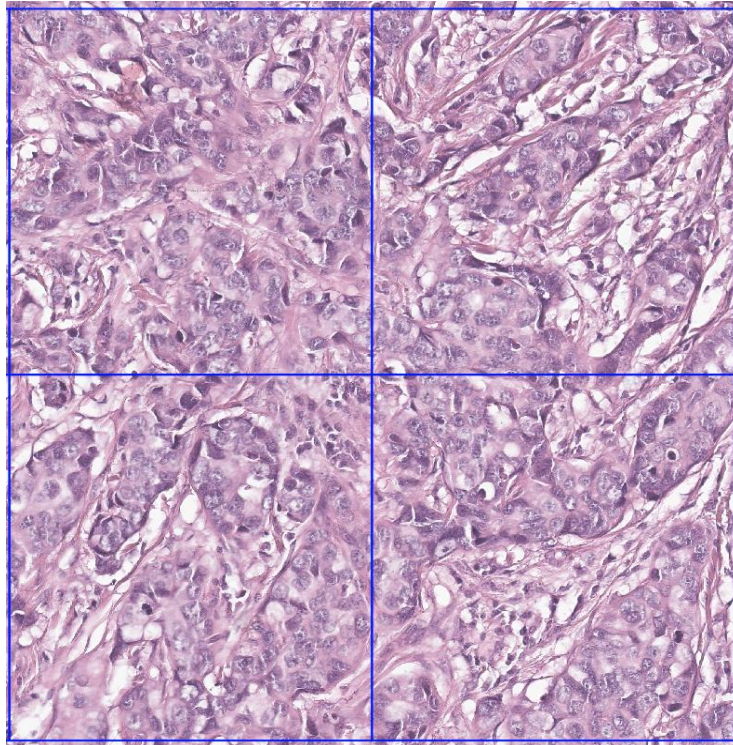
Astromapper 매핑 결과



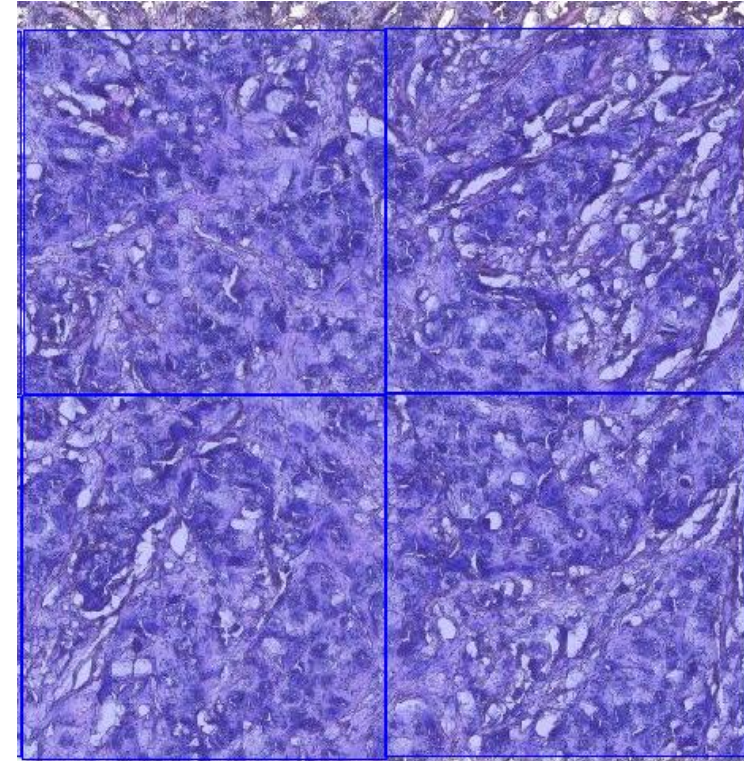


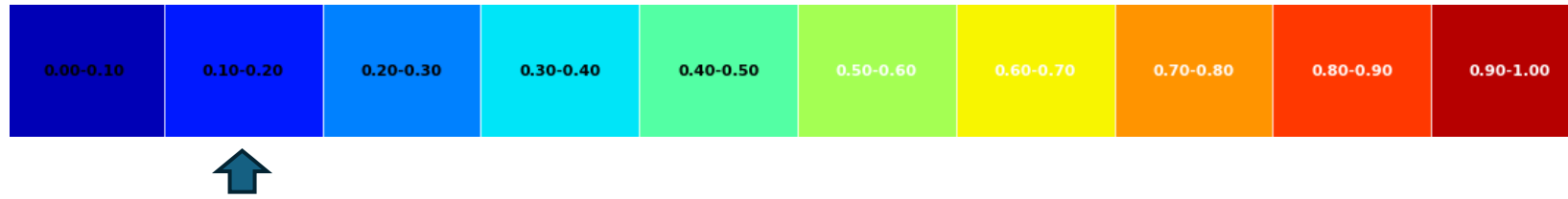
- Section **b6** => Low Score & Tumor

AI 스코어



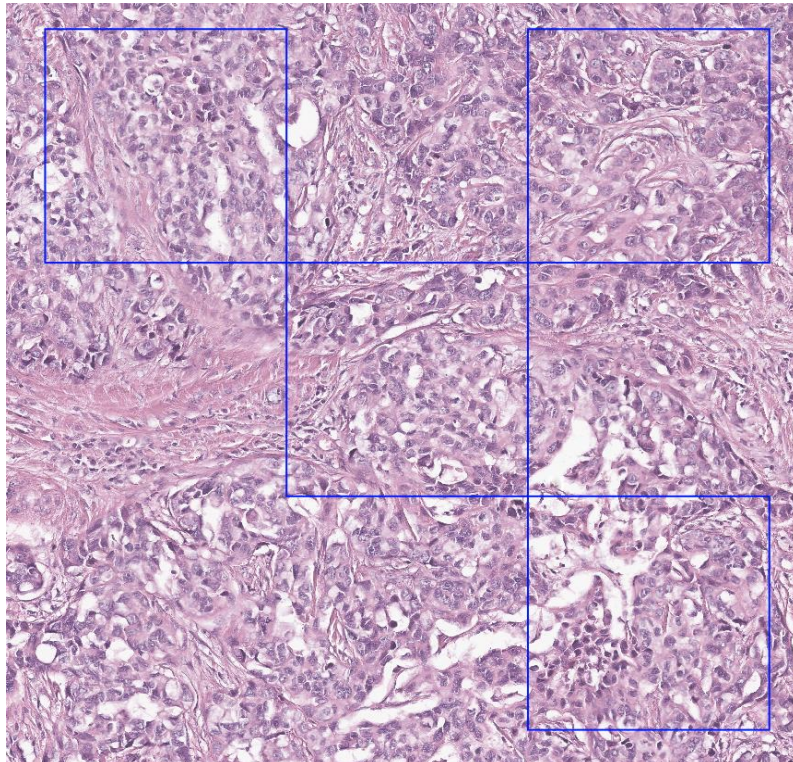
Astromapper 매핑 결과



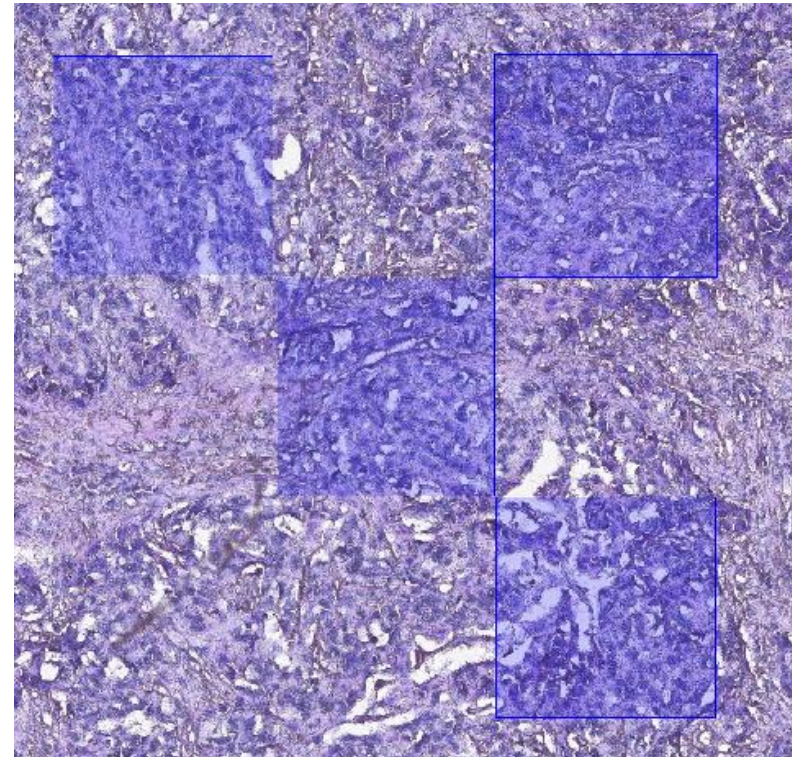


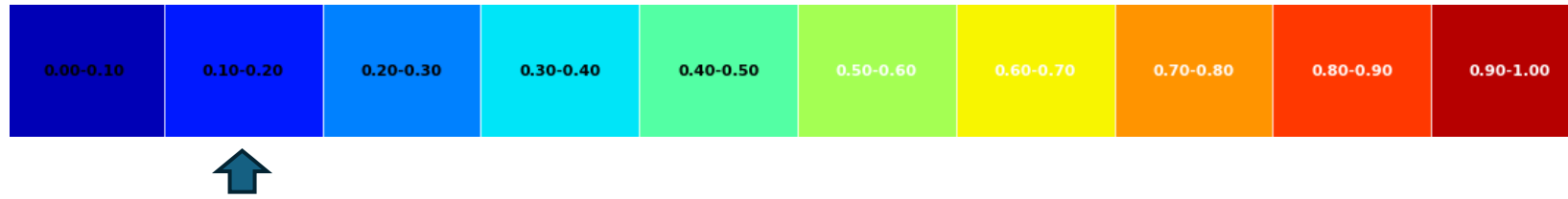
- Section **b7** => Low Score & Tumor

AI 스코어



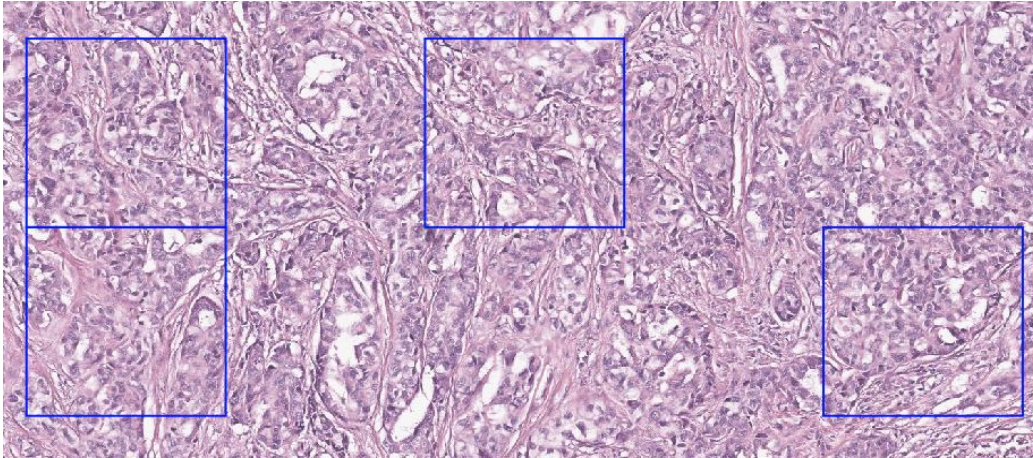
Astromapper 매핑 결과



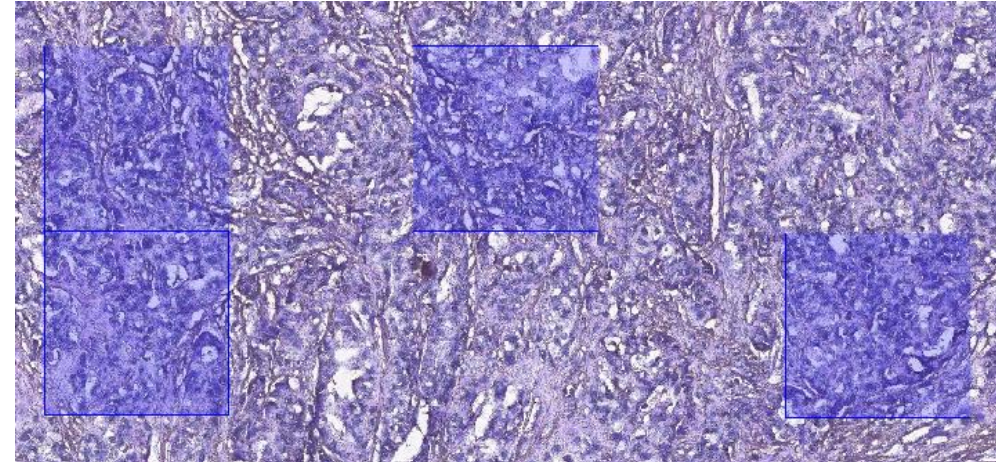


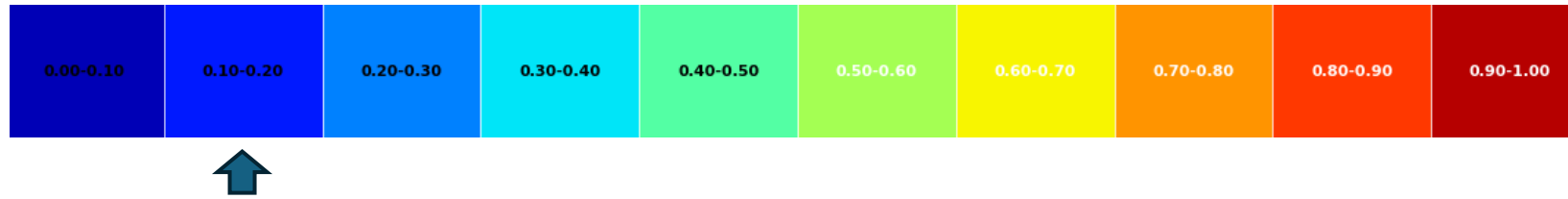
- Section **b8** => Low Score & Tumor

AI 스코어



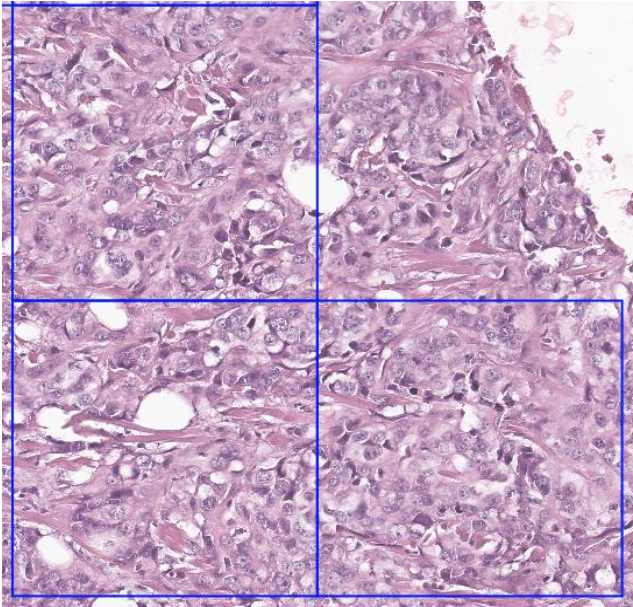
Astromapper 매핑 결과



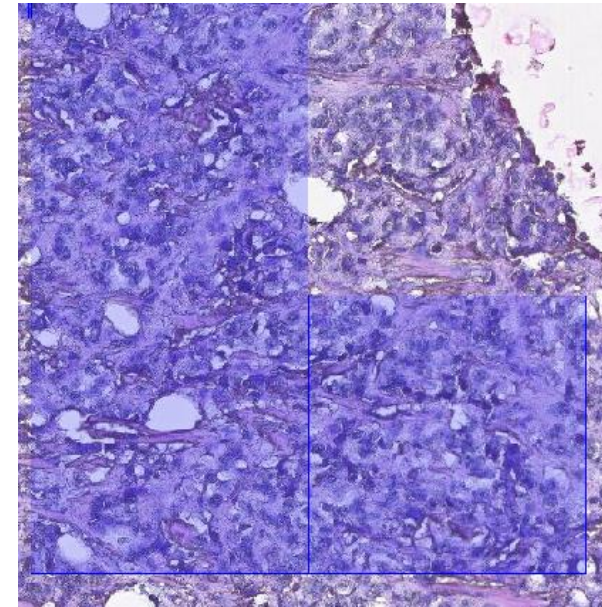


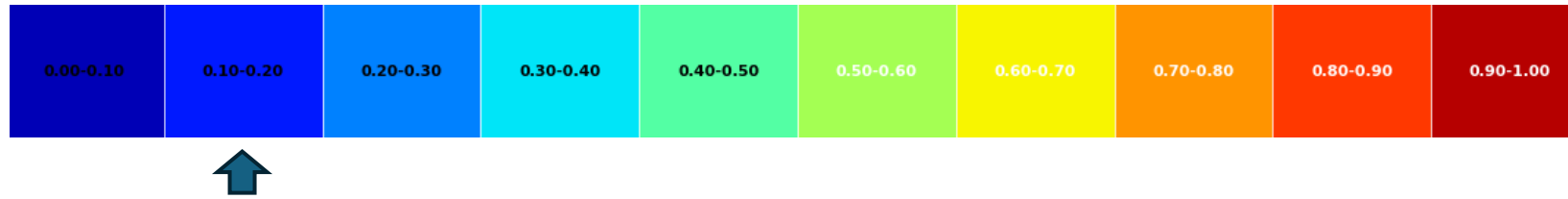
- Section **b9** => Low Score & Tumor

AI 스코어



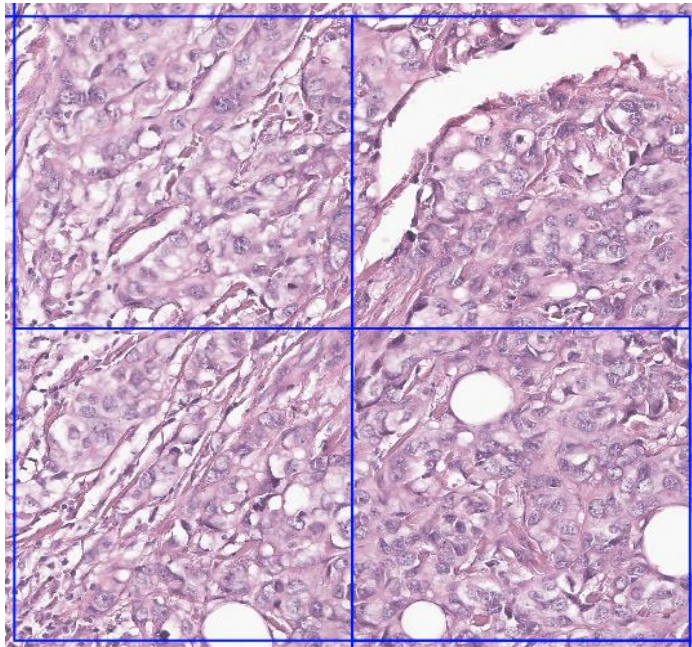
Astromapper 매핑 결과



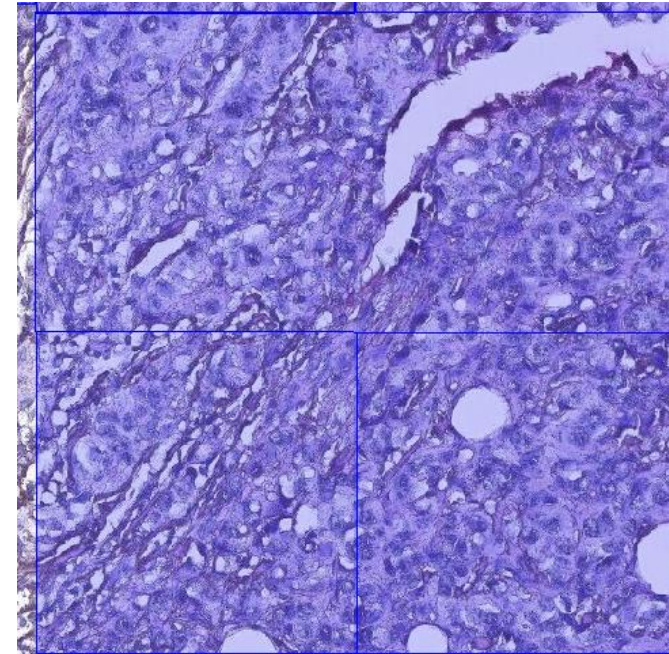


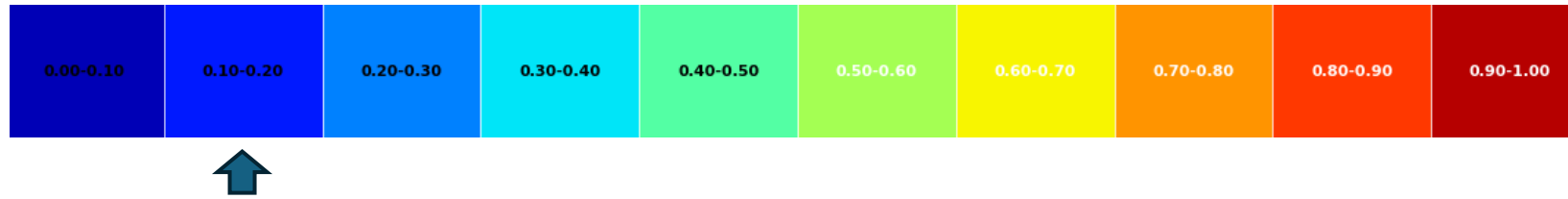
- Section **b10** => Low Score & Tumor

AI 스코어



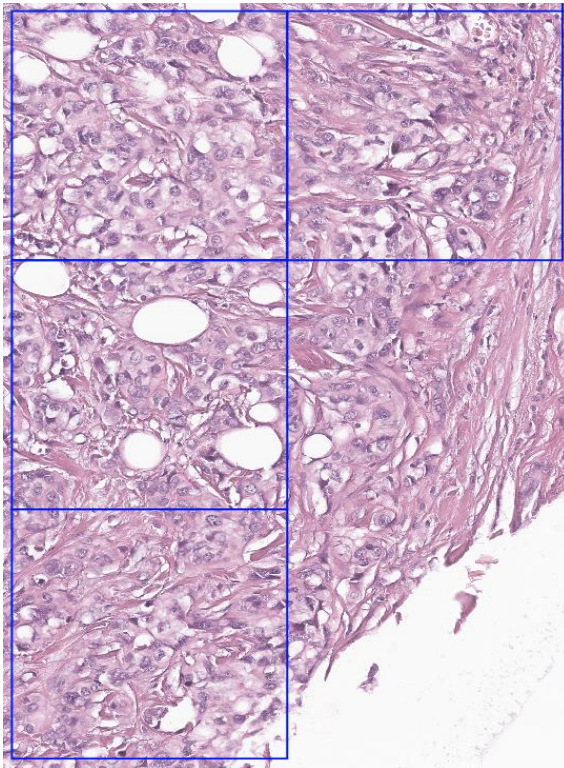
Astromapper 매핑 결과



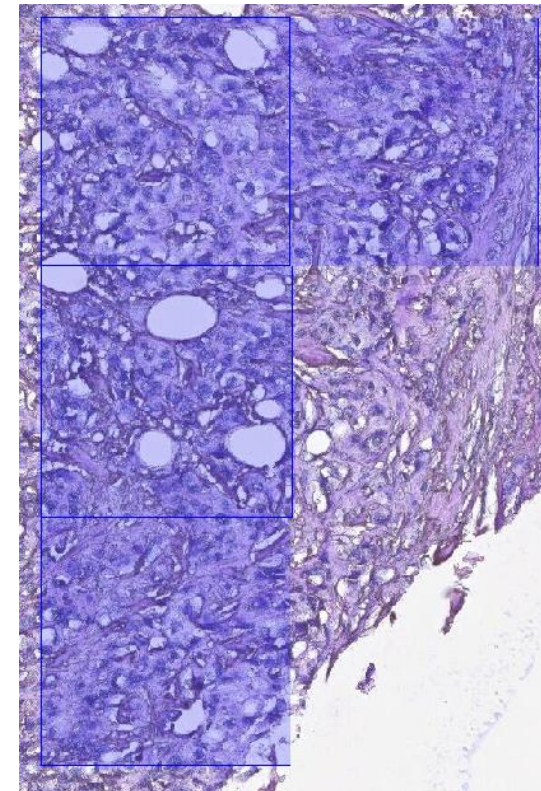


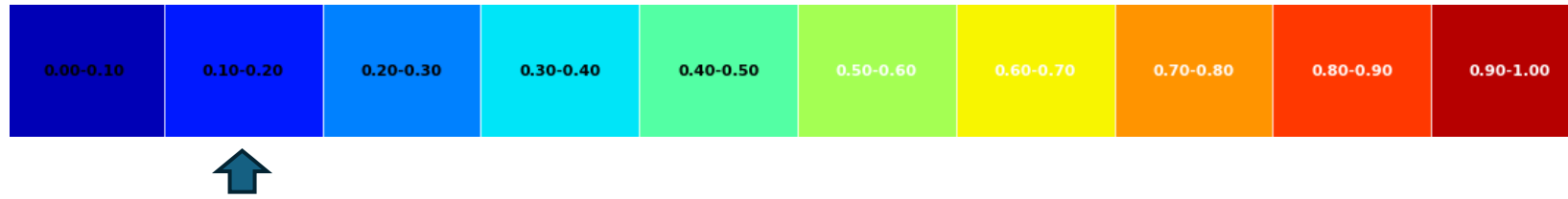
- Section **b11** => Low Score & Tumor

AI 스코어



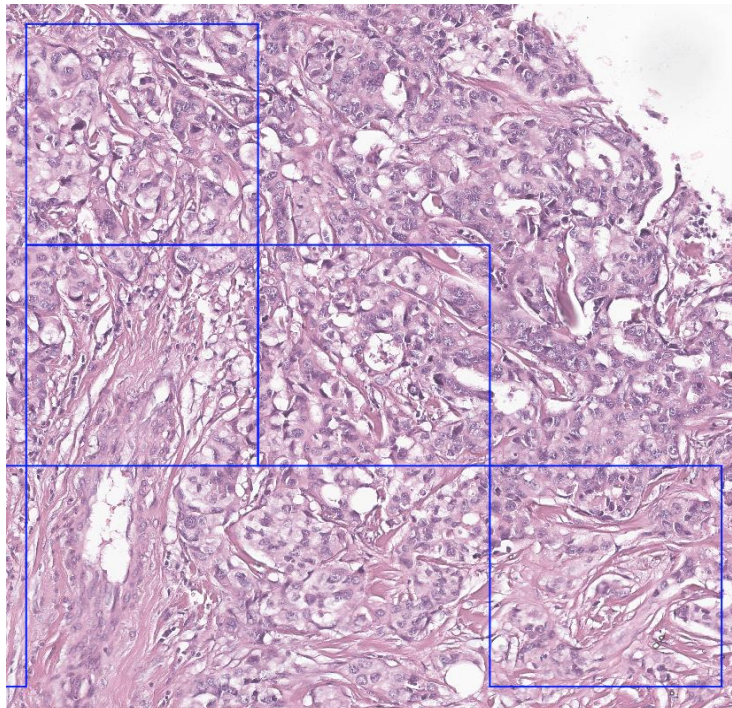
Astromapper 매핑 결과



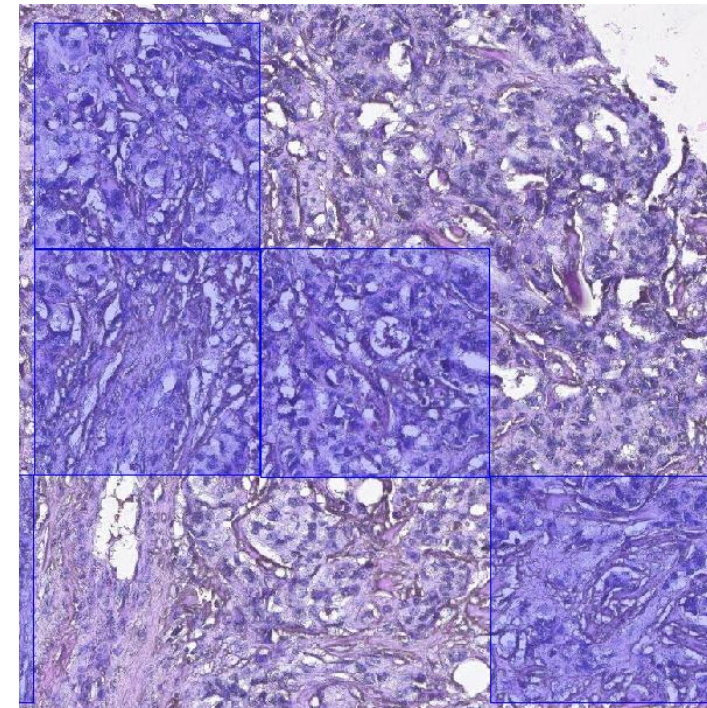


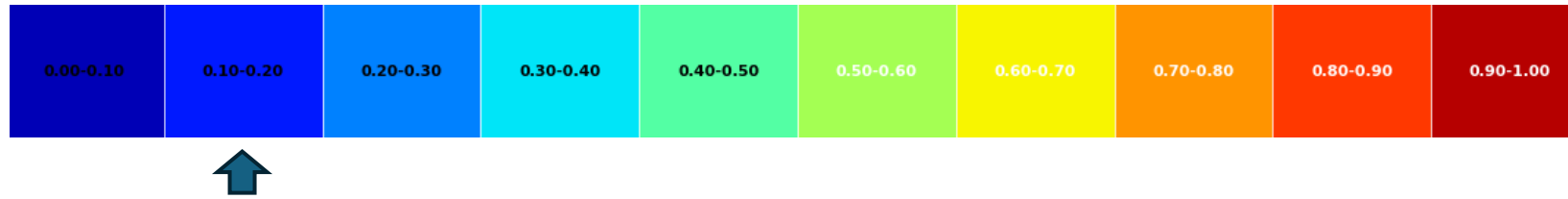
- Section **b12** => Low Score & Tumor

AI 스코어



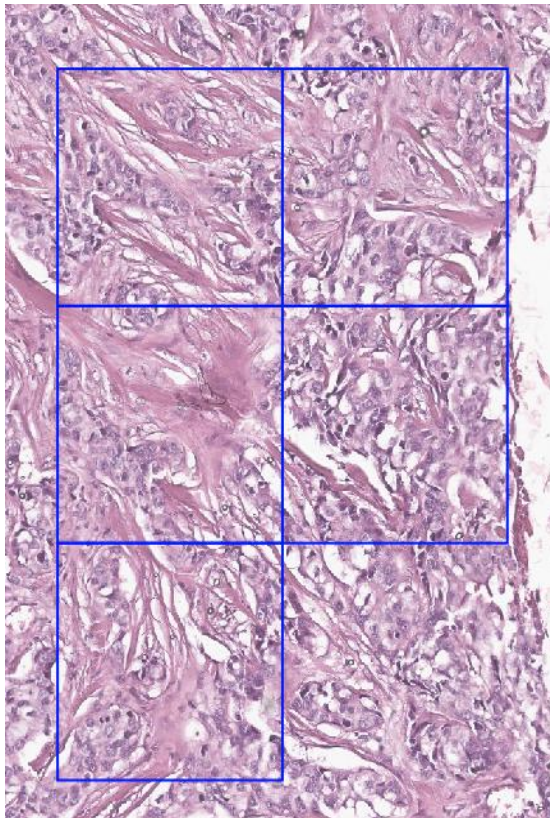
Astromapper 매핑 결과



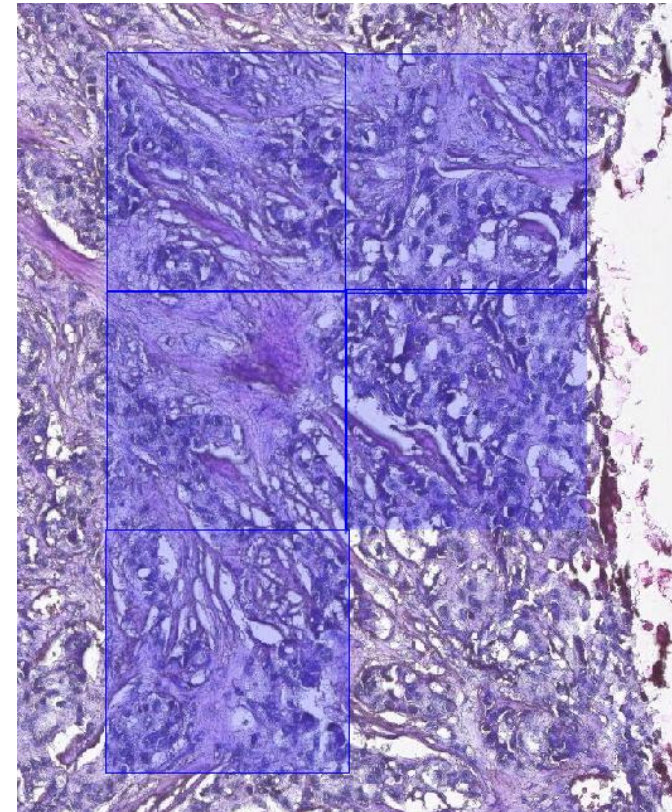


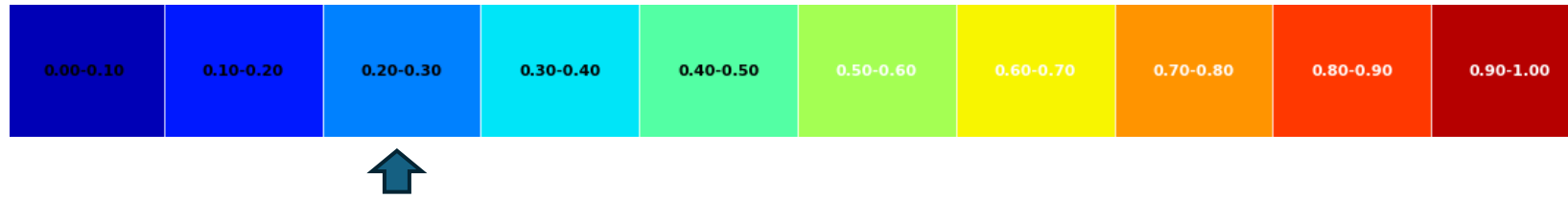
- Section **b13** => Low Score & Tumor

AI 스코어



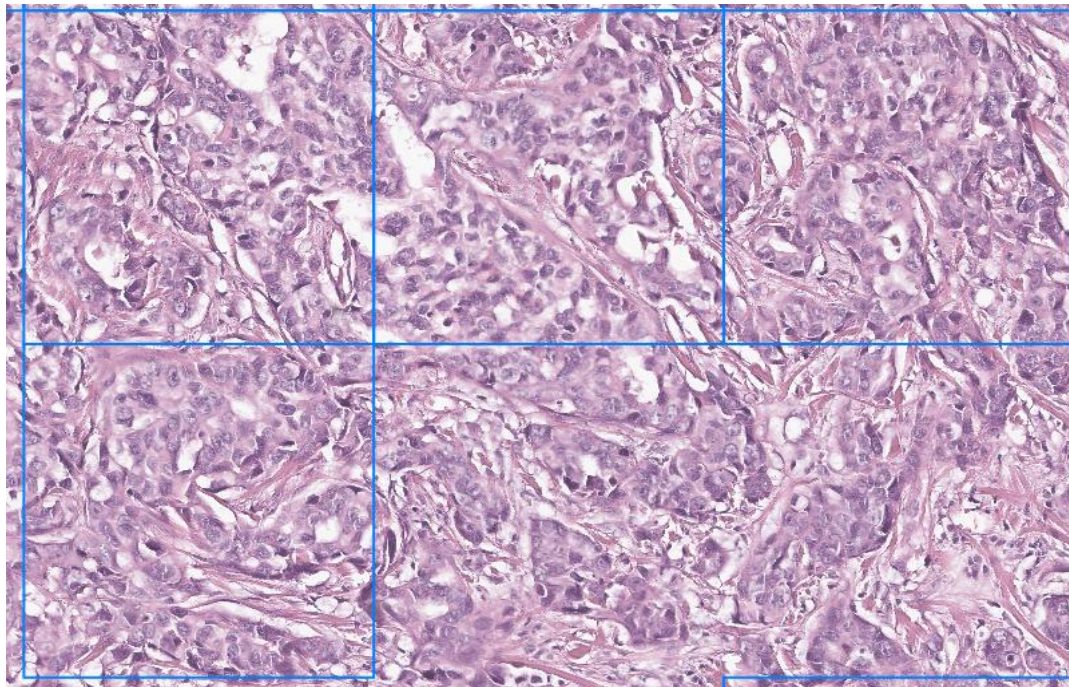
Astromapper 매핑 결과



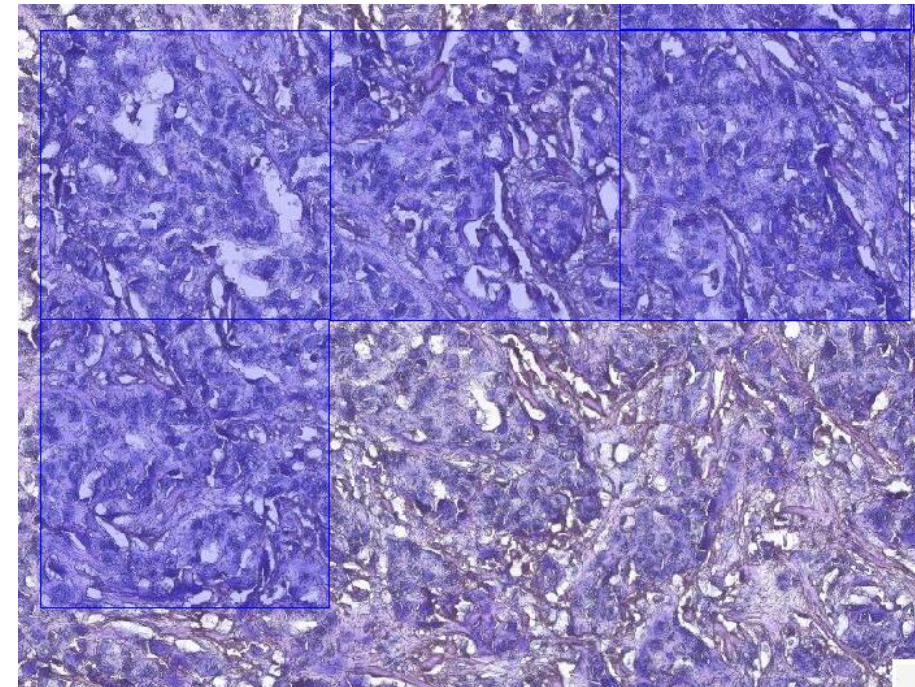


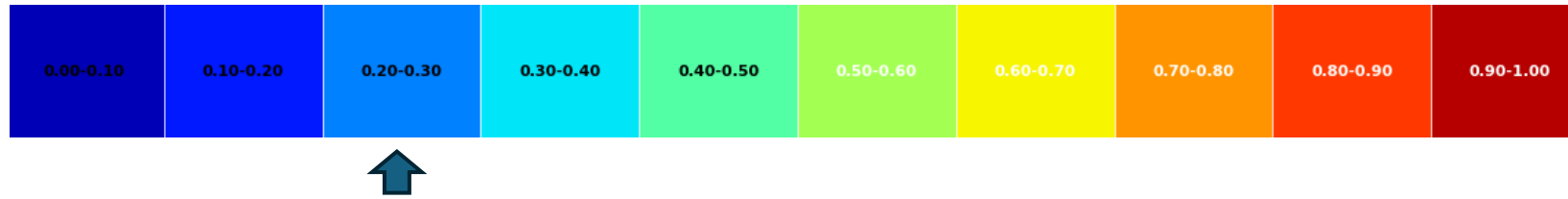
- Section **b14** => Low Score & Tumor

AI 스코어



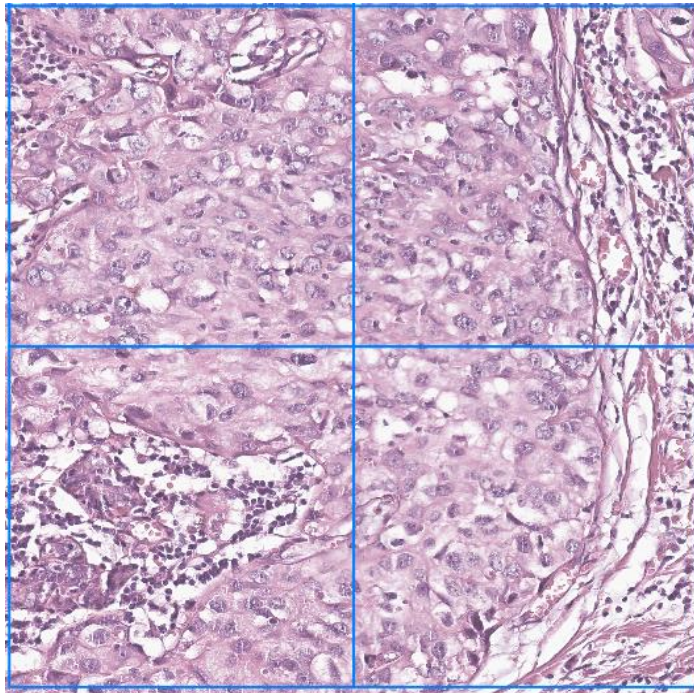
Astromapper 매핑 결과



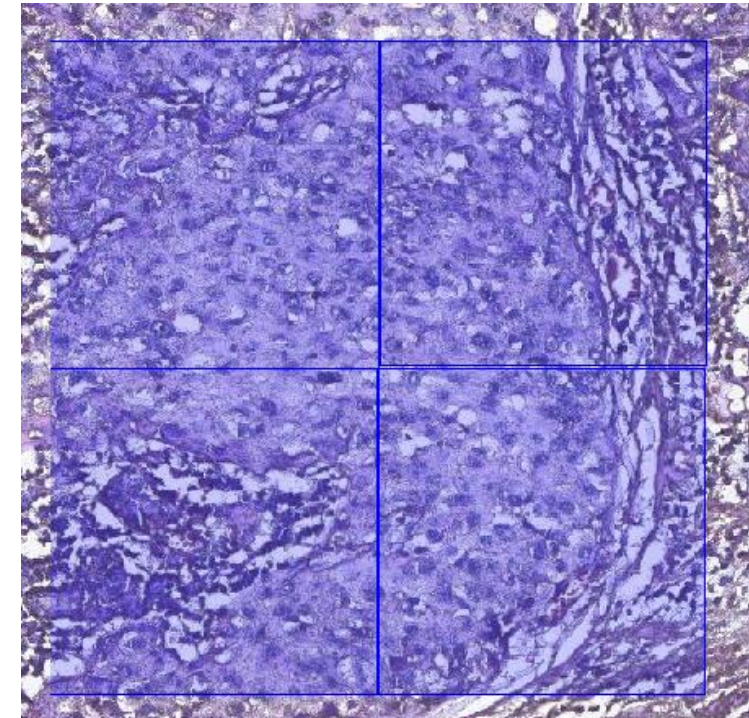


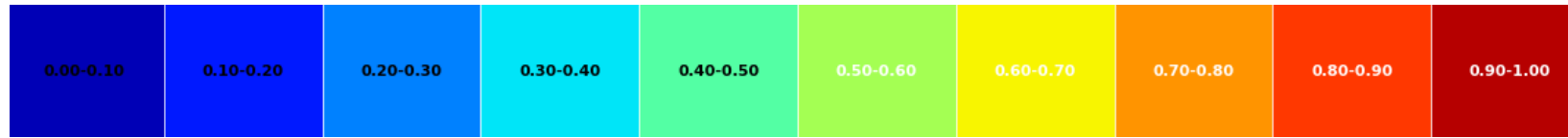
- Section **b15** => Low Score & Tumor

AI 스코어



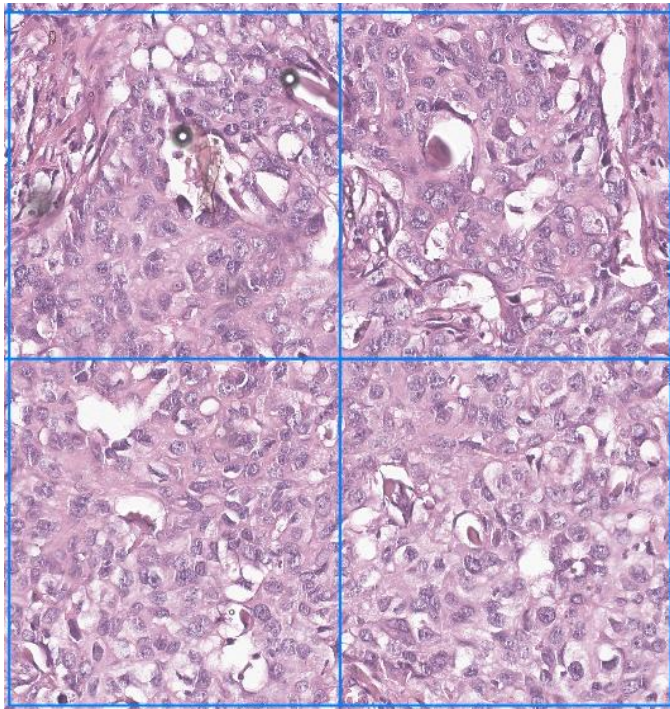
Astromapper 매핑 결과



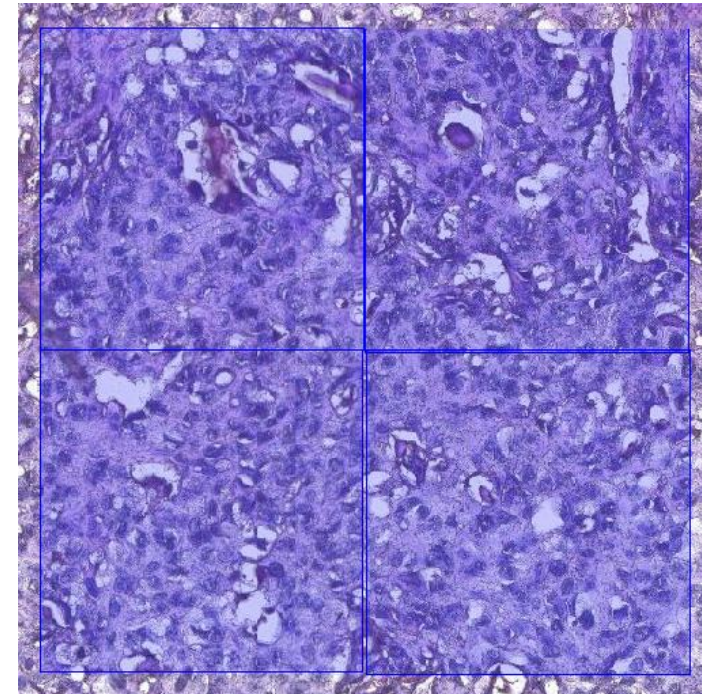


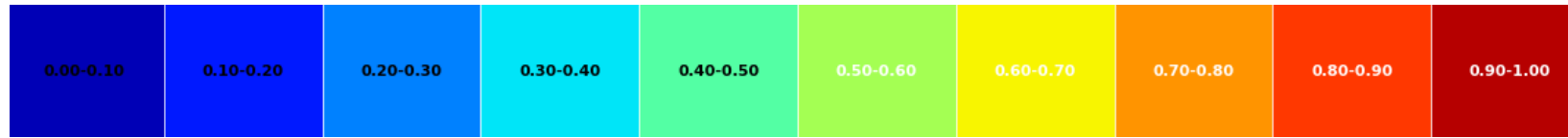
- Section **b16** => Low Score & Tumor

AI 스코어



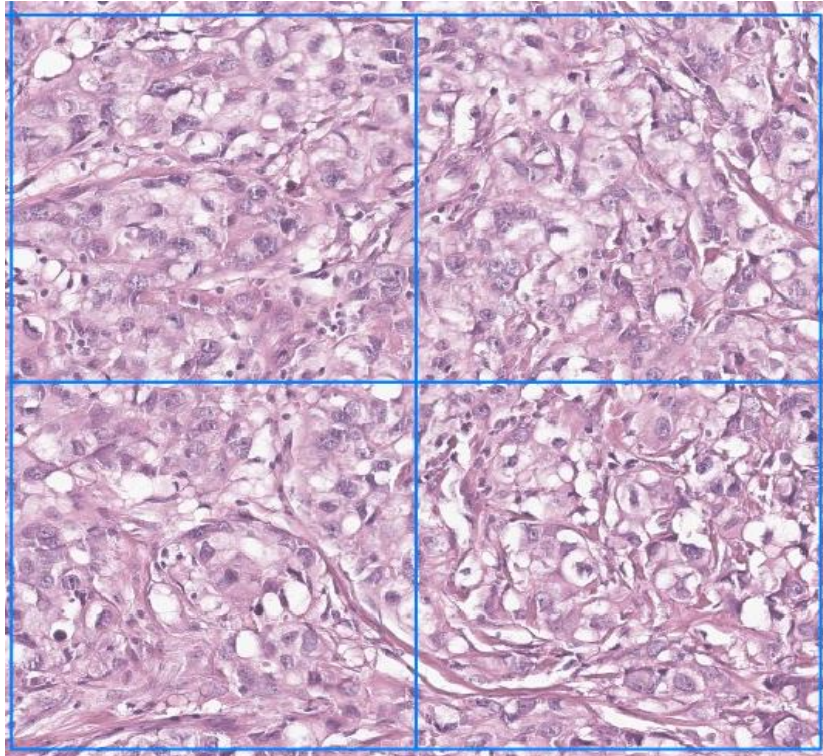
Astromapper 매핑 결과



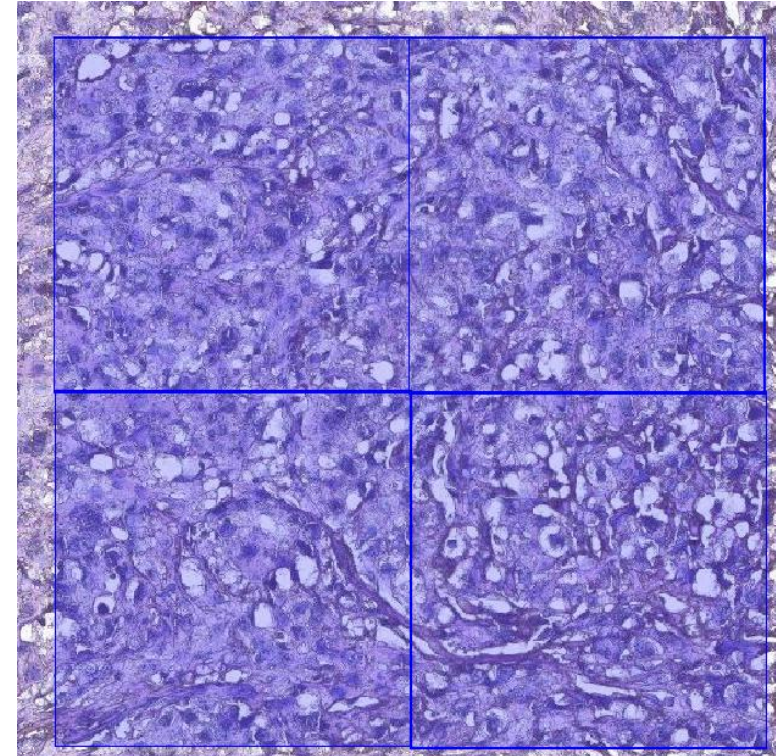


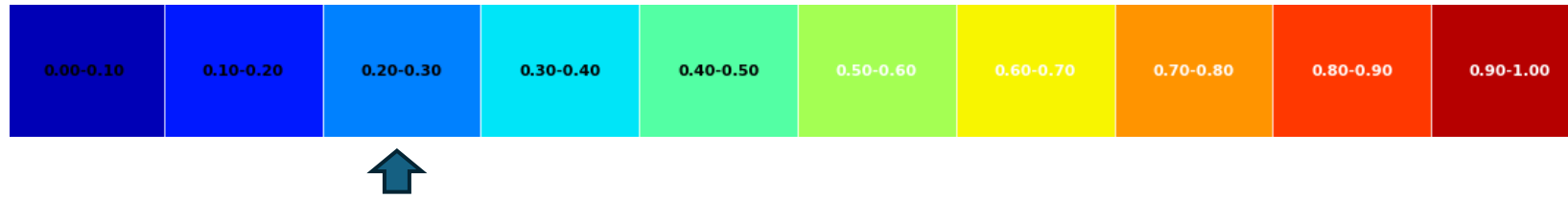
- Section **b17** => Low Score & Tumor

AI 스코어



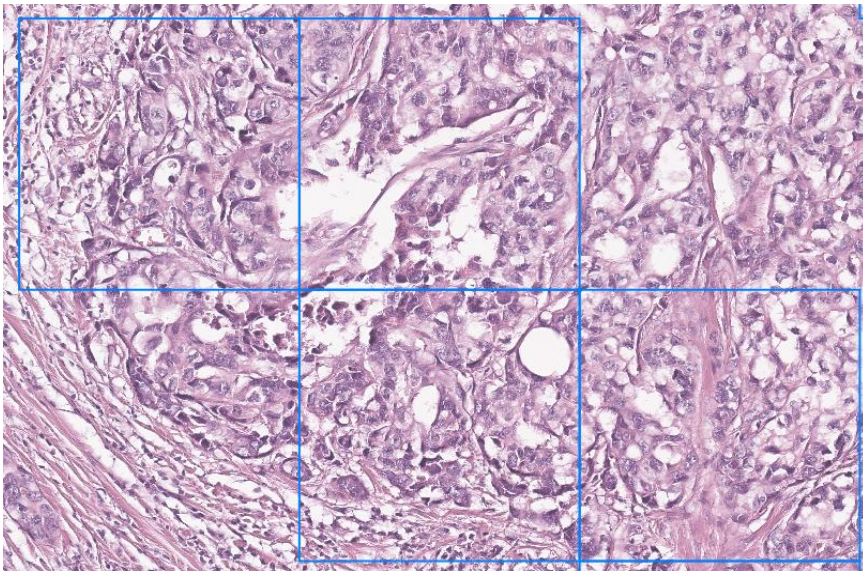
Astromapper 매핑 결과



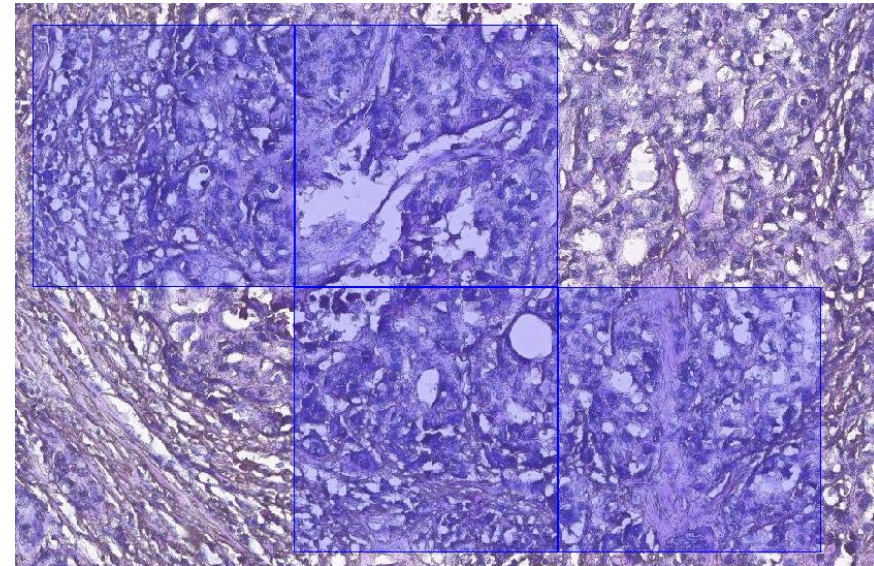


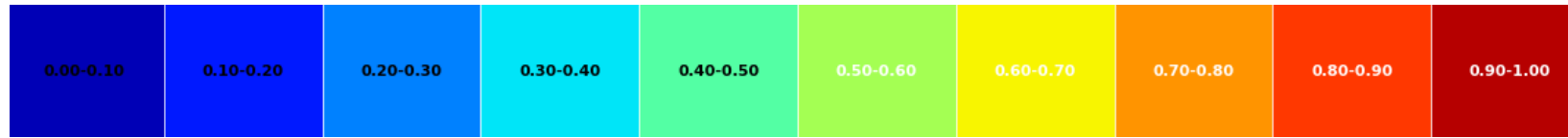
- Section **b18** => Low Score & Tumor

AI 스코어



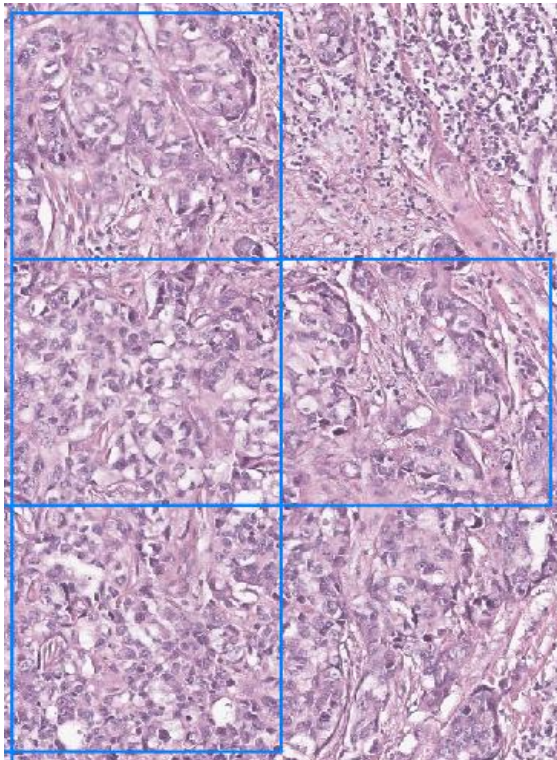
Astromapper 매핑 결과



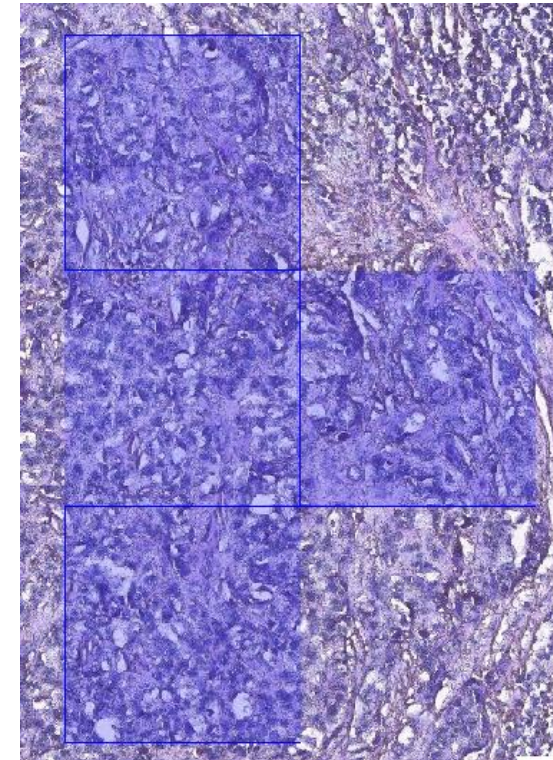


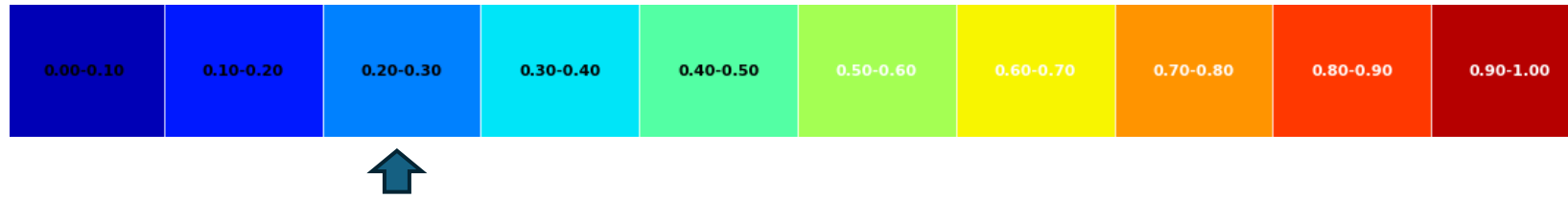
- Section **b19** => Low Score & Tumor

AI 스코어



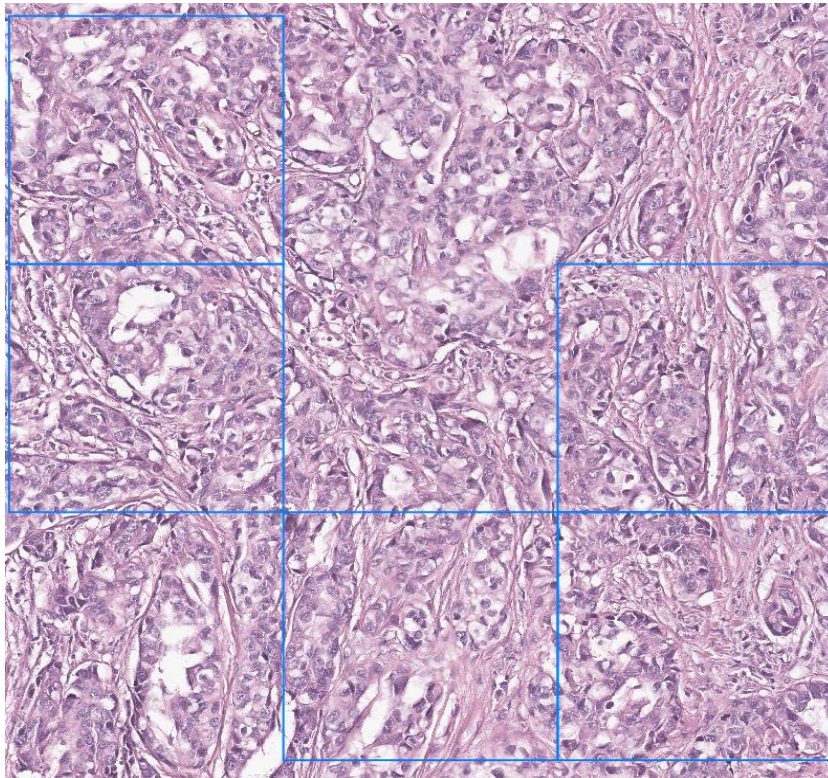
Astromapper 매핑 결과



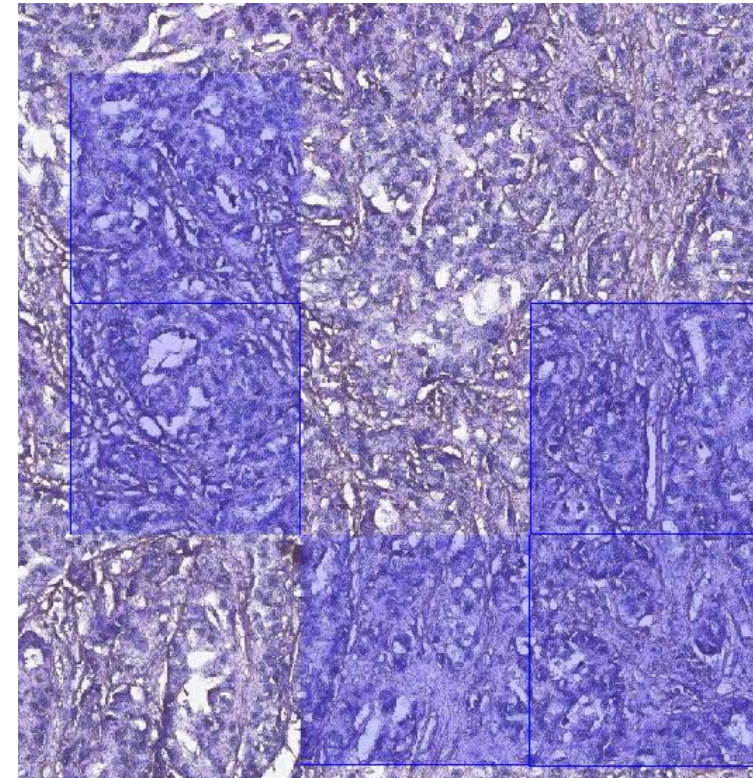


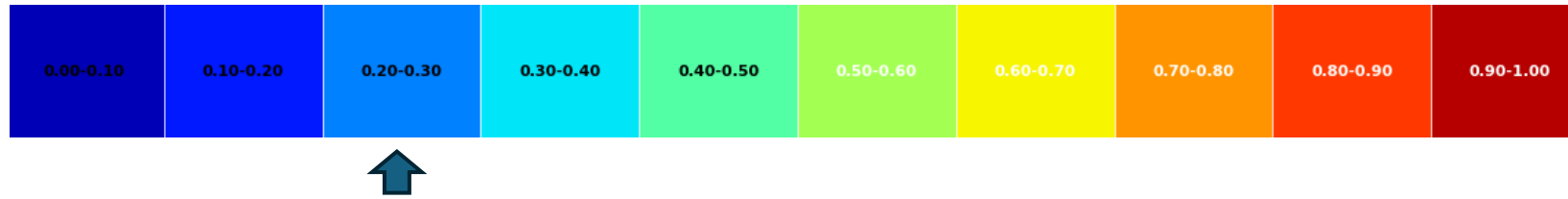
- Section **b20** => Low Score & Tumor

AI 스코어



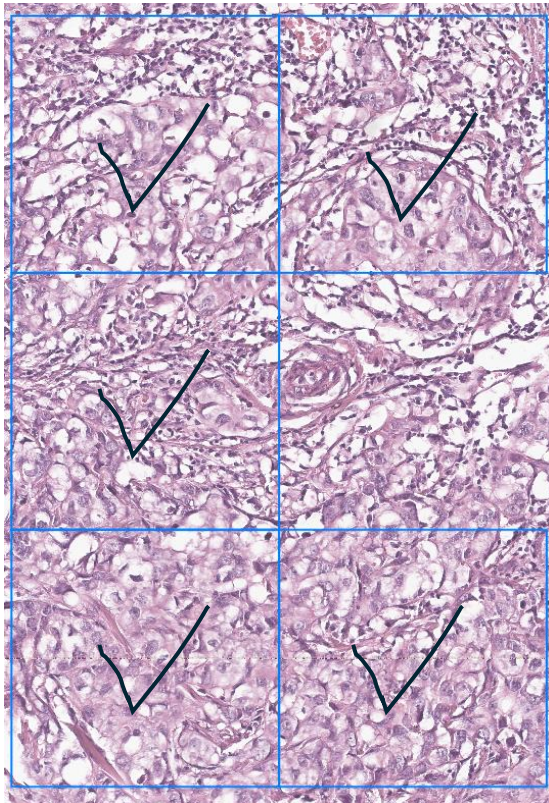
Astromapper 매핑 결과



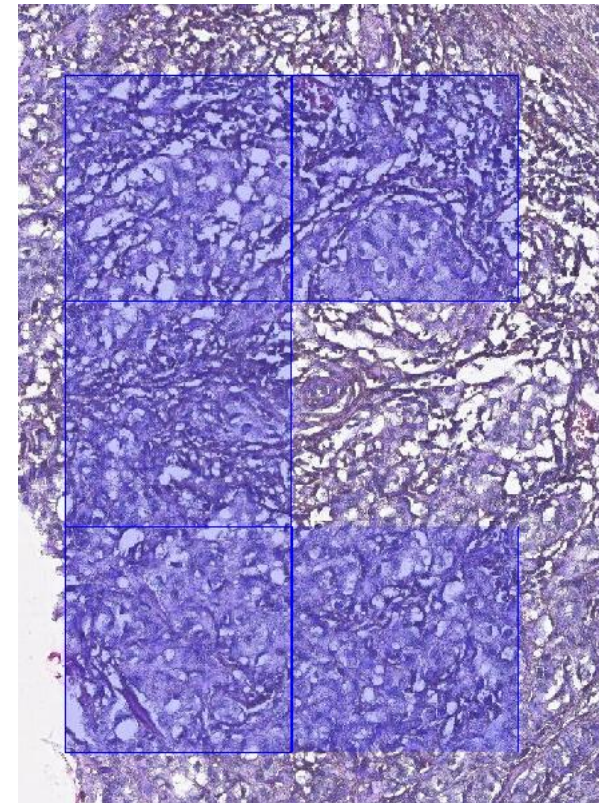


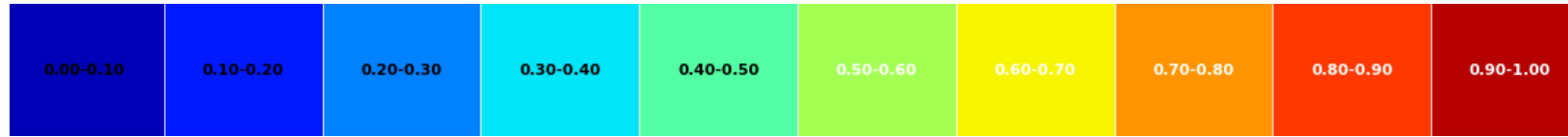
- Section **b21** => Low Score & Tumor

AI 스코어



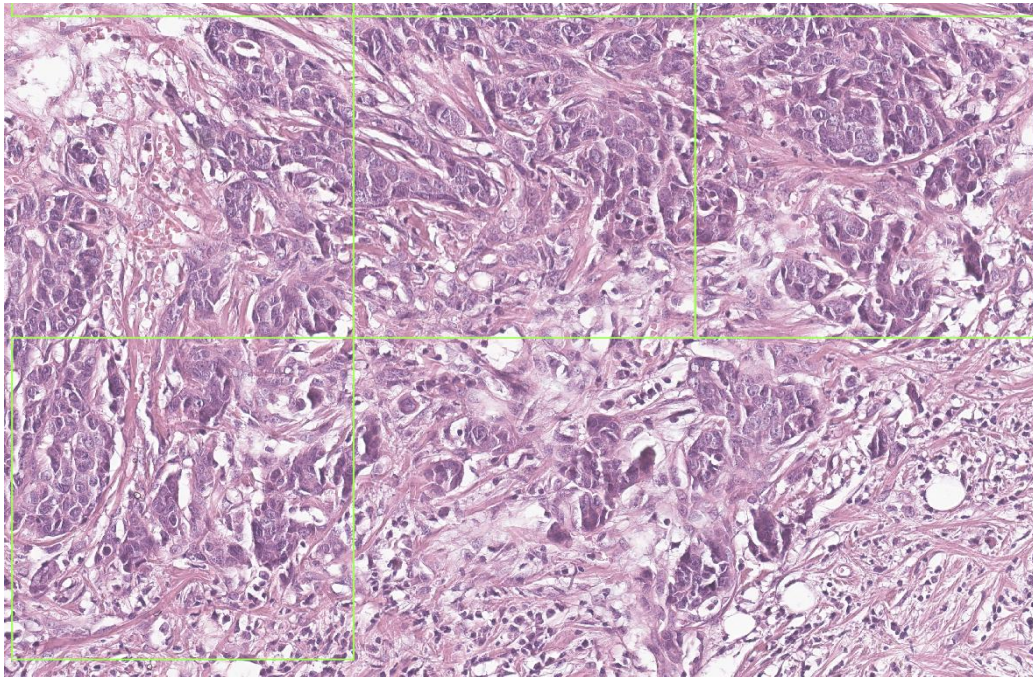
Astromapper 매핑 결과



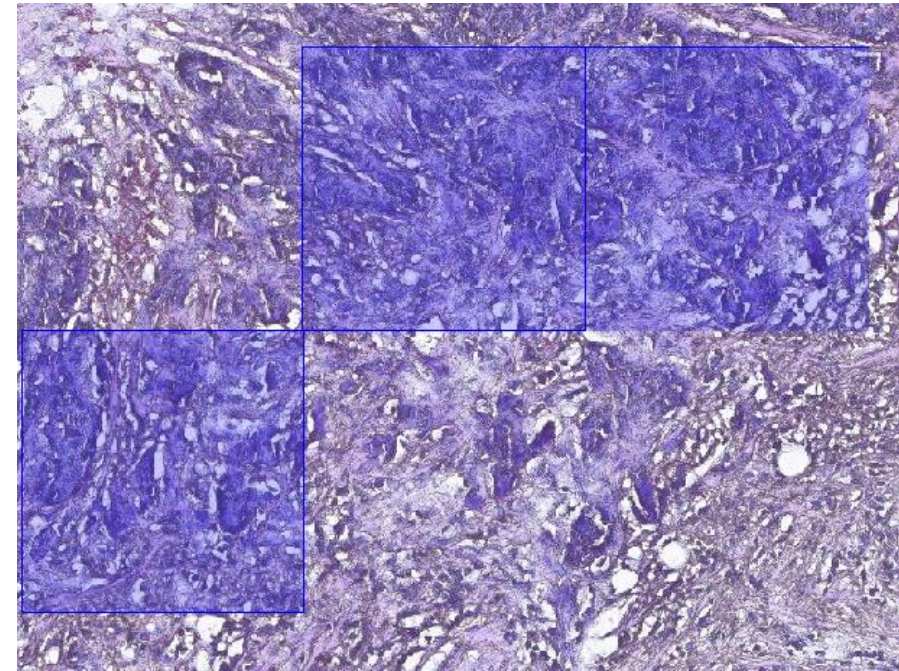


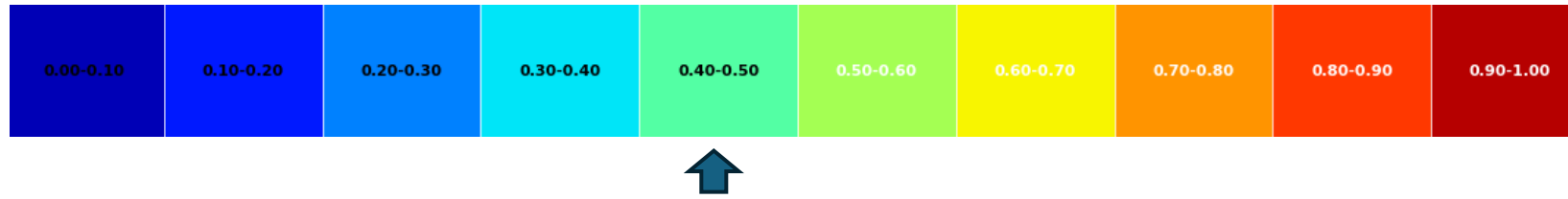
- Section **t1** => Control regions & Tumor

AI 스코어



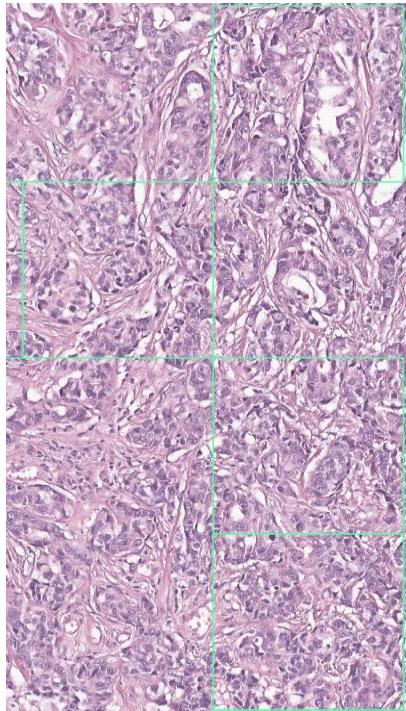
Astromapper 매핑 결과



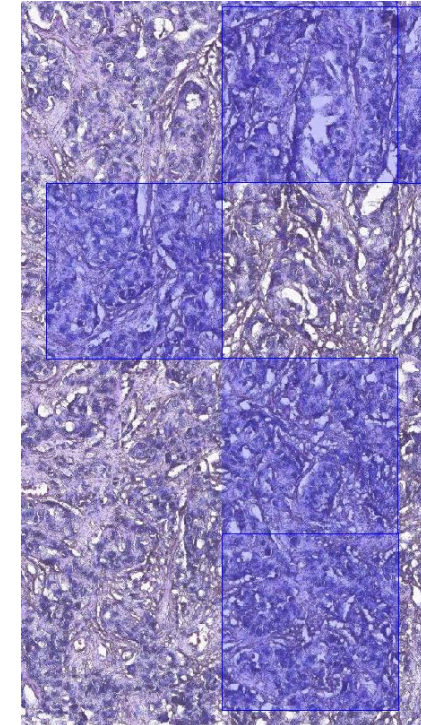


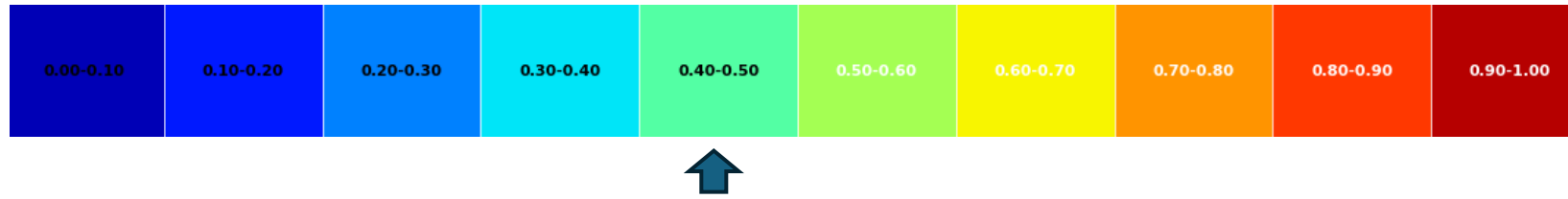
- Section **t2** => Control regions & Tumor

AI 스코어



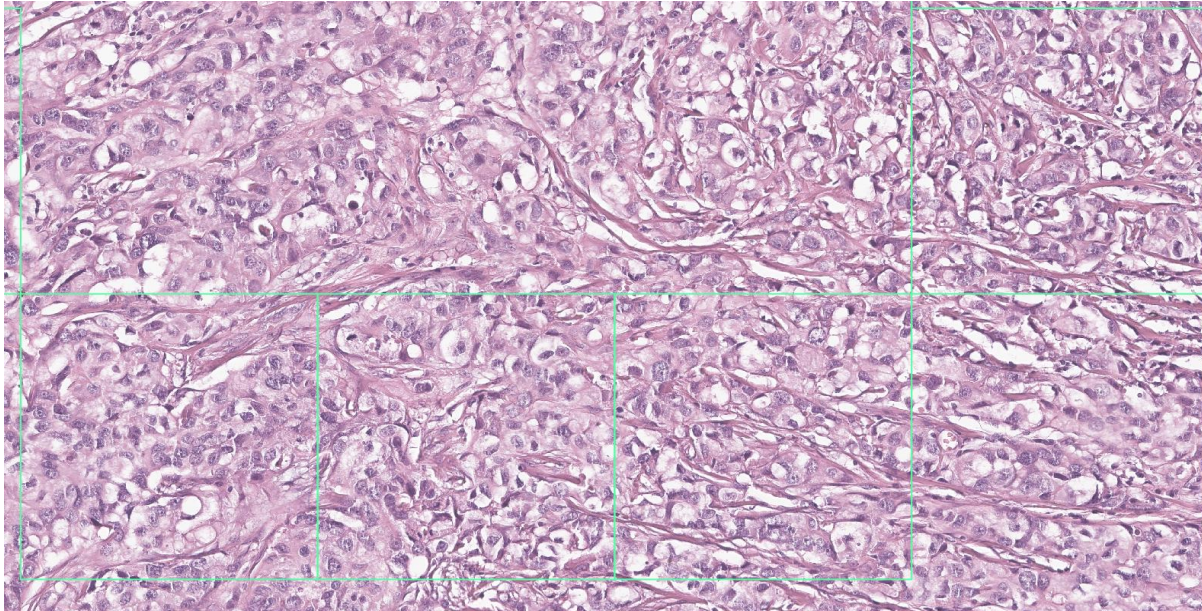
Astromapper 매핑 결과



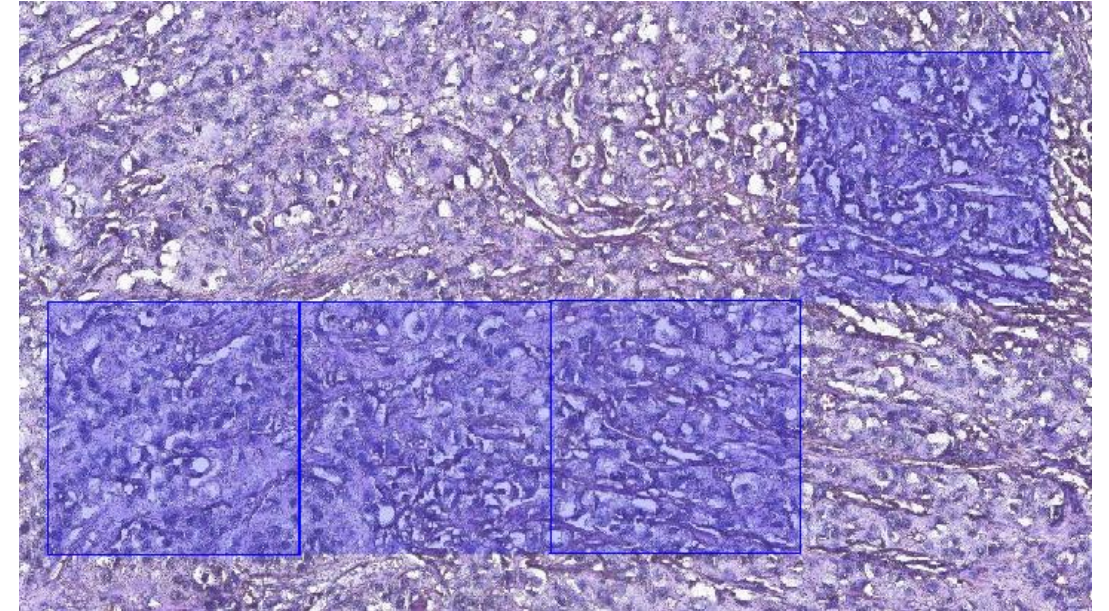


- Section **t3** => Control regions & Tumor

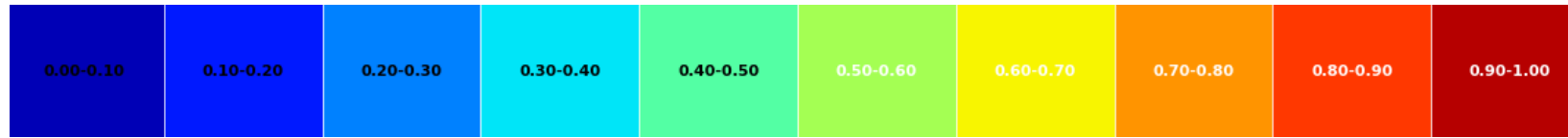
AI 스코어



Astromapper 매핑 결과

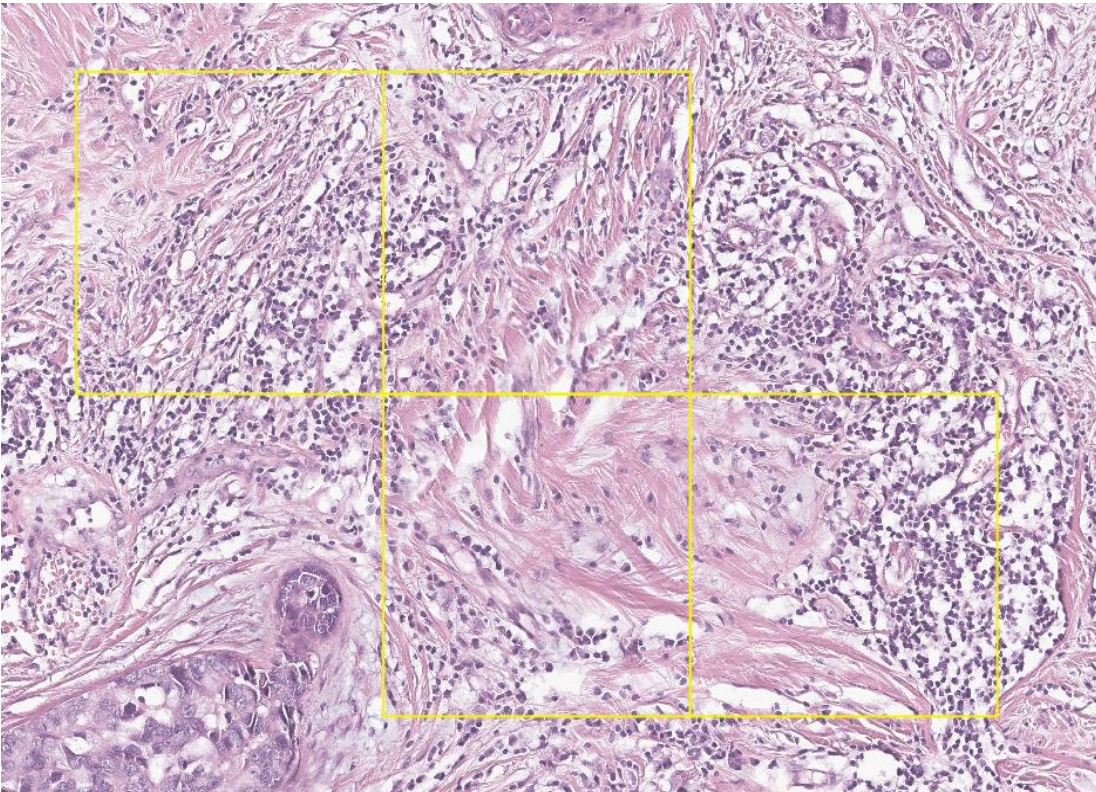


OmixAI 1-085-12 환자 샘플 high-score t-cell regions

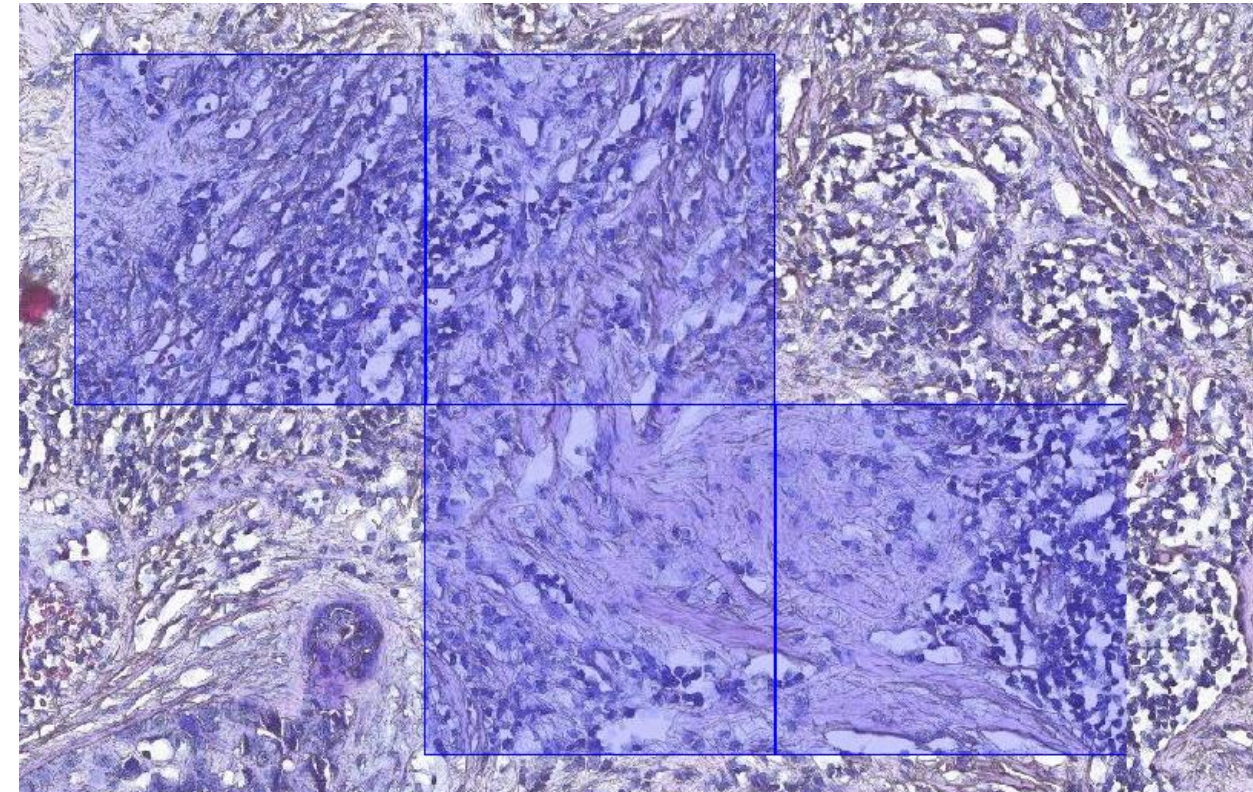


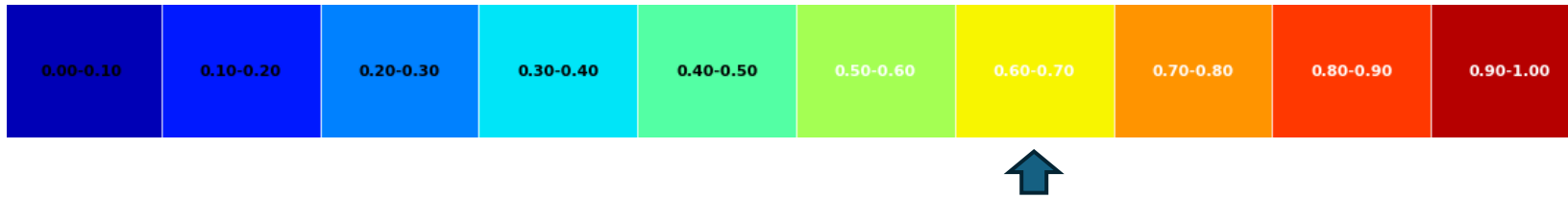
- Section **c1** => **High Score** & **T-cell**

AI 스코어



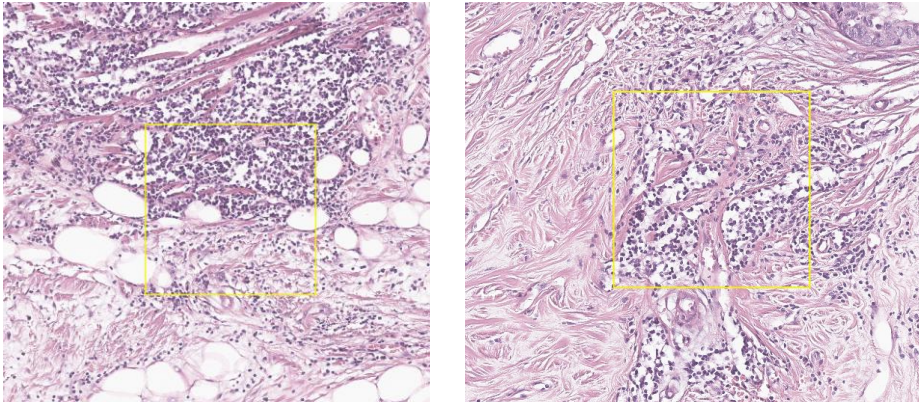
Astromapper 매핑 결과



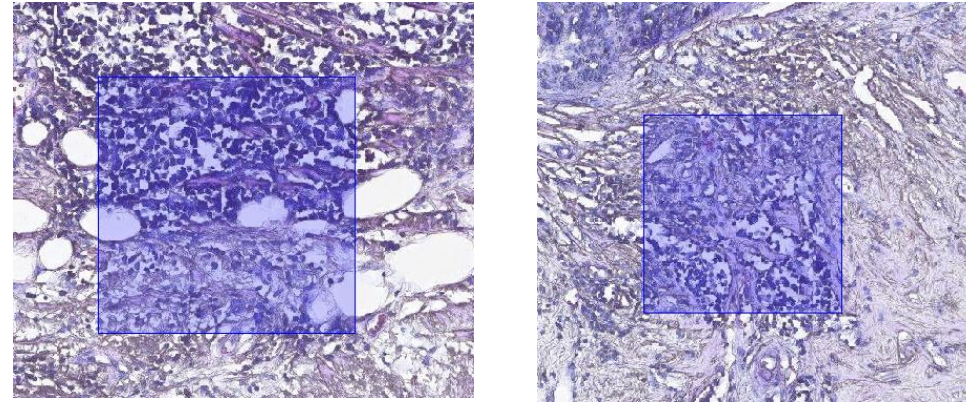


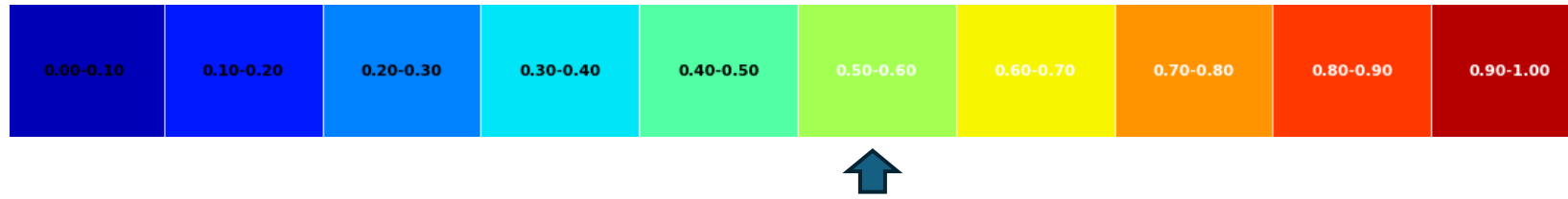
- Section **c2** => **High Score** & T-cell

AI 스코어



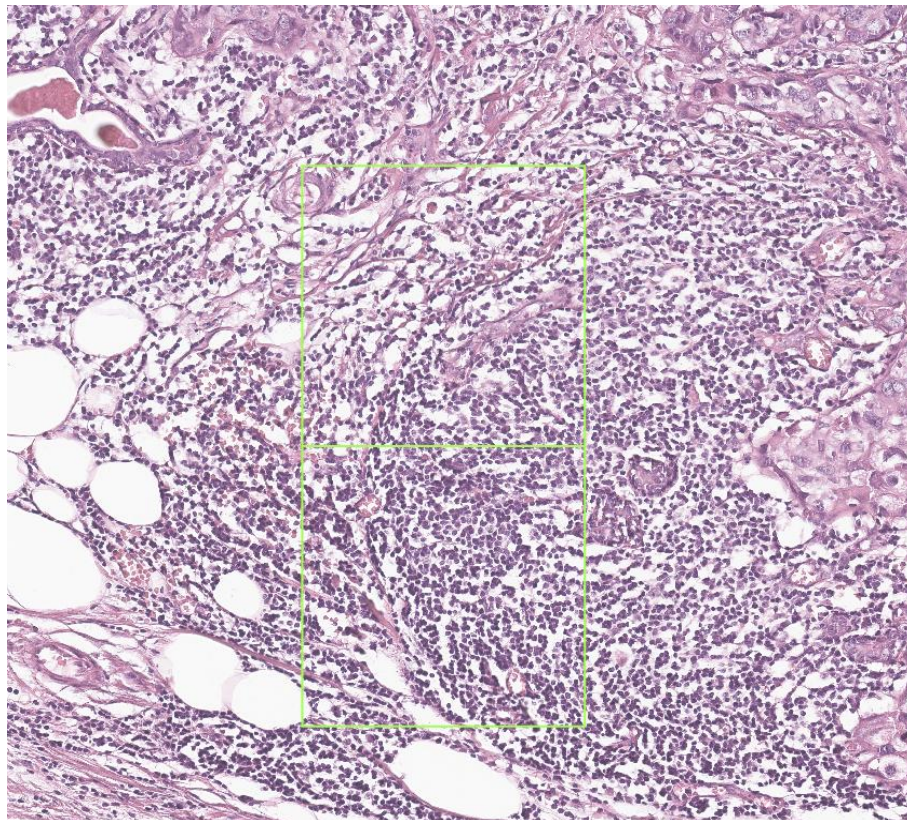
Astromapper 매핑 결과



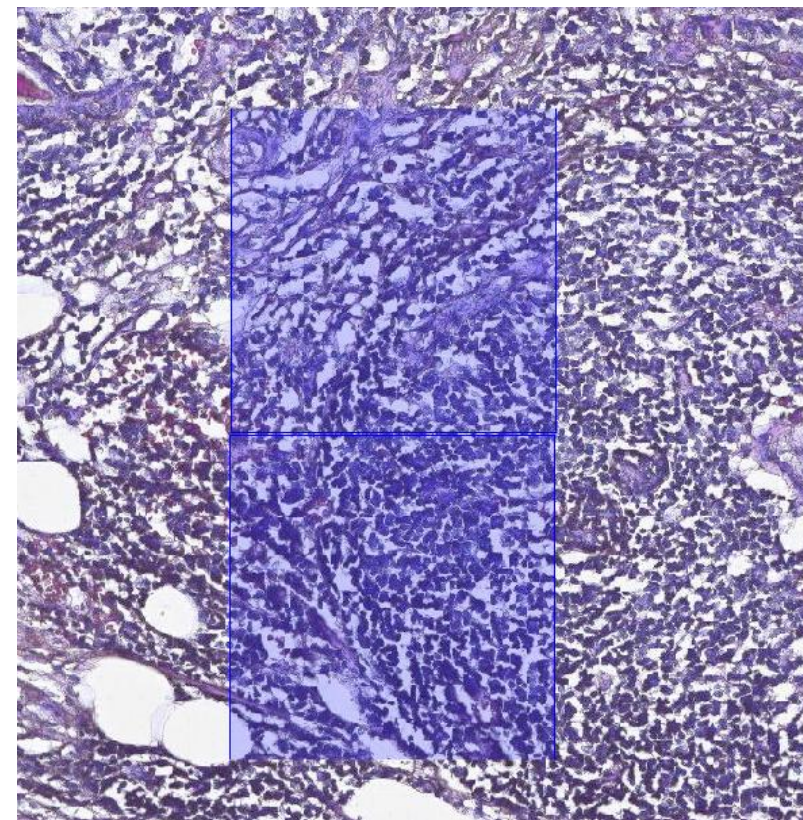


- Section **c3** => **High Score** & **T-cell**

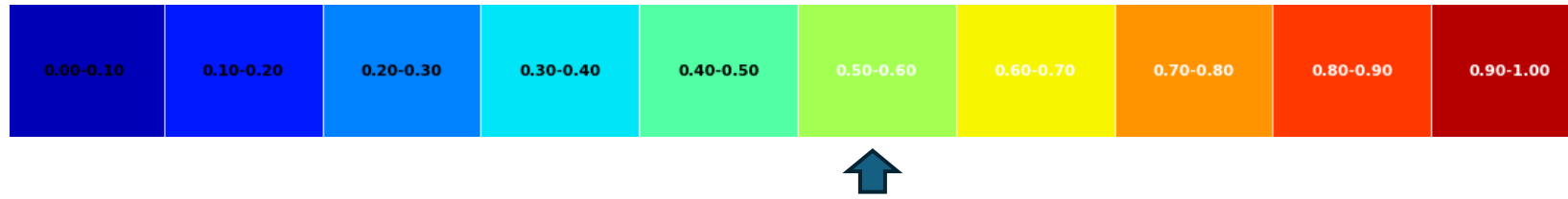
AI 스코어



Astromapper 매핑 결과

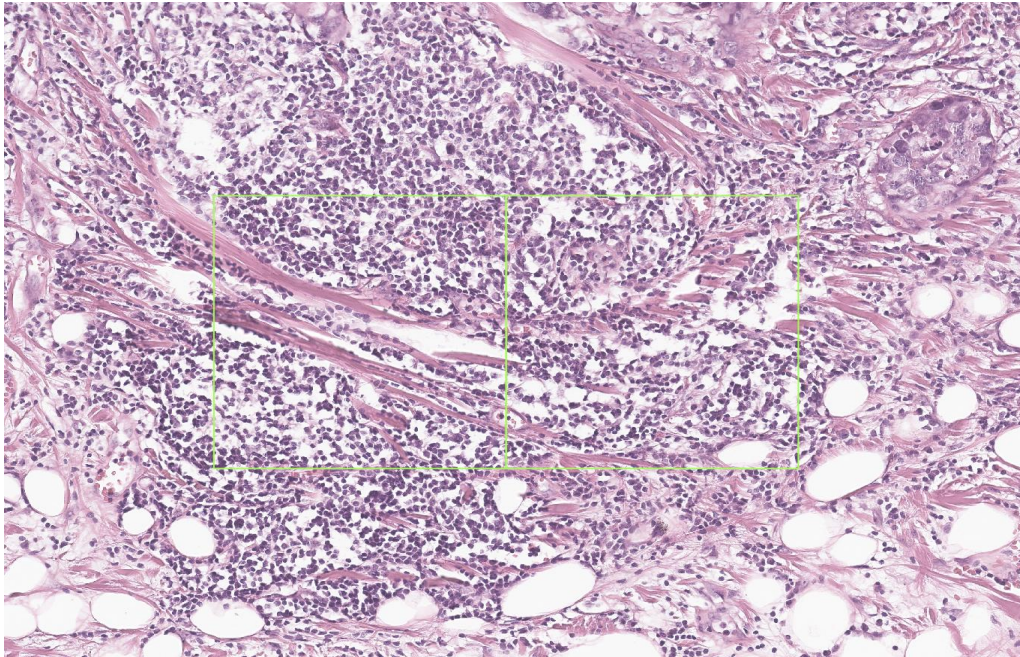


OmixAI 1-085-12 환자 샘플 high-score t-cell regions

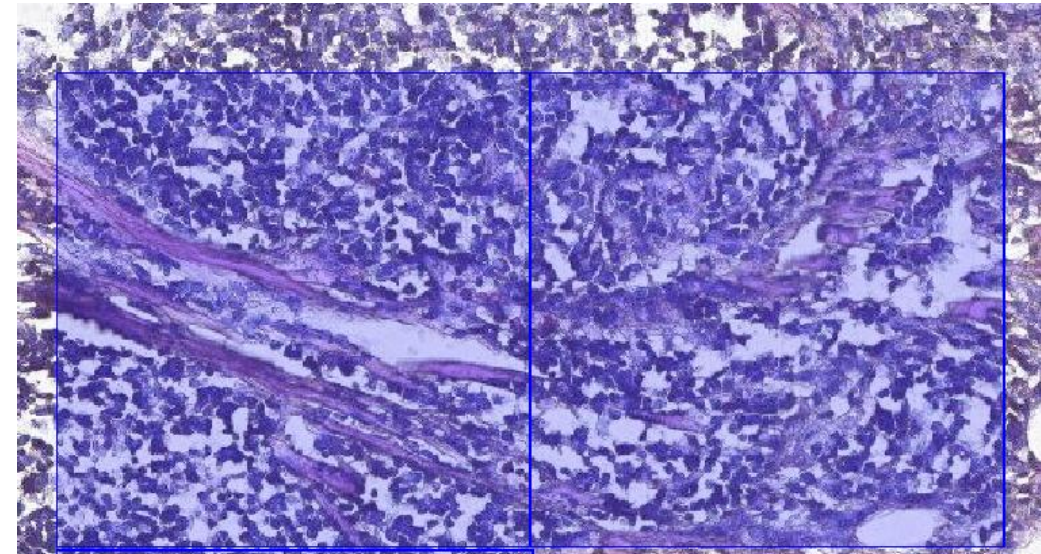


- Section **c4** => **High Score** & **T-cell**

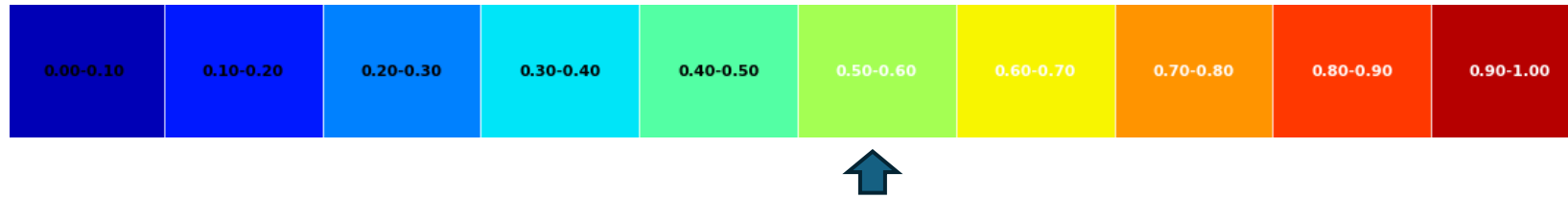
AI 스코어



Astromapper 매핑 결과

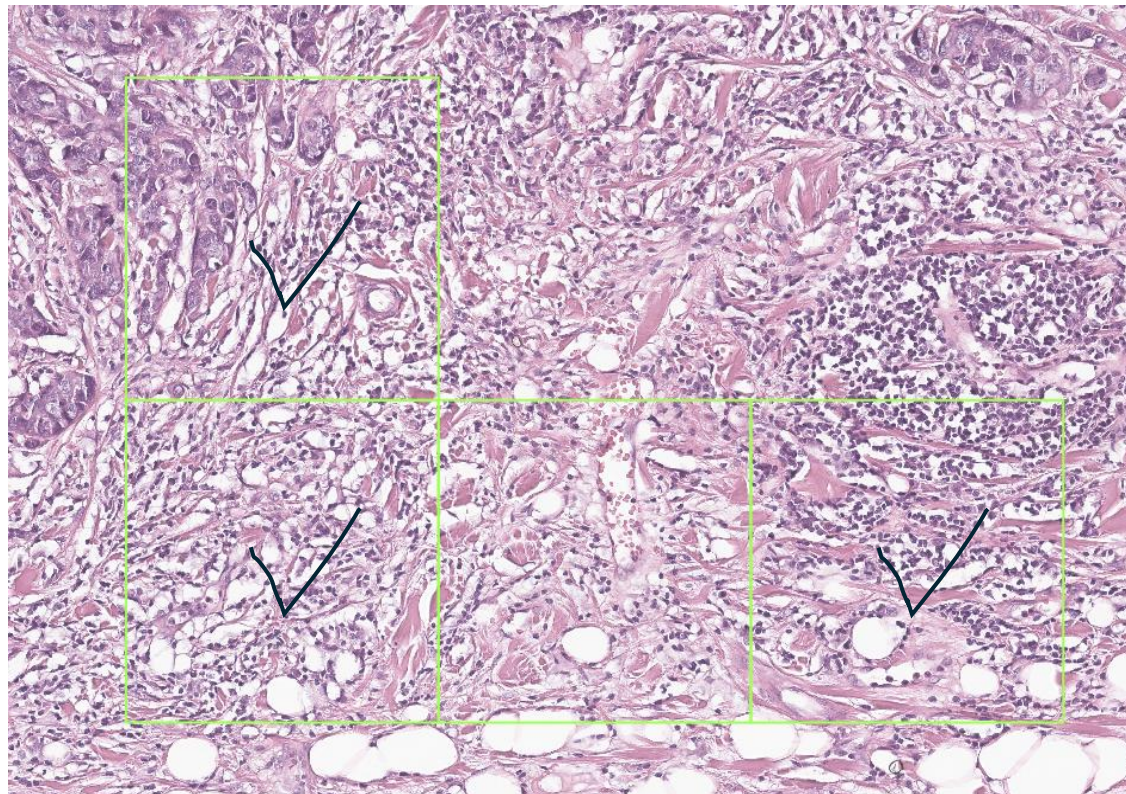


OmixAI 1-085-12 환자 샘플 high-score t-cell regions

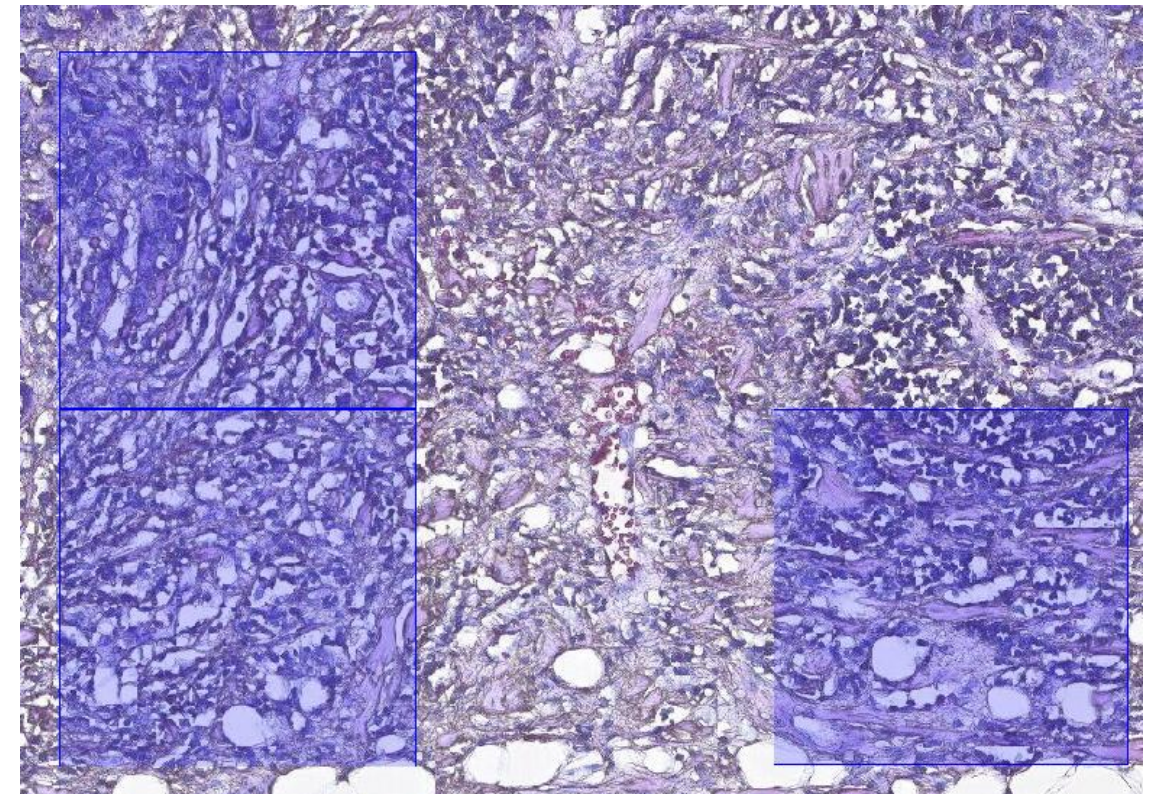


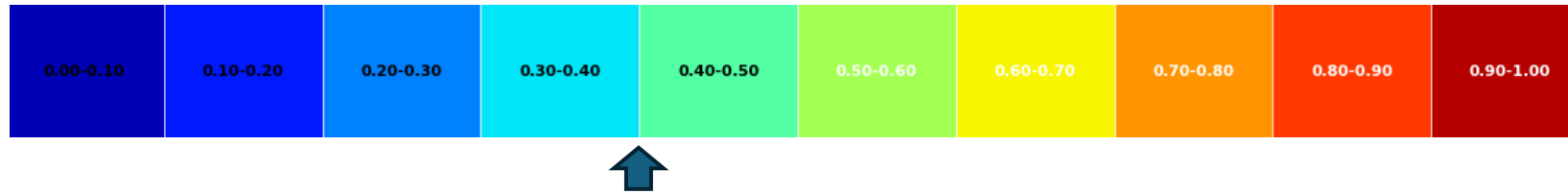
- Section **c5** => **High Score** & T-cell

AI 스코어



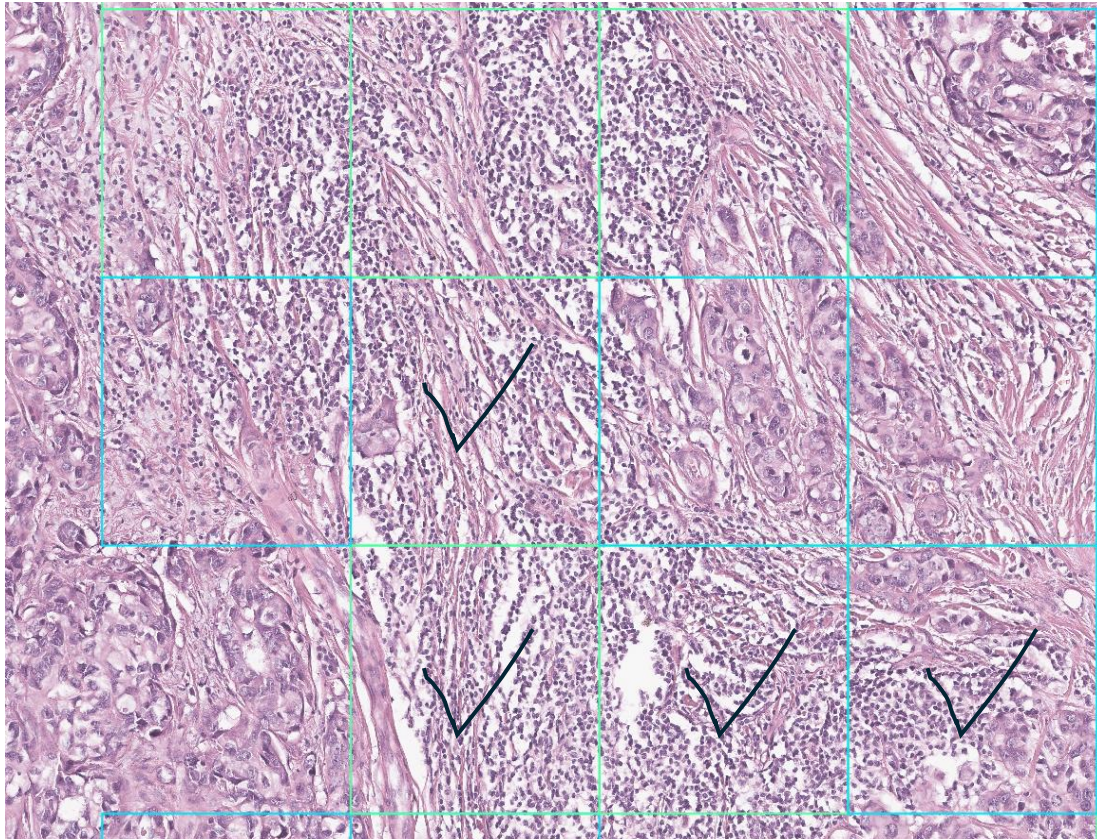
Astromapper 매핑 결과



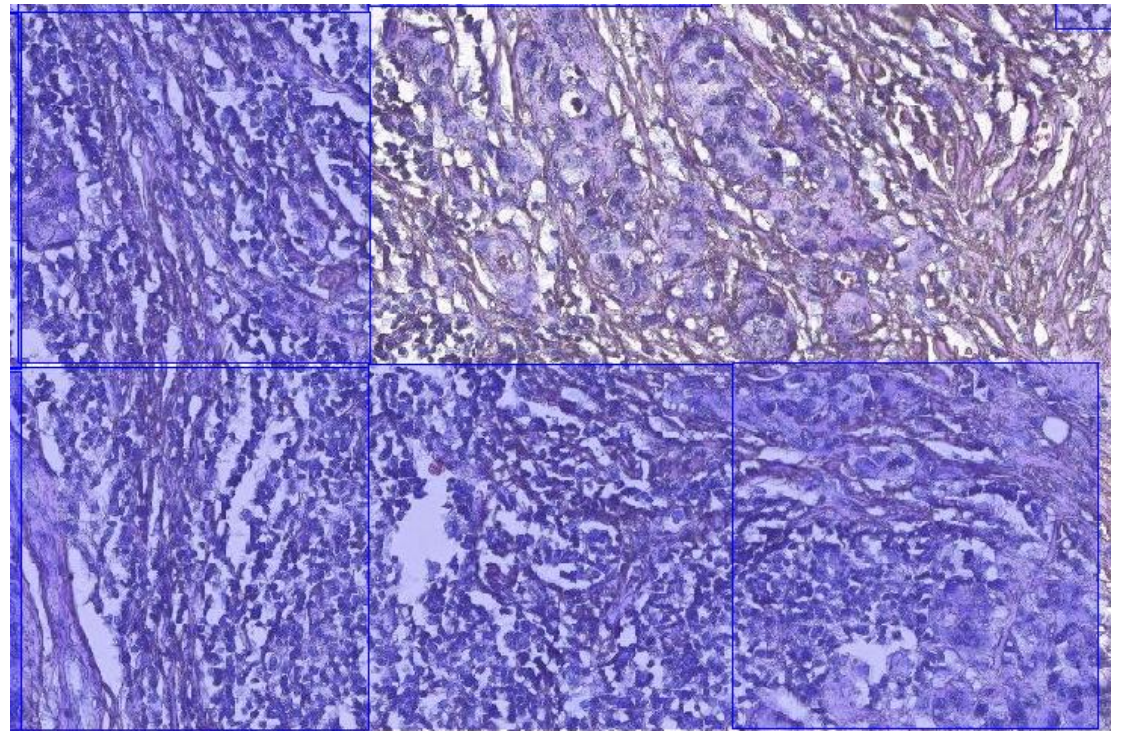


- Section **d1** => Low Score & T-cell

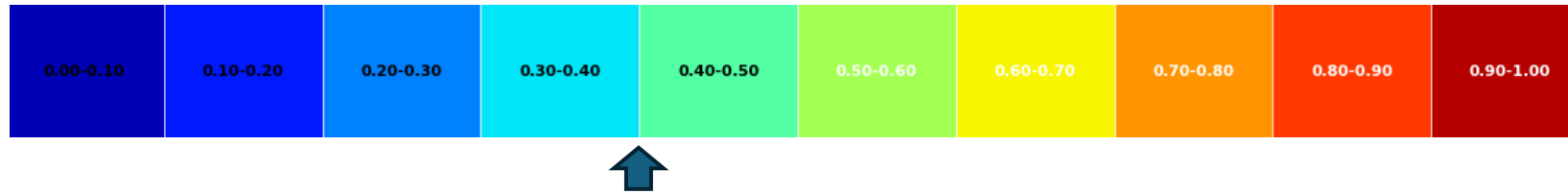
AI 스코어



Astromapper 매핑 결과

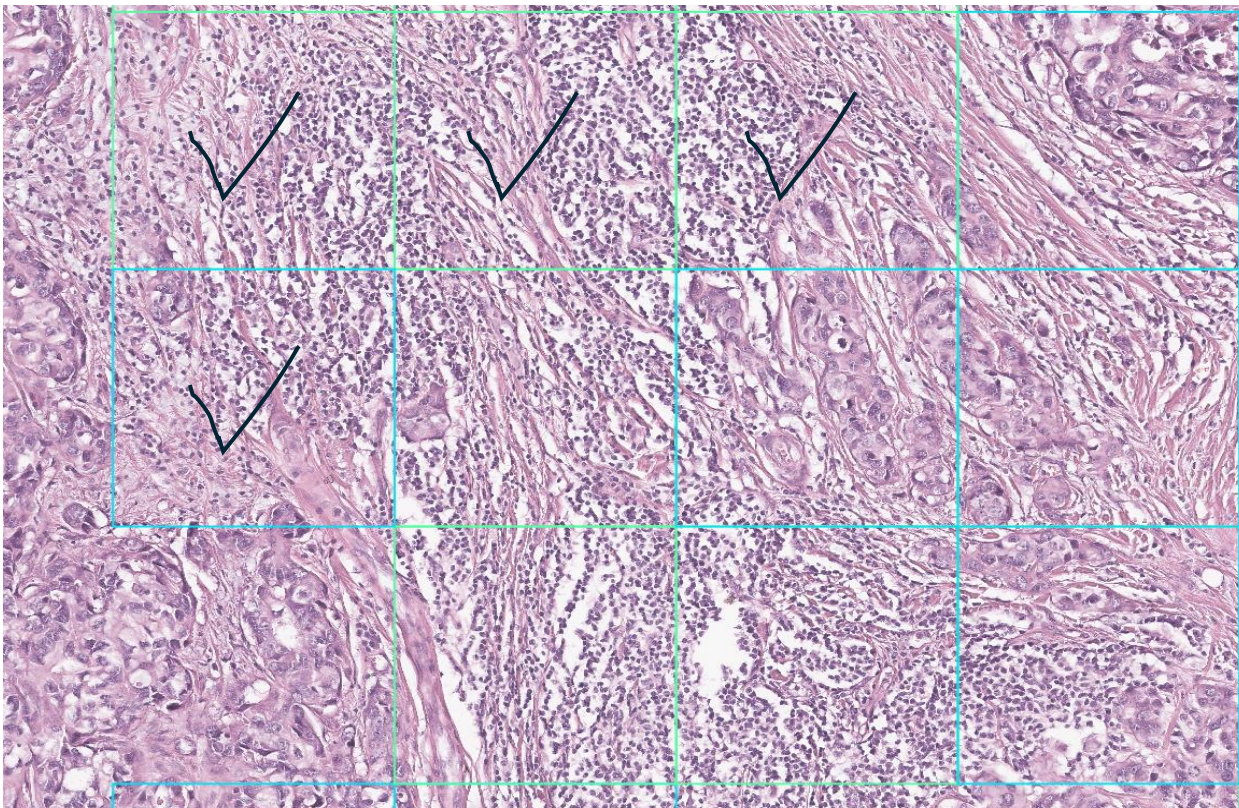


OmixAI 1-085-12 환자 샘플 low-score t-cell regions

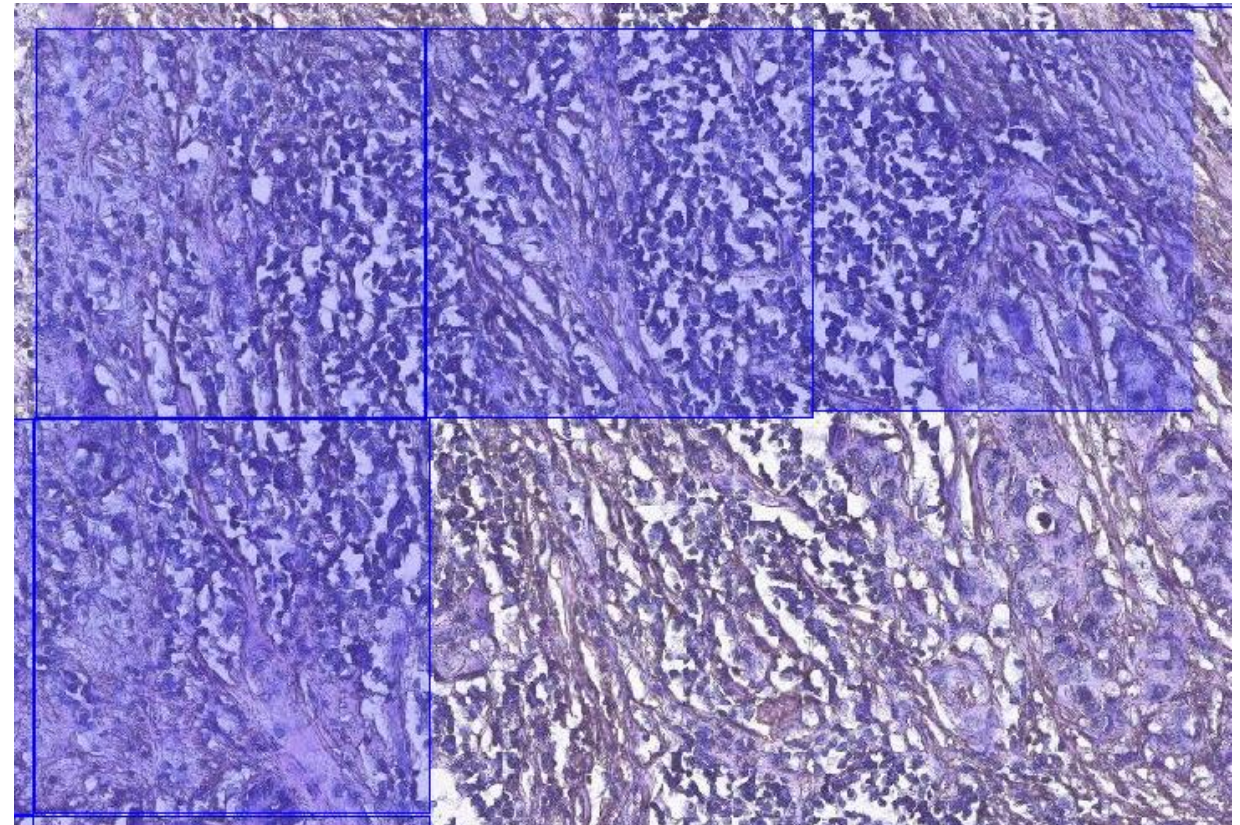


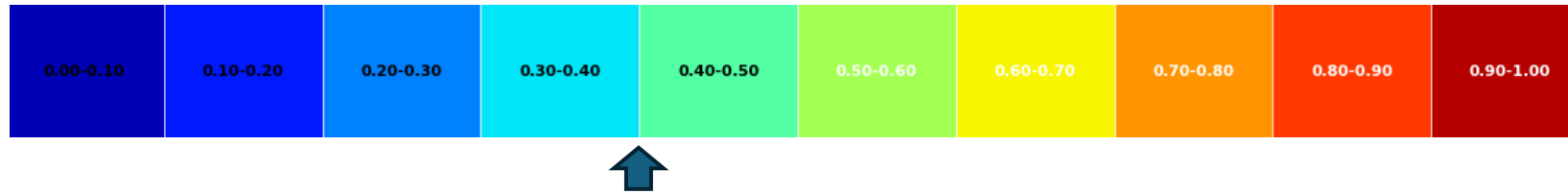
- Section **d2** => Low Score & T-cell

AI 스코어



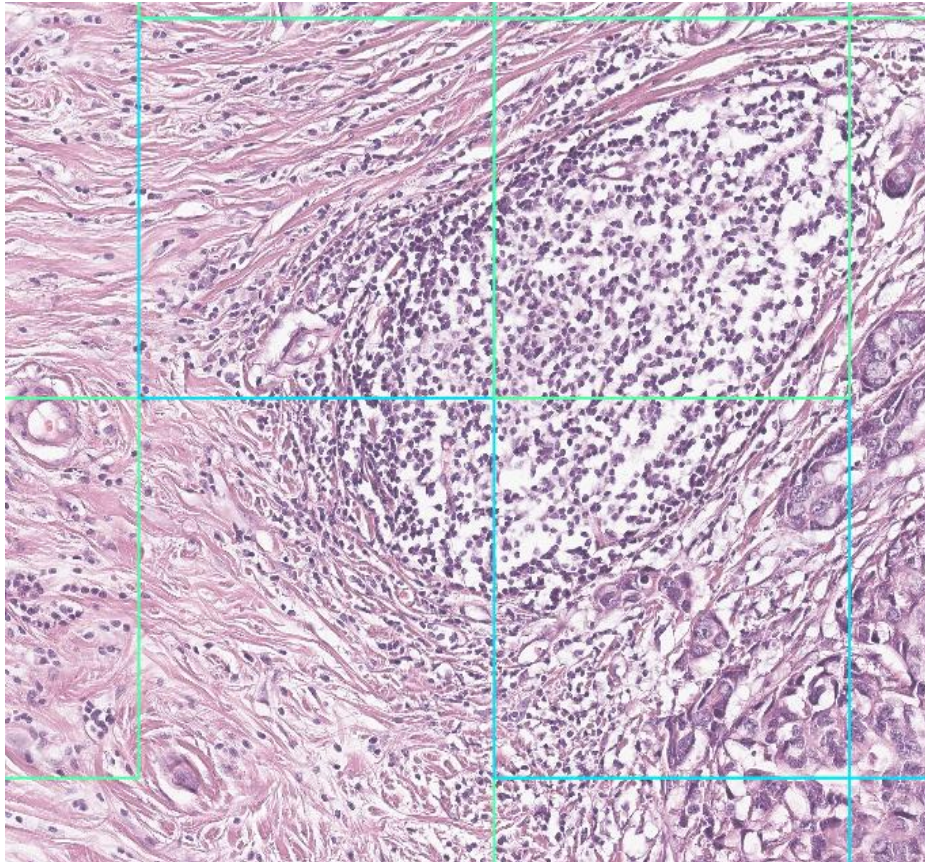
Astromapper 매핑 결과



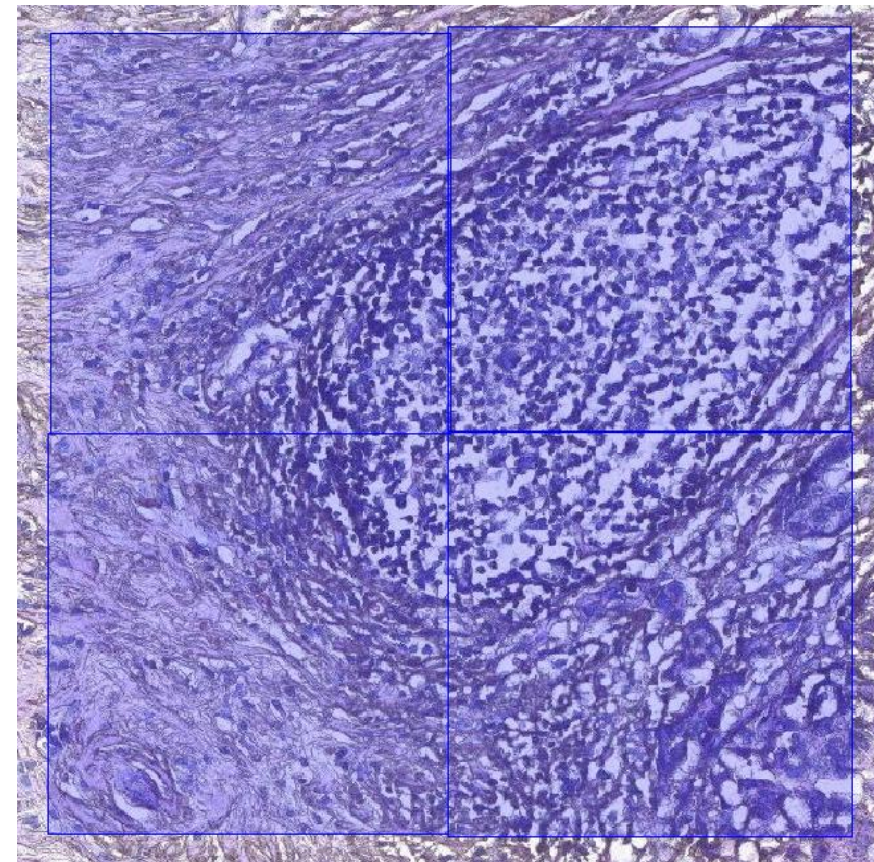


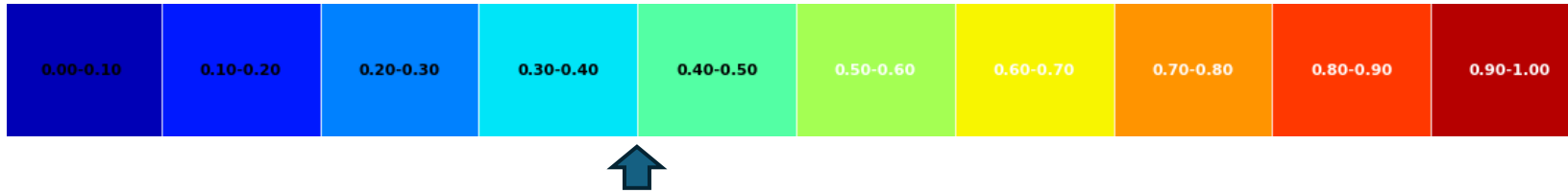
- Section **d3** => Low Score & T-cell

AI 스코어



Astromapper 매핑 결과

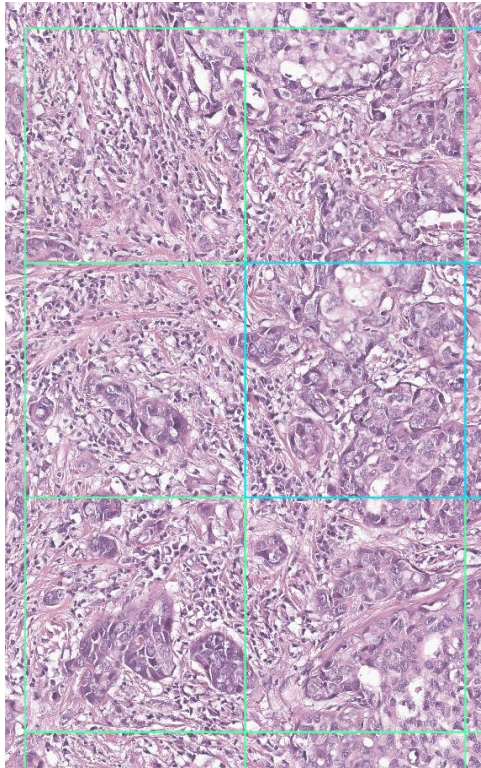




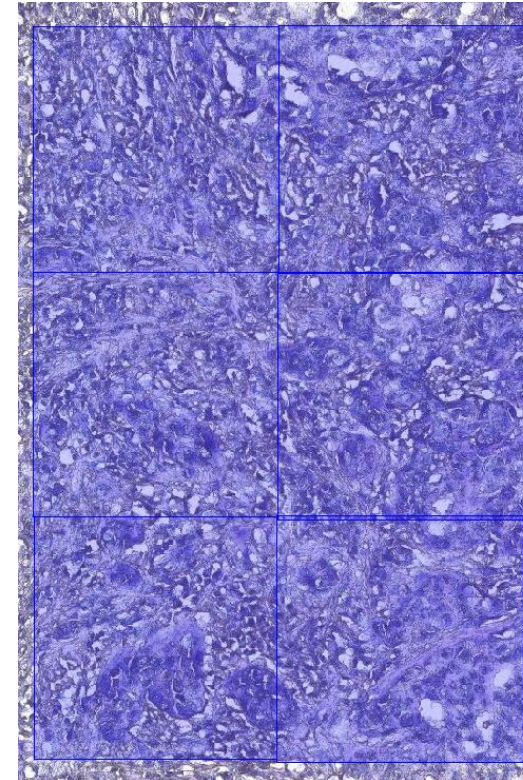
- Section **d4** => Low Score & T-cell

특징 : Tumor의 경계 영역이 아닌, Tumor 안에 있는 Intratumor-TILs (iTILs)

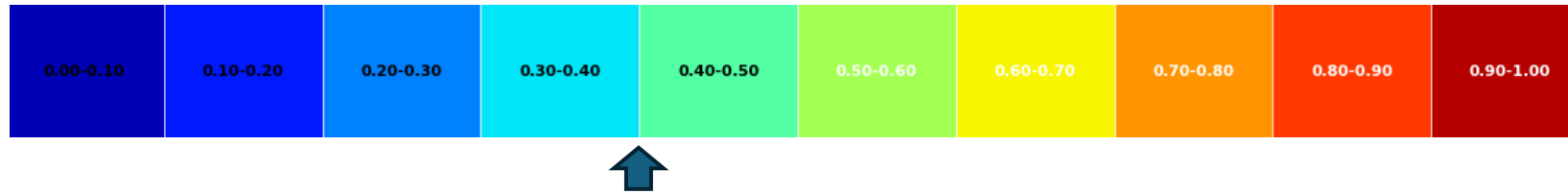
AI 스코어



Astromapper 매핑 결과

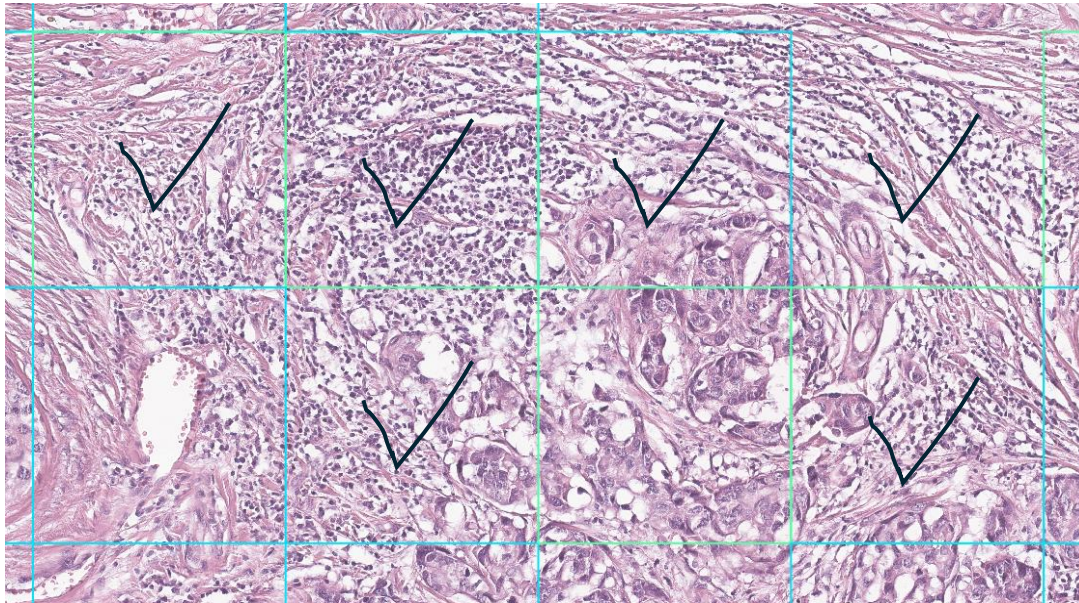


OmixAI 1-085-12 환자 샘플 low-score t-cell regions

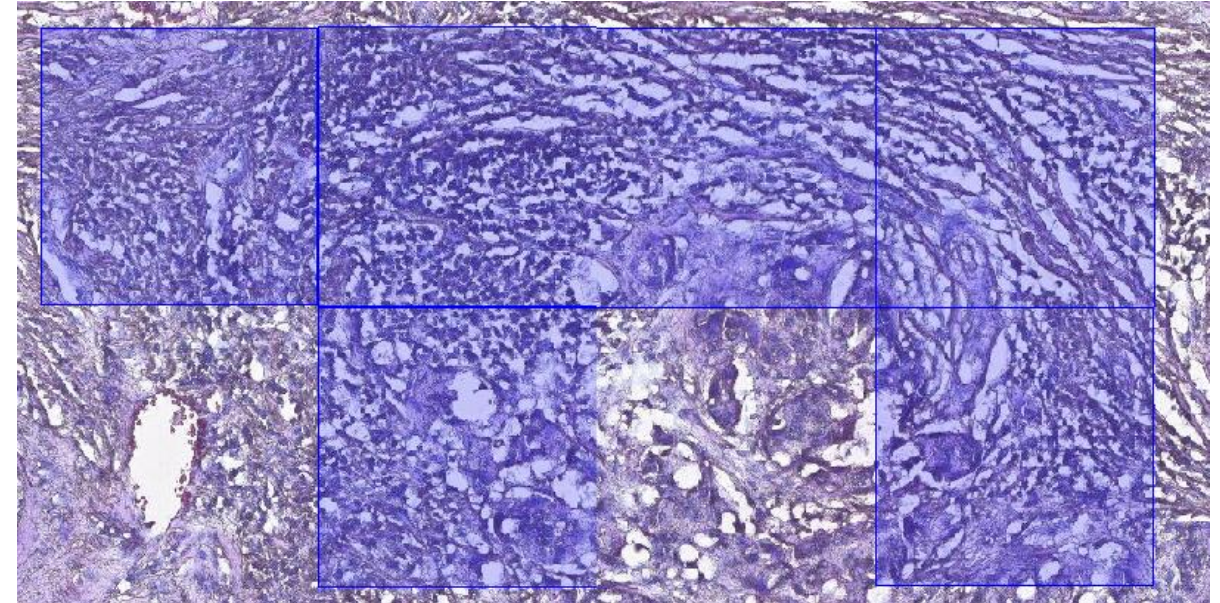


- Section **d5** => Low Score & T-cell

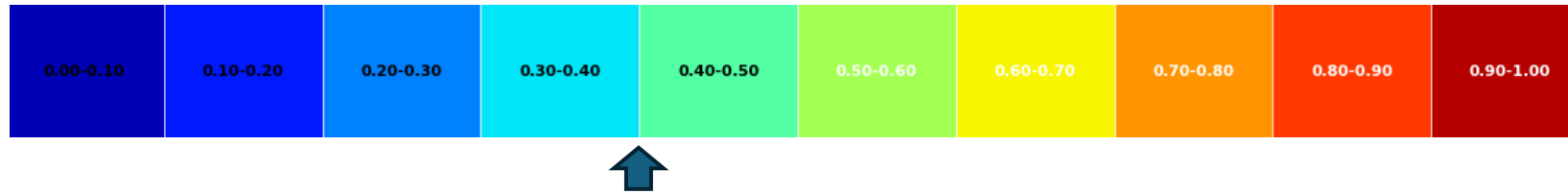
AI 스코어



Astromapper 매핑 결과

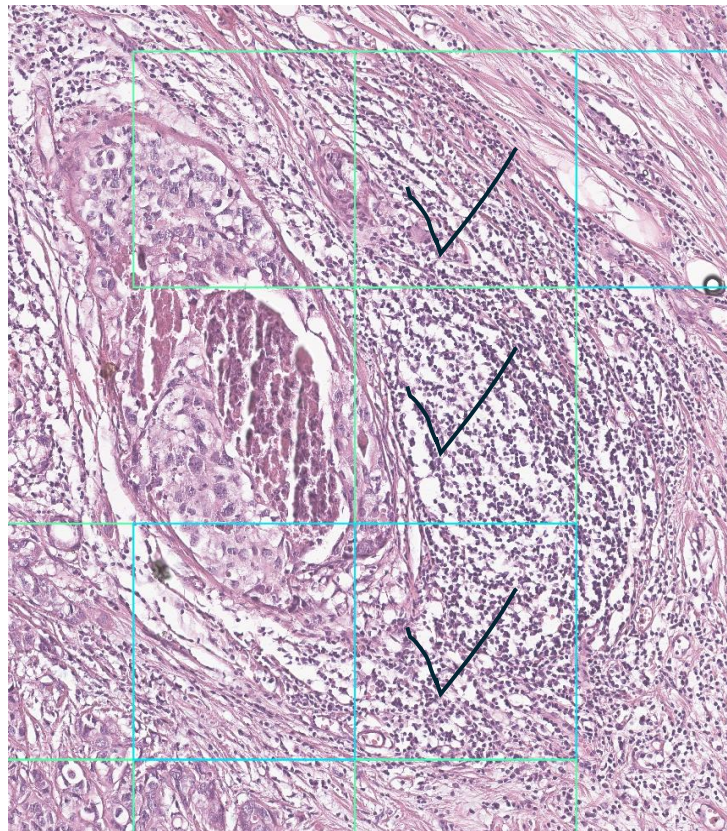


OmixAI 1-085-12 환자 샘플 low-score t-cell regions

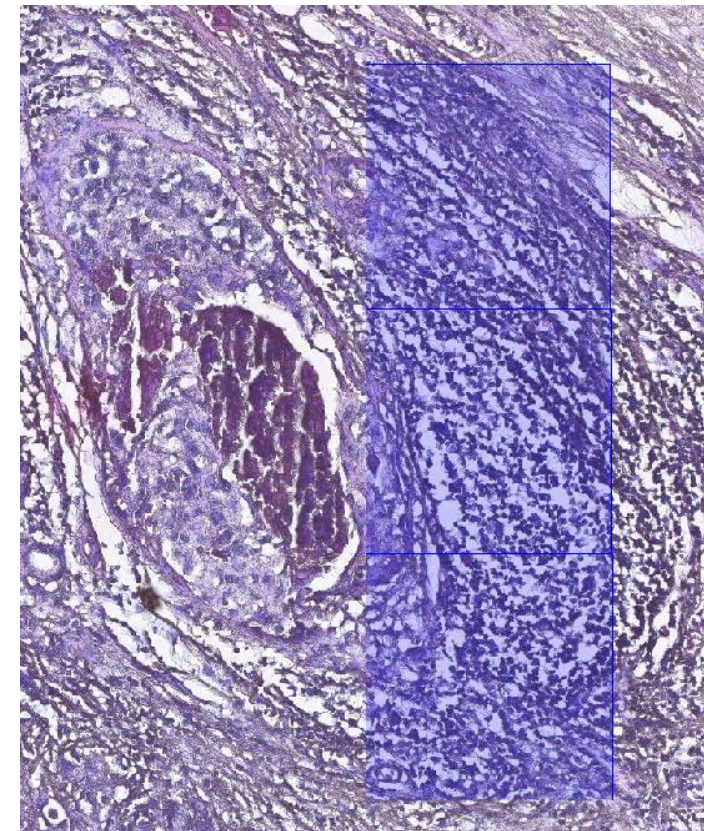


- Section **d6** => Low Score & T-cell

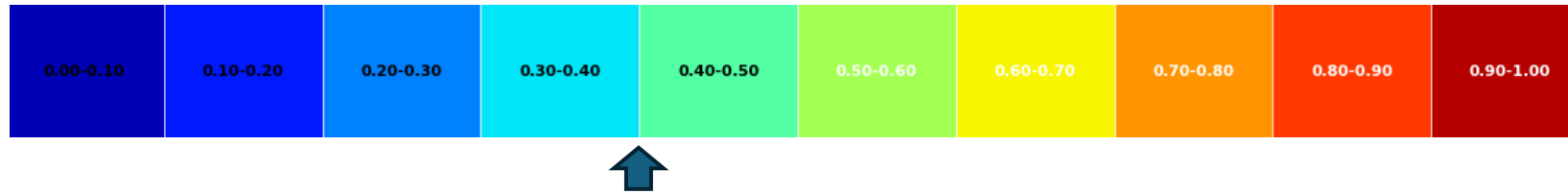
AI 스코어



Astromapper 매핑 결과

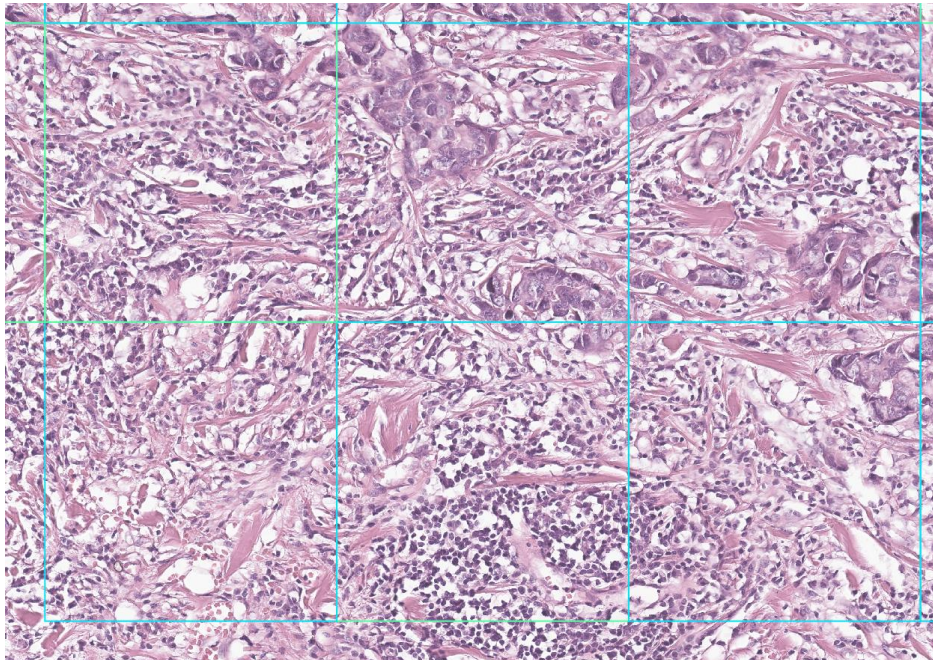


OmixAI 1-085-12 환자 샘플 low-score t-cell regions

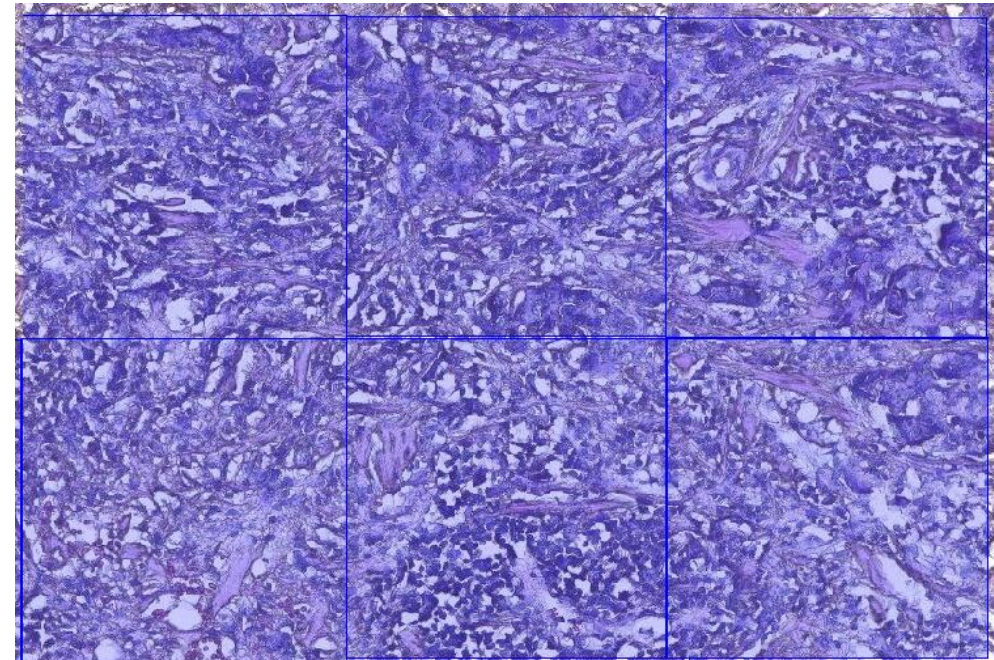


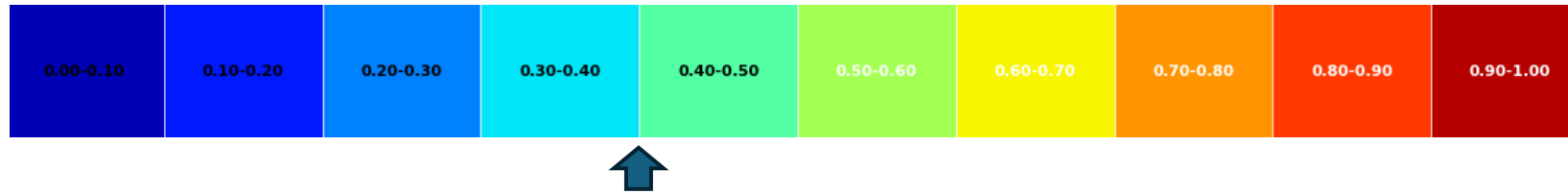
- Section **d7** => Low Score & T-cell

AI 스코어



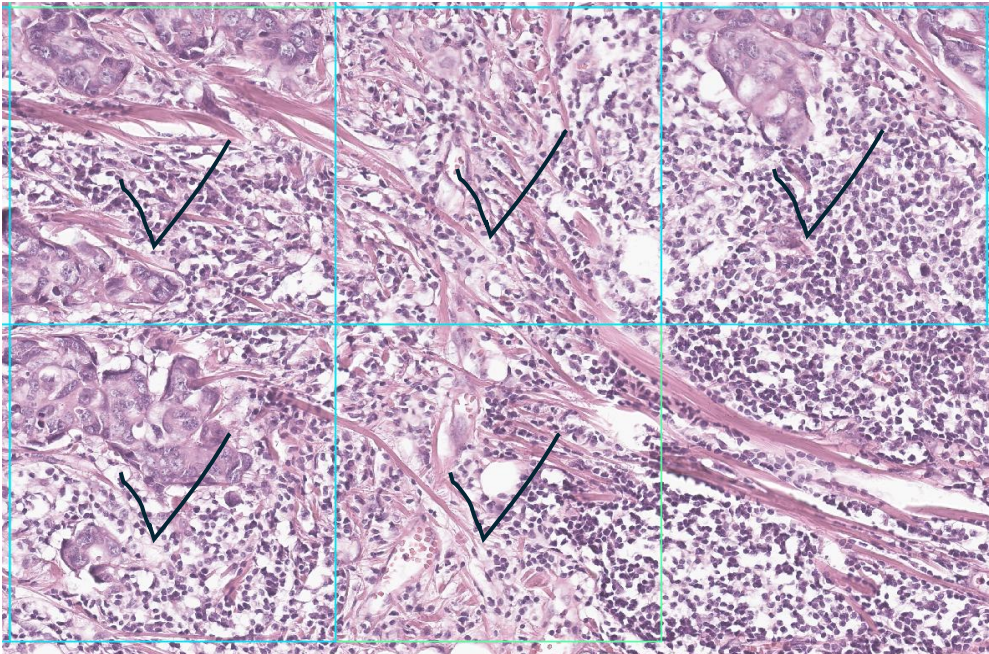
Astromapper 매핑 결과



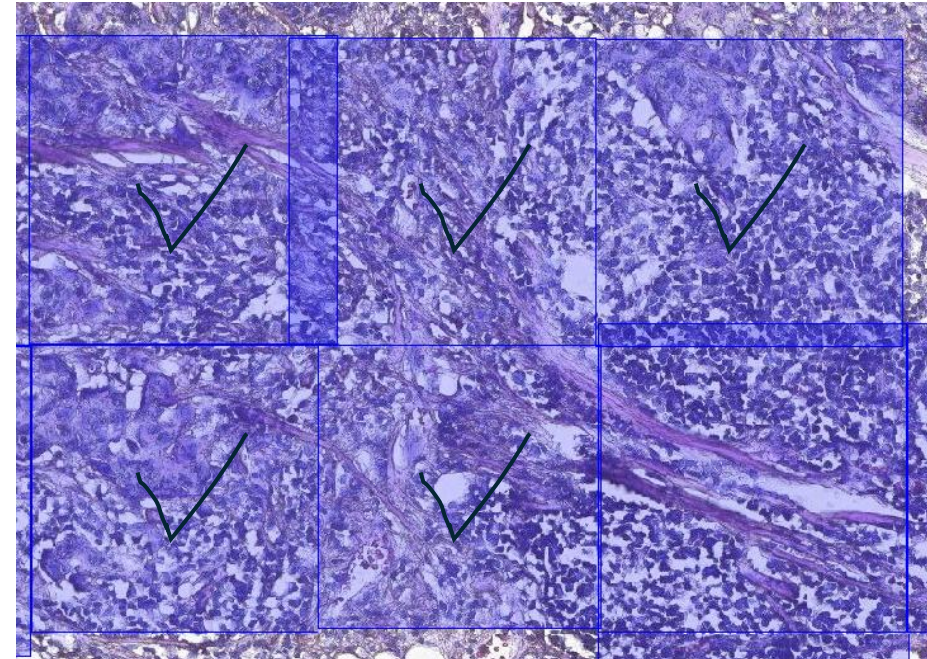


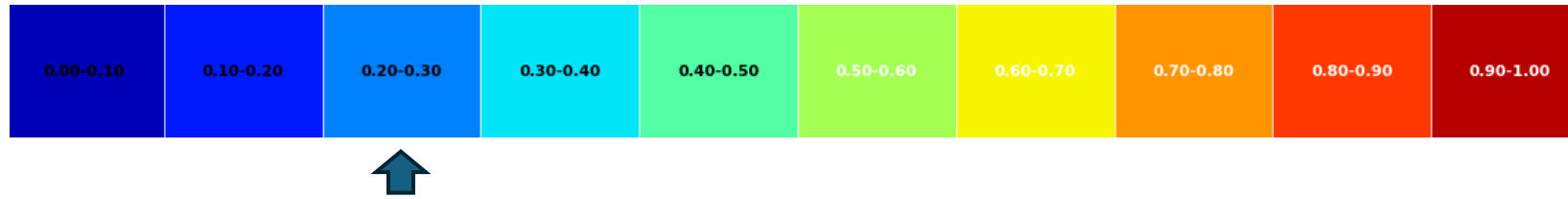
- Section **d8** => Low Score & T-cell

AI 스코어



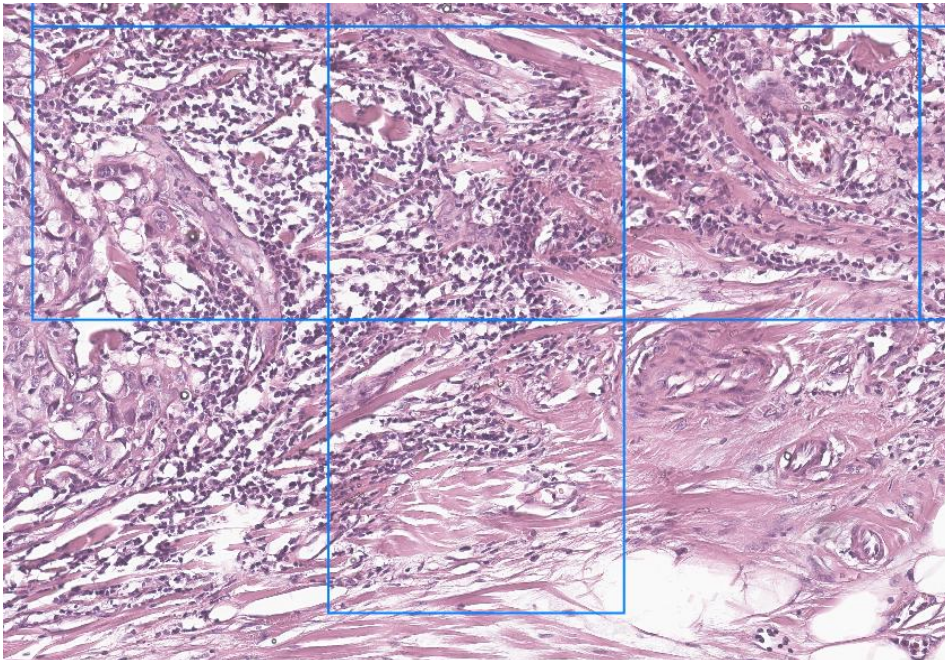
Astromapper 매핑 결과





- Section **d9** => Low Score & T-cell

AI 스코어



Astromapper 매핑 결과

